

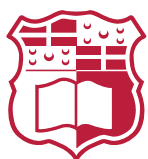
Uncertainty-Conditioned Reinforcement Learning with Evidential Policy Networks for Autonomous Systems

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Abstract

Reinforcement Learning (RL) parking policies are usually developed and tested on the assumption that localisation is reliable. In operational parking areas, however, the accuracy of Real-Time Kinematic (RTK) Global Navigation Satellite System (GNSS) positioning can fall silently from centimetres to metres during a manoeuvre. The pose estimate remains available and plausible throughout, so a vehicle with no channel to carry that change commits to the manoeuvre regardless.

Two uncertainty mechanisms are therefore examined on a Proximal Policy Optimisation (PPO) parking agent, one supplying uncertainty and the other letting the actor use it. On the input side, an Extended Kalman Filter (EKF) fusing RTK GNSS with an Inertial Measurement Unit (IMU) passes its posterior covariance into the observation, quantifying how far the pose estimate can be relied on. On the output side, that signal is consumed by an evidential Normal-Inverse-Gamma (NIG) actor head, which produces per-action epistemic and aleatoric estimates in a single forward pass. Training runs in the CARLA simulator under an always-on GNSS fix-state Markov chain, rewarding the outcome alone. Uncertainty is thus observed but never rewarded, so any uncertainty-dependent behaviour must emerge on its own.

A 2×2 factorial ablation switches each mechanism on and off across three training seeds. Incorporating the covariance into the input of the evidential actor increased parking success by 26.8 to 36.2 percentage points (pp) on the trained lot, more than doubling the rate of the identical actor denied that input, which parked in under 30% of episodes. Each 95% bootstrap Confidence Interval (CI) excludes zero, and all three seeds favour the arm receiving the covariance. Neither mechanism alone explains the gain, as the same input supplied to a standard Gaussian actor yields no measurable benefit, so the signal is made usable by the state-conditioned action variance of the evidential head. The combination operates through emergent caution, with braking intensifying as the filter reports greater positional uncertainty, a Spearman rank correlation of +0.41 against +0.22 for the covariance-blind arm.

The uncertainty the head reports for its own actions is of less use than the uncertainty supplied to it. In a single-head NIG actor the epistemic estimate is the aleatoric one divided by the evidence parameter, so the two channels separate only when that divisor varies across states. Model-free RL contains no term that supervises that divisor, as a sampled action is not its own label, leaving the two channels inseparable in practice.

The contribution is nonetheless the first placement of a NIG evidential head on the actor of a continuous-control policy-gradient agent, alongside the first use of raw EKF covariance as observation input for parking control.

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List of Abbreviations

ADR Automatic Domain Randomisation.

AUC Area Under the Curve.

CAN Controller Area Network.

CEP Circular Error Probable.

CI Confidence Interval.

COG Course Over Ground.

DDS Data Distribution Service.

DER Deep Evidential Regression.

DQN Deep Q-Network.

DRL Deep Reinforcement Learning.

DRMS Distance Root Mean Square.

EDL Evidential Deep Learning.

EKF Extended Kalman Filter.

GAE Generalised Advantage Estimation.

GNSS Global Navigation Satellite System.

GPS Global Positioning System.

GPU Graphics Processing Unit.

HPO Hyperparameter Optimisation.

HUA High Uncertainty Area.

IMU Inertial Measurement Unit.

JSON JavaScript Object Notation.

KL Kullback–Leibler.

LiDAR Light Detection and Ranging.

MDP Markov Decision Process.

MPC Model Predictive Control.

NGC NVIDIA GPU Cloud.

NIG Normal-Inverse-Gamma.

OOD Out-of-Distribution.

pp percentage points.

PPO Proximal Policy Optimisation.

ReLU Rectified Linear Unit.

RL Reinforcement Learning.

ROC Receiver Operating Characteristic.

ROS Robot Operating System.

RTK Real-Time Kinematic.

SAC Soft Actor-Critic.

SB3 Stable-Baselines3.

SLAM Simultaneous Localisation and Mapping.

TD Temporal-Difference.

TSUM Task-Specific Uncertainty Map.

YAML YAML Ain't Markup Language.

ZUPT Zero-Velocity Update.

Glossary of Symbols

$\hat{A}$ Generalised advantage estimate.

α Shape parameter of the Inverse-Gamma component of the evidential head.

β Scale parameter of the Inverse-Gamma component of the evidential head.

c_1 Weighting coefficient of the value-function loss within the PPO objective.

c_2 Weighting coefficient of the entropy bonus within the PPO objective, decayed to a floor within each stage.

d_{along} Signed along-track distance from the parked position, measured along the bay depth axis; the shaping potential uses its magnitude.

d_{cross} Cross-track distance from the bay centreline; the shaping potential uses its magnitude.

ϵ Clip range of the PPO surrogate objective.

f_{align} Corridor heading-alignment factor in $[0, 1]$, at unity when the vehicle is square to the bay axis.

f_{depth} Corridor depth factor in $[0, 1]$, at unity at the parked depth within the bay.

f_{line} Corridor on-centreline factor in $[0, 1]$, at unity on the bay centreline and zero at the corridor half-width.

γ (discount) Discount factor of the Markov decision process, weighting future reward against immediate reward.

γ (evidential) Mean parameter of the Normal-Inverse-Gamma evidential distribution, giving the predicted action mean.

$\mathcal{H}$ Policy entropy, rewarded by the PPO objective to maintain stochasticity.

K Kalman gain.

L^{CLIP} Clipped surrogate objective of PPO.

$\mathcal{L}_{\text{reg}}$ Evidential prior-anchoring regulariser, penalising deviation of α and β from their initialisation prior.

- L^V Value-function loss of the PPO objective.
- λ_{reg} Coefficient weighting the evidential regulariser against the likelihood in the total loss, applied in this work to the prior-anchoring term and raised linearly from zero to its target at the start of each stage.
- ν Virtual-evidence (pseudo-observation) count of the Normal-Inverse-Gamma distribution, governing confidence in the mean.
- P State estimate covariance matrix of the extended Kalman filter.
- ϕ Bay-frame corridor shaping potential: the negative weighted sum of the along-track magnitude, cross-track magnitude and heading error, zero at the perfectly parked pose and negative elsewhere.
- π_{θ} Stochastic control policy with parameters θ .
- Q Process noise covariance matrix.
- r_{clear} Dense clearance nudge: a small penalty for closing on a parked neighbour, gated off once the vehicle is square on the centreline.
- r_{end} Dense endgame bonus: a product of lateral, heading and depth factors with a stop indicator that rewards settling into the bay, bounded below the discounted terminal payment.
- R Measurement noise covariance matrix.
- r_{oob} Dense out-of-bounds charge: a fixed cost per decision spent beyond the drivable boundary, accumulating until the episode terminates.
- r_{prog} Dense progress reward: the per-episode normalised difference of the bay-frame shaping potential between successive decisions.
- R_{target} Progress target, the total dense reward a full clean approach earns across an episode once normalised by the initial potential (value 39.0).
- r_{term} Terminal payment: the sparse end-of-episode reward for success, collision or timeout.
- r_t Total per-decision reward: the additive sum of the progress, endgame, clearance and terminal terms, with no uncertainty term.
- σ_{alea}^2 Aleatoric (data) variance estimate produced by the evidential head.

- σ_{epi}^2 Epistemic (model) variance estimate produced by the evidential head.
- σ_x Posterior standard deviation of the position estimate along the x axis, exposed to the policy as an observation dimension.
- σ_y Posterior standard deviation of the position estimate along the y axis, exposed to the policy as an observation dimension.
- σ_ψ Posterior standard deviation of the heading estimate, exposed to the policy as an observation dimension.
- θ_{err} Absolute heading error from the bay axis, wrapped with 180-degree parking symmetry.
- w_{along} Along-track weight of the corridor shaping potential (value 1.0).
- w_{clear} Coefficient of the clearance nudge, scaling the penalty for approaching a neighbouring parked vehicle (value 0.02).
- w_{cross} Cross-track weight of the corridor shaping potential (value 2.0).
- w_{head} Heading-error weight of the corridor shaping potential (value 3.0).
- x_k State estimate of the extended Kalman filter at filter timestep k .

1 Introduction

Autonomous parking controllers are conventionally designed under the assumption of dependable localisation. The estimator delivering that position also outputs a measure of its reliability, and conventional designs leave that secondary output unused, so putting it to work is the basis of this dissertation. External influences degrade the localisation partway through a manoeuvre, yet the reported position arrives on schedule and in the expected form, carrying no mark of the error it has acquired. Nothing in the observations of a policy developed on that position alone distinguishes an accurate estimate from a wrong one by several metres.

That missing distinction is supplied by the reinforcement learning architecture devised and assessed in this dissertation, which incorporates uncertainty at both ends of the policy. At the input, it receives the posterior covariance of an Extended Kalman Filter (EKF), which quantifies uncertainty in the state estimate independently of the sensors fused to generate it. At the output, an evidential head generates a confidence metric for every selected action. The two mechanisms are subsequently crossed in a factorial ablation under degraded Real-Time Kinematic (RTK) Global Navigation Satellite System (GNSS) positioning, allowing each contribution to be attributed separately. Given both, the vehicle completes the manoeuvre on the trained lot substantially more often than under the identical architecture with either component absent (Section 1.5). The argument for that finding begins with the problem and with the research question that follows from it.

1.1 Problem Definition

Autonomous parking is treated as a continuous control problem in which a vehicle must come to rest within the markings of a conventional perpendicular bay 2.5 m wide. The scenario examined is forward perpendicular entry into a mapped site, so the geometry of the lot is known in advance and no part of the task involves discovering it. The critical factor is instead the accuracy of the vehicle's position estimate at every moment of the manoeuvre.

The estimate is generated by RTK-corrected GNSS fused with an Inertial Measurement Unit (IMU) through an EKF. Its accuracy is not constant, following instead the quality of the satellite solution available at each moment. This study adopts the u-blox ZED-F9P as its target receiver, taking its accuracy figures from the receiver's datasheet [1]. The position is accurate to within 1 cm while the satellite solution holds, then only to within a decimetre and afterwards a metre as that solution is lost. Those

tiers form a hierarchy of fix states, ordered by the quality of the correction the receiver holds.

A policy trained on that estimate alone receives no evidence of a decline along the hierarchy, so the bay entry relies on a position that may be several metres inaccurate. A failure of that kind is judged by more than its consequences, as the hazard analysis mandated by ISO 26262 [2] for a road-vehicle is scored partly on controllability, which asks whether a driver or operator can intervene in time to prevent harm. A decline that goes unregistered is uncontrollable in that sense, as nothing prompts the intervention that a favourable controllability score assumes.

The information the policy can use to mitigate this is nonetheless available, as the filter already reports the covariance of its own estimate, a direct measure of how reliable that position is. Supplying that quantity to the policy as an observation therefore closes the gap at the input. A second gap remains at the output, where a standard actor emits a control carrying no indication of how reliable it is. An evidential head, by contrast, attaches a measure of reliability to each action. Both remedies are available in principle, yet neither has been tested on a parking policy while the satellite solution degrades mid-manoeuve (Section 2.3), leaving the following question open:

Does conditioning a continuous-control parking policy on its localisation uncertainty and incorporating an evidential output head measurably improve parking success and precision as localisation quality degrades, relative to the same architecture with each mechanism removed?

The inquiry is intentionally comparative: the focus is on whether two separable mechanisms produce a measurable improvement over controls that differ from them by exactly one characteristic. The answer presented in Chapter 4 therefore takes the form of controlled contrasts rather than a single performance metric.

1.2 Motivation

Conditioning a parking policy on its localisation uncertainty, the first mechanism named in the research question, has precedent, although the systems discussed in Section 2.2 each derive that signal from an indirect source. The covariance reported by the localiser at every step therefore remains unutilised. This study draws on it directly, deriving three features from that covariance to form an input available on any vehicle equipped with the filter.

The uncertainty a policy reports over its own chosen actions is a separate quantity from the one the filter supplies, and its two sources call for different responses. Noise inherent to the sensors is best answered by slowing the vehicle, since

no amount of further data removes it, whereas a situation the policy has seldom met in training warrants handing control to an operator instead. A handover needs someone to receive it, and the proposed deployment environment is an instrumented outdoor parking facility, where an operator is on hand to take over. The dependability of the reported estimate then governs whether it can trigger that handover, a question assessed in Section 4.8.2.

Reporting both components is nonetheless constrained by what a parking controller can afford to compute. Decisions are issued at a fixed rate on embedded hardware, where the available compute is shared with perception, planning and the rest of the control stack. The sampling estimators discussed in Section 2.1.2 obtain their uncertainty from repeated forward passes, which multiplies the inference cost of every decision by the number of samples drawn. Evidential Deep Learning (EDL) instead requires only a single pass [3], making it affordable at that rate, and the head constructed upon it is detailed in Section 3.5.

1.3 Aims and Objectives

The filter covariance at the input and the single-pass evidential head at the output together shape the aim of this work, which is to develop and evaluate an uncertainty-conditioned reinforcement learning architecture that enables an autonomous parking agent to degrade gracefully as localisation quality declines. Four objectives decompose that aim, each depending on its predecessor and progressing from the components developed to the experiments that isolate their contributions.

- O1** *Honest localisation uncertainty.* An EKF localisation framework is implemented in the simulator, integrating RTK-GNSS and IMU measurements under datasheet-derived noise models, such that the covariance it reports is an honest predictor of its true pose error. The objective is assessed by a calibration analysis in the evaluation (Section 4.2), where the reported position standard deviation is compared with the actual position error.
- O2** *An evidential actor.* An evidential variant of Proximal Policy Optimisation (PPO) is developed in which the actor produces Normal-Inverse-Gamma (NIG) parameters, providing the single-pass uncertainty estimates motivated in Section 1.2. Training such a variant to convergence first requires the variance-collapse pathology of this design to be mitigated, which the methodology addresses through a purpose-built head construction (Section 3.5). Convergence alone is an insufficient criterion, since the quantities the trained head reports must also be informative, so the objective is assessed on three

counts. The training record is first checked for stability across all seeds and curriculum stages, alongside the curriculum that produced it in Section 3.7. The uncertainty reported at runtime is then examined for how it behaves in Section 4.3, and the epistemic and aleatoric estimates are finally tested for separability in Section 4.3.1.

- O3** *Isolated contributions.* The contribution of each uncertainty layer is isolated through a two-by-two factorial ablation that crosses covariance observation with the evidential head, producing four de-confounded arms, each trained under three seeds. Every arm is then evaluated according to parking success, final position error and an outcome taxonomy. The objective is therefore assessed by pooling those arms into covariance contrasts reported with bootstrap confidence intervals in Section 4.4, and the same contrasts are then broken down by seed in Section 4.7 to test whether the intervals survive retraining.
- O4** *The mechanism.* Any performance gain is shown to operate through uncertainty-conditioned behaviour, in that the vehicle measurably slows, brakes harder and turns less sharply as EKF uncertainty rises. Such behaviour is not instructed by the reward, which scores only the outcome of the manoeuvre and contains no term referring to uncertainty, so any conditioning of this kind is necessarily emergent rather than imposed. The claim is assessed by an analysis of correlation and level presented in Section 4.6.1, where brake command, speed, yaw rate and action jerk are each grouped into bands according to the EKF standard deviation and then compared across arms.

1.4 Proposed Solution

The architecture that realises the four objectives above is a dual-layer uncertainty architecture, depicted in Fig. 1.1 and developed in full in Chapter 3. Its two layers align with the two mechanisms motivated in Section 1.2, as the input layer incorporates localisation uncertainty into the observation, while the output layer represents uncertainty over the selected action.

On the input side, three of the thirteen dimensions of the observation vector are pose standard deviations derived from the EKF covariance diagonal. The standard deviation along each positional axis gives σ_x and σ_y , and the standard deviation of the heading gives σ_ψ . The remaining ten convey the vehicle state, the relative target pose and five Light Detection and Ranging (LiDAR) obstacle clearances. These three deviations arise from a standard Robot Operating System (ROS) EKF node operating live within a training loop, fusing the noisy GNSS and IMU streams rather than

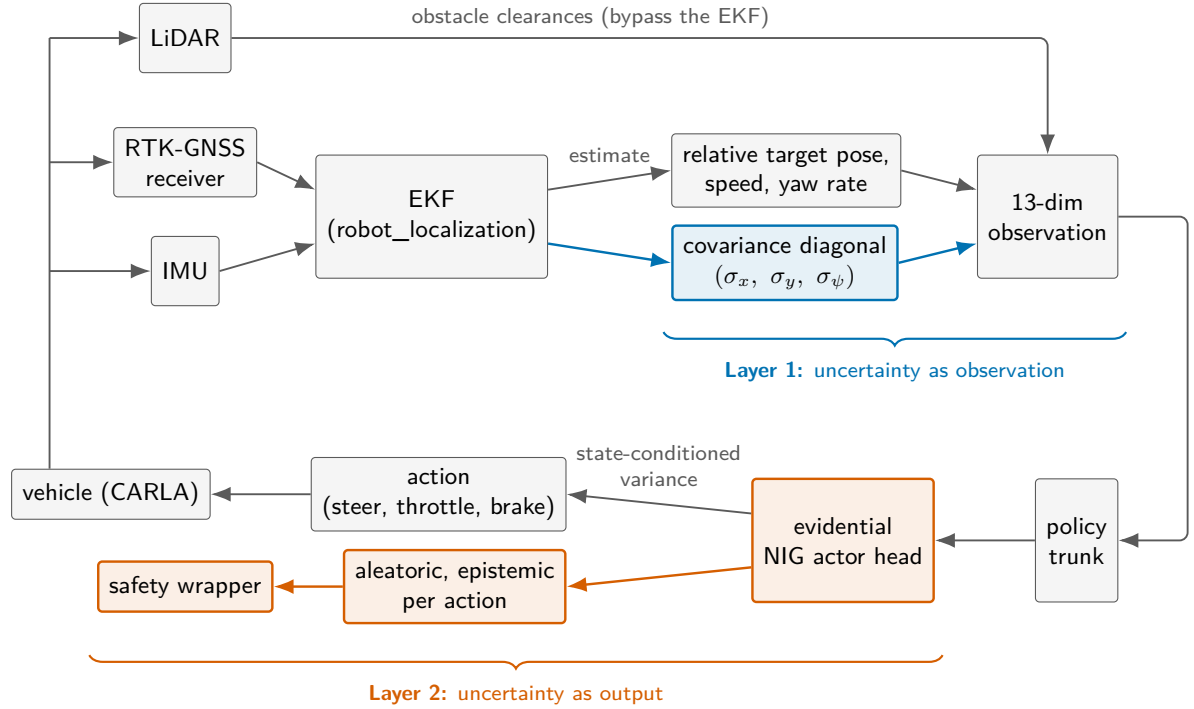


Figure 1.1 The two-layer uncertainty architecture.

sampling a prescribed noise model. The covariance observed by the policy is consequently generated by the same estimation machinery that a deployed vehicle would carry. The complete observation specification and the localisation pipeline behind it are detailed in Sections 3.3 and 3.4.

On the output side, the standard Gaussian actor of PPO is set against an evidential head [3], a construction described in Section 2.1.3. Its output is the four parameters of a NIG distribution for each of the three action dimensions, which are steering, throttle and braking. Under the evidential arm, the variance used to sample actions is aleatoric and conditioned on the state, and therefore it varies from one state to the next, a capability the Gaussian formulation lacks. The same forward pass simultaneously produces an epistemic estimate, making both quantities available to the runtime safety wrapper. Adapting the evidential formulation to model-free reinforcement learning, in which no ground-truth action target exists, required a redesigned regulariser and a new initialisation protocol besides the variance-collapse mitigation already noted, all documented in Section 3.5.

Both layers are trained together in CARLA 0.9.16 [4] within a bespoke flat parking environment, under a six-stage single-phase curriculum (Section 3.7) that widens one difficulty axis at a time. Every observation channel, including the GNSS fix state, is nonetheless present from the initial stage, meaning that successive stages increase the difficulty of the manoeuvre rather than the information available to the policy. Both uncertainty layers are nonetheless observed rather than rewarded during

training, as the reward is based solely on outcomes (Section 3.6), and the safety wrapper that acts on their output is active only during evaluation and deployment (Section 3.9).

1.5 Summary of Results and Contributions

Incorporating the EKF covariance into the evidential actor enhanced parking success by 26.8 to 36.2 percentage points (pp) on the trained lot, and every seed favoured the covariance, with a margin of +16.0 pp in the closest pairing. The same input, meanwhile, yielded no measurable benefit on a standard Gaussian actor, a null result reported with the headline contrasts in Section 4.4 and with the interaction analysis in Section 4.4.2. The uncertainty that the same head reports for its own actions proves less tractable than the uncertainty supplied to it, as the epistemic and aleatoric estimates remain inseparable in practice under model-free reinforcement learning. A failure-prediction test reported in Section 4.3.1 attributes the cause to the construction of the head rather than to the conditions under which it was evaluated. The remedy is consequently architectural as well, requiring a head grounded in the training distribution, which is proposed as future work in Section 5.2.

Each objective is assessed against the evidence obtained for it, and the five contributions are stated in full in Sections 5.1.1 and 5.1.2.

1.6 Document Structure

The foundational theory for each layer of the architecture is established in Chapter 2, which addresses reinforcement learning, uncertainty quantification and EKF localisation. Against that foundation, three strands of existing work are evaluated: uncertainty-aware reinforcement learning for driving, evidential approaches within reinforcement learning, and reinforcement learning for autonomous parking. None of these strands couples a localisation-uncertainty observation with an actor capable of responding to it, and that deficiency is the research gap addressed by the proposed architecture.

The design decisions mandated by the gap are set out with their justifications in Chapter 3. The input layer is introduced first, covering the parking environment and the localisation pipeline that supplies its covariance signal. The output layer follows in the form of the evidential actor that consumes that signal. Both layers are then trained together under the curriculum, and the ablation design isolates the contribution of each. The runtime safety wrapper is addressed last, as it is the only component established around deployment rather than training.

In Chapter 4, the assembled system is assessed against objectives O1 to O4 across four conditions and three seeds. Two uncertainty channels are scrutinised before any performance claim rests on them, followed in turn by task performance, a degradation analysis, a behavioural mechanism and an evaluation based on consistency across seeds. A closing section bounds what has been demonstrated, stating each limitation together with its cause.

The work as a whole is assessed in Chapter 5, where every objective is evaluated against the evidence reported for it. Five contributions are derived from that evidence, two of them claims of novelty addressing the research gap established in Chapter 2, while the remaining three rest solely on the measured results. The same results leave specific questions unanswered, which form the closing proposals for further work.

2 Background and Literature Review

Six pillars support every component of the system, with the basis for each set out in Section 2.1. The learner is considered first, founded on policy-gradient reinforcement learning and PPO. Uncertainty in its predictions then divides into aleatoric uncertainty, which concerns noise in the data, and epistemic uncertainty, which concerns the model’s own ignorance, with EDL estimating the epistemic component in a single pass. A separate form of uncertainty concerns the vehicle’s actual location, reported as a covariance through EKF localisation using RTK satellite positioning. Curriculum learning organises the learner’s exposure to that signal, while the evaluation methodology of Deep Reinforcement Learning (DRL) determines how far the resulting measurements may be trusted.

Those same pillars provide the axes for comparing prior work in Section 2.2, enabling systems that address different aspects of the problem to be read against one another. Omissions on the input and output sides of a learned controller emerge from that comparison, and both are consolidated into a single gap statement in Section 2.3.

2.1 Background

2.1.1 Reinforcement Learning and Proximal Policy Optimisation

Reinforcement Learning (RL) treats sequential decision-making as a Markov Decision Process (MDP), characterised by a state space $\mathcal{S}$, an action space $\mathcal{A}$, dynamics $p(s', r \mid s, a)$ that determine the transition probabilities and expected rewards, and a discount factor γ constrained by $0 \leq \gamma \leq 1$ [5]. The discount factor is maintained strictly below one throughout, which keeps the return finite even when the sum runs over an unbounded number of steps. A policy $\pi_\theta(a \mid s)$ maps the states of the process to actions through neural-network weights θ , which are optimised to maximise the expected discounted return $\mathbb{E}[\sum_t \gamma^t r_t]$. Policy-gradient techniques achieve this by modifying θ according to an estimated return gradient, computed with respect to the action representation the network parameterises. In the continuous-control setting adopted here, that representation takes the form of a distribution over actions rather than a single probability for each action. Steering and throttle commands are therefore selected on a continuous scale rather than from a predetermined set of levels. The primary vulnerability of these techniques is that the gradient is derived from data gathered by the policy itself: the estimate is valid only while θ stays near its value at collection time, a constraint that a substantial update contravenes.

This instability is mitigated in PPO [6] using a clipped surrogate objective. The probability ratio $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ quantifies the extent to which an update has altered the policy. It equals one before any modification and diverges from one as θ evolves. The objective limits the benefit obtained from that deviation:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (2.1)$$

where $\hat{A}_t$ estimates the advantage of action a_t , and ϵ defines the range $[1 - \epsilon, 1 + \epsilon]$ to which the probability ratio $r_t(\theta)$ is confined. The clipped and unclipped terms are identical for any ratio within that range, so the update continues unchanged. They diverge once the ratio moves outside it, where the lesser of the two is taken: no further gain is credited once the ratio moves out, while the associated penalty still applies in full. A batch of data can thus be reused across several gradient epochs without the policy deviating significantly from the distribution that generated it.

The advantage $\hat{A}_t$ entering Eq. (2.1) is itself produced by Generalised Advantage Estimation (GAE) [7], which blends multi-step Temporal-Difference (TD) errors into a single exponentially weighted sum:

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \quad (2.2)$$

The term δ_{t+l} is the TD error l steps after t , combining the reward with V_ϕ , a learned value function estimating the expected return from a given state. The weight $\lambda \in [0, 1]$ determines how many of these terms the outer summation of Eq. (2.2) retains, trading the bias introduced by reliance on V_ϕ against the variance introduced by further reward terms. Setting $\lambda = 0$ occupies one end of that trade-off, reducing the sum to the single term δ_t , while $\lambda = 1$ occupies the other, leaving the full discounted reward sequence from t onward, offset by the baseline $V_\phi(s_t)$. Between those two extremes, an intermediate value of λ retains every term but discounts each successive one by a further factor of $\gamma \lambda$. A TD error occurring many steps after t therefore contributes almost nothing to the sum, while one occurring close to t dominates it. This weighting matters in long-horizon tasks for which a terminal outcome must propagate back through numerous decisions. Every setting of λ nonetheless depends on V_ϕ , which is not supplied but learned alongside the policy, its weights ϕ adjusted to minimise the following regression loss against a return target:

$$L^V(\phi) = \mathbb{E}_t \left[(V_\phi(s_t) - \hat{R}_t)^2 \right] \quad (2.3)$$

where $\hat{R}_t = \hat{A}_t + V_\phi(s_t)$ is the return implied by the advantage estimate, so the same λ governs both the advantage and the regression target. The three components are

subsequently combined into the full PPO objective:

$$L(\theta, \phi) = \mathbb{E}_t [L^{\text{CLIP}}(\theta) - c_1 L^V(\phi) + c_2 \mathcal{H}[\pi_\theta](s_t)] \quad (2.4)$$

Policy entropy is denoted by $\mathcal{H}$, while c_1 and c_2 are the weighting coefficients. Once assembled, the objective is optimised as a whole, so the policy parameters θ and the value parameters ϕ are updated in one gradient step rather than in alternating phases.

Two mechanisms beyond the clipping in $L^{\text{CLIP}}(\theta)$ restrict how far an update can move the policy. The first is the entropy term, introduced by Williams and Peng [8] and incorporated into deep actor-critic training by Mnih et al. [9], which rewards a policy for maintaining stochasticity and prevents early convergence to a suboptimal deterministic policy. The clip nonetheless bounds the probability ratio of each action independently, so numerous individually permitted shifts can still accumulate into a large change in the policy as a whole. The second mechanism closes that gap from the training code rather than the objective. In the Stable-Baselines3 (SB3) implementation [10, 11], any remaining minibatches of an update are discarded once the approximate Kullback–Leibler (KL) divergence between the data-collecting and updated policies exceeds 1.5 times a configured target.

That safeguard exists because minibatches from a single collection phase are applied across several optimisation epochs before the data is discarded, an on-policy approach that trades sample efficiency for stability [5, 6]. The trade-off is modest for the training runs of this study, which draw on a single high-fidelity CARLA instance [4]. It dominates the wall-clock cost and reduces the gradient steps to a marginal share of it, so revisiting a batch would deliver negligible savings. Several passes over each one nonetheless remain valid under the clipped objective, recovering a portion of the lost efficiency. The number of passes retained is accordingly treated as a hyperparameter, set to the value reported in Table 3.13 and constrained by the KL target so that the stale-data instabilities of off-policy replay are avoided. The clip threshold ϵ is the one quantity governing the update that remains unspecified. A large-scale empirical study by Andrychowicz et al. [12] found a moderate value for ϵ to perform reasonably well across five distinct continuous-control environments, supporting the decision made in Chapter 3 to fix a single threshold rather than tune one per curriculum stage.

Two further findings from the same body of empirical work bear directly on the design developed in the next chapter (Chapter 3), both concerning the configuration surrounding the objective rather than the objective itself. The first comes from Engstrom et al. [13], who ascribed a significant portion of the reported performance gap between PPO and its predecessors to details of the implementation, such as the annealing of the learning rate. Andrychowicz et al. [12] subsequently trained over 250 000 agents to determine which components contribute to the measured

differences. Normalising the observations proved crucial in almost every environment, and the learning rate ranked among the most influential parameters evaluated individually. Its decay schedule, by comparison, had substantially less impact than the rate itself. Such parameters must therefore remain fixed across policy variants, which motivates the shared normalisation and learning-rate settings adopted in the next chapter (Section 3.3.2). The same finding on the learning rate applies at a curriculum boundary, where its schedule, rather than its value, should characterise what configuration carries across a stage, an arrangement the curriculum of Section 3.7 later follows. The second finding moves past configuration to the objective itself, establishing a provably safe form of reward shaping. Ng et al. [14] showed that an extra reward term F leaves the optimal policy of the underlying task unchanged whenever F is built from some potential Φ defined over states, taking the form $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$ at a transition from state s to state s' under action a . Shaping of this form can expedite learning without modifying the criteria for success. The same analysis establishes that a bonus outside the potential-based framework can instead admit cyclic trajectories of positive net return, allowing a policy to accumulate reward indefinitely without ever reaching termination. Precluding that outcome is the rationale behind the design of the progress reward in Section 3.6.

Beyond the conduct of training lies a constraint on what the actor can express, since the action spread is determined by the architecture of the standard actor rather than by the state. In the default Gaussian head of SB3 [10, 11], the action mean is a function of the state, whereas the log standard deviation is a single learned parameter vector independent of the input. A single global variance therefore governs exploration across all states, preventing the head from narrowing its action distribution in familiar states while widening it in unfamiliar ones. The limitation is a property of this particular head, not of policy-gradient methods as a class. Soft Actor-Critic (SAC) [15] conditions both moments of its action distribution on the input, so its spread varies from one situation to the next. A spread that responds to the vehicle’s circumstances is thus available only by replacing the head, which is the purpose of the evidential actor developed in Section 3.5. The standard head is nevertheless retained as the comparison arm of a factorial ablation set out in Section 3.8, since holding it fixed while the input varies is what isolates the contribution of the replacement. A localisation covariance supplied to that unchanged head has nothing to act upon, and the consequences of the pairing are measured in Section 4.3.3.

2.1.2 Aleatoric and Epistemic Uncertainty in Deep Learning

Predictive uncertainty in deep learning is categorised into aleatoric and epistemic components. The distinction was initially established in risk and reliability analysis [16]

and subsequently integrated into deep vision models by Kendall and Gal [17], who first estimated both within a single network. Aleatoric uncertainty refers to the intrinsic noise present in observations, which originates from sensors and stochastic disturbances in a driving context and cannot be reduced by gathering additional data. Epistemic uncertainty instead refers to the model’s lack of knowledge regarding the function that fits the data, diminishing as more data is gathered and increasing for inputs that deviate from the training distribution. Both components are set out in Table 2.1 alongside their sources and the response each warrants. The division bears on the present task precisely because it dictates the appropriate response. Irreducible noise can only be accommodated in the manoeuvre itself, whereas a gap in the model’s knowledge can be closed with further experience.

Table 2.1 The aleatoric-epistemic taxonomy.

Type	Source	Reducible	Operational response
Aleatoric	Noise inherent in the observations and dynamics (sensor noise, stochastic disturbance)	No	Act more conservatively; slow down; widen margins
Epistemic	Model ignorance; inputs outside the training distribution	Yes, with data	Defer or hand off the decision; collect the missing experience

For a regression output y , the two components add to the total predictive variance:

$$\sigma_{\text{total}}^2 = \underbrace{\sigma_{\text{alea}}^2}_{\text{data noise}} + \underbrace{\sigma_{\text{epi}}^2}_{\text{model ignorance}} \quad (2.5)$$

In conventional estimators, the epistemic component is obtained through model sampling. Under Monte Carlo dropout [18], the random masks that silence a fraction of units during training are left active at test time, and T stochastic forward passes are interpreted as samples from an approximate posterior distribution over the weights. This sampling accounts for the epistemic component alone, so the aleatoric noise term remains a single calibrated constant rather than an estimate that varies with each input. Deep ensembles [19] instead recover both components by training K independently initialised networks, extracting the epistemic component from disagreement among their predicted means and the aleatoric component from the variance predicted by each member. Disagreement obtained this way carries no uncertainty beyond what the observed data supplies, so Osband et al. [20] add a fixed random prior network to each member and train it against the residual. The networks are driven into agreement wherever data constrains them, while retaining the diversity of their priors elsewhere. The contrast between agreement on constrained data and divergence away from it

makes the spread of such an ensemble a specifically epistemic quantity. Both methods incur the same structural cost, in that uncertainty characterised across samples cannot be assessed in fewer than T or K forward passes, whereas a deterministic network requires only one. For a controller operating at a fixed decision rate on embedded hardware, the sample count multiplies the per-decision inference budget. This cost is the primary rationale for the single-pass approach adopted in Section 3.5.

2.1.3 Evidential Deep Learning

EDL acquires the same two components through a single forward pass by predicting the parameters of a higher-order distribution over the likelihood. No model sampling is necessary, so the cost of obtaining an uncertainty estimate reverts to that of a deterministic network. Sensoy et al. [21] proposed a classification framework that trains a network to produce evidence, which determines the concentration parameters of a Dirichlet distribution over class probabilities. A large total evidence thus sharpens the distribution and signals confidence, while a small one keeps it broad and signals ignorance. This total nonetheless collapses to a single scalar uncertainty mass, so the Dirichlet formulation leaves the two components of Eq. (2.5) unseparated. Amini et al. [3] closed that gap by extending the mechanism to continuous targets as Deep Evidential Regression (DER), imposing a NIG prior on the parameters of a Gaussian likelihood to reinstate the separation:

$$y \sim \mathcal{N}(\mu, \sigma^2) \quad \mu \sim \mathcal{N}\left(\gamma, \frac{\sigma^2}{\nu}\right) \quad \sigma^2 \sim \Gamma^{-1}(\alpha, \beta) \quad (2.6)$$

where μ and σ^2 represent the unknown mean and variance of the likelihood over the target y , and Γ^{-1} denotes the inverse-gamma distribution.

A network trained with this prior generates the four NIG parameters $(\gamma, \nu, \alpha, \beta)$ for each output dimension, which are interpreted as virtual observations. The predicted mean is γ , distinct from the discount rate in Eq. (2.2), and the observations supporting that mean are counted by ν . A further pair of parameters describes the noise variance itself, α counting the observations supporting it and β setting its scale. Since counts and a scale cannot take negative values, γ alone is produced by a linear unit, while the remaining three pass through softplus activations that constrain them to their valid ranges ($\nu > 0, \alpha > 1, \beta > 0$). A softplus returns a positive value that can fall below one, so the activation on α carries an additive offset to meet its lower bound.

The two counts combine into the total evidence $\Phi = 2\nu + \alpha$, from which the prediction and both uncertainty components are derived in closed form as the NIG

moments:

$$\mathbb{E}[\mu] = \gamma \quad \underbrace{\mathbb{E}[\sigma^2] = \frac{\beta}{\alpha - 1}}_{\text{aleatoric}} \quad \underbrace{\text{Var}[\mu] = \frac{\beta}{\nu(\alpha - 1)}}_{\text{epistemic}} \quad (2.7)$$

Aleatoric uncertainty is therefore the expected data noise, while epistemic uncertainty is the variance of the predicted mean. The two differ by a factor of $1/\nu$, so the epistemic component diminishes as ν increases while the aleatoric estimate remains unaffected. All four parameters are emitted from a shared feature trunk, as illustrated in Fig. 2.1.

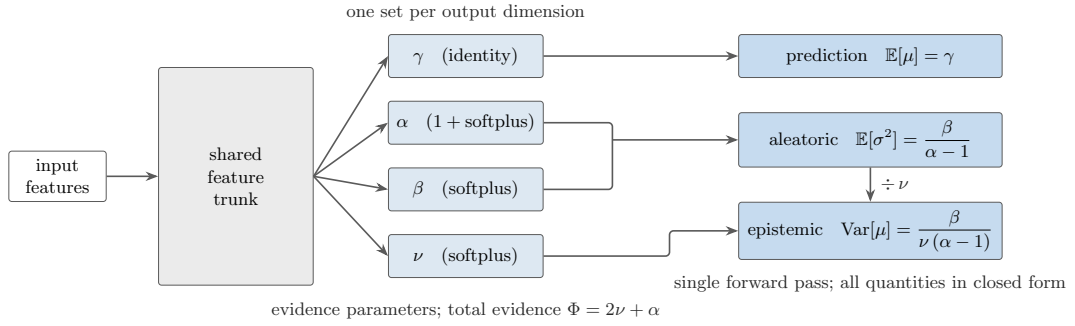


Figure 2.1 The NIG evidential regression head, after Amini et al. [3].

Fitting that head requires a target to regress against, which the supervised setting of Amini et al. [3] supplies as a label. Training then optimises the marginal likelihood of the label under the NIG prior, available in closed form as a Student- t density. A supplementary evidence regulariser then penalises confidence in incorrect predictions:

$$L^R(\theta) = |y - \gamma| (2\nu + \alpha) \quad (2.8)$$

The total evidence Φ consequently scales the prediction error, and this scaled penalty enters the total loss weighted by a coefficient λ_{reg} [3]:

$$L(\theta) = L^{\text{NLL}}(\theta) + \lambda_{\text{reg}} L^R(\theta) \quad (2.9)$$

where $L^{\text{NLL}}(\theta)$ denotes the negative log of the Student- t marginal likelihood, and λ_{reg} sets how strongly the regulariser is weighted against it. This setting governs the balance between uncertainty inflation and model fit. A value of zero yields an overconfident estimate, whereas an excessive one inflates the predictive uncertainty beyond what the data warrants. The likelihood and the regulariser nonetheless share a common dependence on the ground-truth target y , since the likelihood assesses the fit of the predicted distribution to the label and the regulariser withdraws evidence in proportion to the prediction error. Such a target is therefore a structural premise of the framework, and its absence proves decisive in the reinforcement-learning adaptation examined in Section 3.5.2.

Three acknowledged limitations restrict how the NIG output may legitimately be used. Meinert et al. [22] establish the first of these by identifying three distinct grounds on which DER functions as a heuristic rather than an exact method for quantifying uncertainty. The first concerns clean disentanglement, which the marginal likelihood alone cannot provide because it collapses the four NIG parameters into a single Student- t density. Several different divisions between the aleatoric and epistemic components can therefore fit any given label equally well. Recovering a unique separation between them would require assuming a degree of smoothness in the input and possessing prior knowledge of at least one component, neither of which is supplied by the training loss alone. The second concerns the quantity represented by the epistemic estimate, since the regulariser’s gradient dynamics determine how quickly ν converges during training. The estimate consequently reflects the point-wise convergence rate of the model rather than a principled posterior, so its magnitude cannot be interpreted in absolute terms and supports only comparisons between inputs observed in the same pass. The third concerns calibration, which remains unattainable because epistemic uncertainty is a property of the model rather than the data, leaving no ground truth against which it can be calibrated. The same authors observe that the method remains useful when the response to a flagged instance depends only on the estimate being large rather than on its nominal value being correct. The approach then retains practical value because its convergence-rate proxy is available for any input at inference time and flags slow-converging instances as uncertain without requiring labelled out-of-distribution examples during training. The evidential output is therefore treated as a single-pass, relative uncertainty signal rather than an exact Bayesian quantity, and its reliability in conveying this relative information is tested in Section 4.3.1.

The second limitation is that the objective favours small predicted variance, causing training to drive the evidence parameters towards degenerate values that minimise it rather than values that reflect the data. Optimisers push ν to within machine precision of zero even under small values of λ_{reg} [22], which makes the epistemic proxy meaningless. The concern extends beyond this particular result, as the survey by Gao et al. [23] reports that later evidential formulations deprecate the variance-minimising regularisation term to alleviate overconfidence. The consequences of this drift depend on the role assigned to the variance. In a supervised regressor, the damage is confined to the reported uncertainty, which understates the model’s ignorance while the predictions remain as accurate as before. Once the predicted variance also sets the exploration noise, as it will for the actor introduced in Section 3.5, the shrinking variance narrows exploration itself, a collapse the adaptation in Section 3.5.2 mitigates.

The final limitation is that the gradient vanishes at both boundaries of the evidential parameter space, preventing a network from being trained out of either extreme once it arrives there. The most severe case is the High Uncertainty Area (HUA) characterised by Ye et al. [24], in which the gradient of the loss with respect to the pre-activations vanishes as α approaches its lower bound of one. At this limit, the aleatoric estimate of Eq. (2.7) also diverges, so training stalls at the exact point where the reported uncertainty becomes least trustworthy. Oh and Shin [25] identified a similar failure at the opposite extreme, where evidence growing large vanishes the gradient with respect to the predicted mean instead, so training terminates without correcting the prediction. Their remedy alters the objective rather than the output range, thereby penalising the region instead of excluding it. Pandey and Yu [26] found the same failure in classification, among the activations that maintain non-negative evidence. Such activations return exactly zero over part of their input range, so samples mapped to that region contribute neither evidence nor gradient. Collectively, these findings establish the boundaries of the NIG parameter space as a region any architecture producing these parameters needs to stay clear of. Two devices are introduced in the next chapter to enforce exactly that, a head construction that bounds the outputs at source (Section 3.5.1) and a pair of anchoring and flooring terms added to the objective (Section 3.5.2).

Beyond these three limitations, disentangled uncertainty in RL has a precise definition. Charpentier et al. [27] propose four desiderata for uncertainty in a reinforcement-learning agent. Two of them directly govern the epistemic estimate, which should diminish as training accumulates experience of a fixed environment and rise on inputs drawn from conditions outside that training distribution. The remaining two concern the utility of the estimate, requiring that actions sampled from the epistemic distribution rather than the aleatoric one improve sample efficiency during training and generalisation at test time. Meeting all four is an architectural matter, and Charpentier et al. evaluate four uncertainty methods against the desiderata, of which only an evidential model satisfies the concentration criterion. In their evaluation, the estimates of dropout and of an ensemble either rise or plateau without converging during training, whereas the evidential estimate decreases towards zero as training on their task progresses. Detection of departed states additionally requires the estimate to be tied to a density or a kernel over the encoded state. The estimate is attached to the predicted return in every variant they examine, whereas the head developed in Section 3.5 attaches it to the action distribution, so their conclusions carry over to the present design only as far as that change of target permits. These desiderata motivate the aleatoric-epistemic separation examined in Section 4.3.1, and the placements that have been attempted against the four criteria are surveyed in Section 2.2.2.

2.1.4 Extended Kalman Filter and Satellite Positioning

State estimation provides a confidence metric alongside each pose estimate, so the structure of that signal determines what the proposed architecture can condition on. The Kalman filter is the optimal recursive estimator for linear-Gaussian systems. The EKF broadens its application to non-linear motion and measurement models by linearising each around the current state estimate through a first-order Taylor expansion, in which the linear term is the model's Jacobian [28]. Two phases alternate, the first being prediction, in which the motion model advances the state x_k while the process noise Q inflates the error covariance P according to the level of trust placed in that advancement:

$$P_k^- = F_k P_{k-1} F_k^\top + Q_k \quad (2.10)$$

where k denotes the filter timestep, F_k is the motion-model Jacobian, and the superscript minus indicates a quantity that has been predicted but not yet corrected by a measurement. The second phase incorporates a measurement with noise covariance R_k , weighting it against the predicted covariance through the Kalman gain K :

$$K_k = P_k^- H_k^\top (H_k P_k^- H_k^\top + R_k)^{-1} \quad (2.11)$$

with H_k representing the measurement Jacobian. This gain determines the extent of the resulting contraction, with a larger gain assigning greater weight to the measurement and therefore contracting the covariance more sharply:

$$P_k = (I - K_k H_k) P_k^- \quad (2.12)$$

The covariance therefore expands between position fixes according to Eq. (2.10) and contracts at each accepted fix according to Eqs. (2.11) and (2.12), producing a sawtooth profile that is traced empirically for the configured filter in Section 3.4.2 and plotted in Fig. 3.5. The height of each tooth is set by the quality and recency of the measurements, so the profile records the fix state prevailing at every moment of a manoeuvre.

Such covariance is available to any policy at no additional inference cost, since it is computed as a by-product of estimation. Its informativeness depends on how the filter is configured. A study of a wheeled-robot EKF that varied the noise matrices and the parameters governing how uncertainty propagates found that a tuned filter substantially reduced position error relative to a fixed-parameter baseline [29]. That result motivates treating Q as a tuning variable for the present system in Section 3.4.2. Such tuning is exposed by the `robot_localization` package of Moore and Stouch [30, 31], the de facto standard EKF implementation in the ROS ecosystem. An arbitrary set

of pose, velocity and inertial inputs is fused there at a fixed rate, and the covariance is published alongside each pose.

GNSS supplies the position fixes that contract this covariance, so the magnitude of each reduction is governed by the accuracy of the measurement that produces it. A standalone receiver computes position from code-phase pseudoranges with metre-level errors, whereas RTK processing uses carrier-phase measurements relative to a nearby base station and achieves centimetre accuracy once the integer carrier ambiguities are resolved [32, 33]. The ambiguity-resolution status is reported alongside every fix, creating a discrete quality ladder: an ambiguity-fixed RTK solution at centimetre level, an ambiguity-float solution at decimetre level, and a standalone fallback at metre level, the three regimes into which measured highway error distributions separate [34]. Manufacturer datasheets specify representative accuracies for the fixed and standalone modes [1], quoted as Circular Error Probable (CEP), the radius within which half of the fixes fall. Recovering a per-axis standard deviation from that radius requires a circularly symmetric Gaussian assumption, a relation derived together with the margin adopted for each rung of the ladder in Appendix A.1. Each level can therefore parameterise a Gaussian noise model directly. This symmetry idealises the error of a real fix, which is elliptical rather than circular. The arrangement of the visible satellites orients that ellipse, and the error is smallest in the direction that the arrangement constrains most tightly [35, sec. 6.1.3]. The east and north components of a reported covariance are consequently unequal in general.

The accuracy a pose must hold before it can be treated as reliable is set independently of the receiver, so the tiers of the ladder can be assessed against it. Reid et al. [36] derive localisation alert limits for automated vehicles from road geometry, obtaining bounds of 0.57 m laterally and 1.40 m longitudinally on freeways and a tighter 0.29 m in both directions on local streets, where the narrower lane leaves less room for error. Each bound is paired with an integrity risk, the permissible rate at which it may be exceeded. Commercial aviation suffers roughly one hundredth the fatalities per mile travelled that road transport does, and that rate is adopted as the target for a self-driving vehicle since it represents a reasonable safety standard. It implies a fixed budget of failures per mile across the whole vehicle, and dividing that allowance among the subsystems leaves localisation a budget of 10^{-9} exceedances of the alert limit per mile. An ambiguity-fixed RTK solution errs by centimetres and so breaches the limit rarely enough to stay within that budget, whereas the lower two tiers breach it far more often. A float solution errs by a decimetre and a standalone fallback by a metre, both against limits of tens of centimetres. Descending the ladder therefore carries the pose beyond the alert limit on a growing share of fixes, and the consequences of that for a learned controller are taken up in Section 2.2.3.

A receiver ordinarily maintains the fixed tier of this ladder throughout a manoeuvre, but the centimetre accuracy required for parking is lost whenever that tier is lost. Reflections from structures and large vehicles are the principal cause of this degradation, interfering with satellite signals in two distinct ways. Under multipath interference, the directly received signal is distorted by a reflected copy, whereas under non-line-of-sight reception, the blocked direct path is replaced by the reflection alone. The latter condition produces large positive ranging errors that conventional multipath mitigation does not correct [32]. The same obstructions also reduce the number of satellites a receiver can track, weakening the geometry from which the solution is computed, and may interrupt the correction stream on which RTK depends [32]. These disruptions do not stop positioning outright. The reported fix state instead descends the quality ladder while an obstruction persists and recovers once it clears, making the degradation observable to any system that consumes the fix. The appropriate response to a descent is for the vehicle to slow while its position is poorly known, or to stop until the fix recovers. An agent trained exclusively on a fixed solution would have no opportunity to learn either response because it would never encounter a state in which continuing at full speed was incorrect. Training therefore requires exposure to the entire ladder, which the Markov fix-state chain in Section 3.4.1 provides instead of a per-episode constant.

Reproducing that ladder in simulation does not require modelling the signal path of a GNSS receiver. Two bodies of work instead model the error statistically at successively later stages of the receiver chain. Rankin [37] focused on the pseudorange, representing the ephemeris, atmospheric and multipath contributions as first-order Gauss-Markov terms, adding Gaussian white measurement noise, and then scaling the result to a position error using the dilution of precision of a simulated constellation. Karlsson and Mohammadiha [38] focused on the published fix, fitting a state-based stochastic error model for automated driving. A discrete state selects the error statistics, and an autoregressive Gaussian-mixture model parameterises the error within each state. Both works establish statistical injection rather than signal-path modelling as an accepted simulation practice, and they differ in where the error is injected. Rankin’s pseudorange-level treatment would require a simulated constellation that the driving environment does not expose, whereas the fix-level formulation of Karlsson and Mohammadiha acts on the quantity a vehicle actually consumes. The discrete-state formulation is adopted for that reason.

Position is not the sole observable such a model must cover, as a receiver reports more than one. Velocity derived from Doppler measurements is considerably more precise than differencing successive position fixes, reaching centimetre-per-second accuracy even for a standalone receiver [39]. The Course Over Ground (COG) is the direction of that same Doppler velocity vector and inherits its precision, though it is

defined only while the antenna is in motion. Position, velocity and COG are therefore available to a fusion filter as separate observables with distinct error budgets [32].

2.1.5 Curriculum Learning and Domain Randomisation

Exposing a policy to the full difficulty of a task from the outset can leave it without a viable path towards competence. Curriculum learning addresses that risk by arranging training experience from simpler to more complex tasks, a principle formalised by Bengio et al. [40] and further developed into a framework for RL by Narvekar et al. [41]. The same survey identifies the associated risk: each transition between stages alters the training distribution, leaving a policy that has converged at one stage vulnerable to catastrophic forgetting once the meaning of its inputs changes at the next. This mechanism is well established, documented by French [42] and addressed in contemporary networks by Kirkpatrick et al. [43]. The principal curriculum-related mitigation is to vary every input channel from the first step, ensuring that a stage transition changes the degree of variation in a channel rather than whether it varies at all.

A second reason to vary every channel from the first step concerns plasticity, a network’s capacity to keep adapting its weights as training progresses. A complementary body of research shows that holding a channel at zero spread, meaning its value does not vary across training samples, degrades that capacity. Sokar et al. [44] show that neurons falling below an activation threshold become dormant and remain so for the rest of training, while Abbas et al. [45] demonstrate the resulting loss of plasticity across sequential tasks. In each case the damage is done by inputs that stop driving a unit’s activation, either because a task change has made the features it had specialised in irrelevant or because those features no longer vary. Both studies induce that condition by shifting a channel from constant to varying partway through training, whereas a channel held at zero spread from the first step never drives activation to begin with, so the same failure to drive activation carries across. A dimension with no variance gives the units reading it nothing to fit, diminishing their capacity to use that channel once it begins to vary after millions of training steps.

Domain randomisation addresses the same problem from the opposite perspective. Tobin et al. [46] vary every randomised quantity for each training sample across the full distribution of environment parameters, so no channel becomes degenerate and the plasticity loss described above cannot set in. The cost is that a policy must cope with the entire spread before it has learnt the task, a situation that a curriculum is designed to avoid. Automatic Domain Randomisation (ADR), introduced for training in-hand manipulation at scale [47], reconciles the two approaches. The initial distribution is concentrated on a single calibrated environment, and each

randomisation range is expanded independently as measured performance permits [47, 48]. Difficulty is therefore introduced progressively, although the narrow initial distribution reinstates the degeneracy avoided by the fixed scheme. The rule governing when each range expands is itself a pacing decision, and Narvekar et al. [41] treat pacing as a question distinct from task ordering, separating curricula that advance on a predetermined schedule from those that advance only once the agent meets a performance threshold. Florensa et al. [49] show that the latter pacing scheme learns faster than a predetermined schedule, holding the agent at a stage until its success rate enters a target band and advancing it only then. A stage is therefore neither left before it is mastered nor repeated once it is.

The training protocol adopted in Section 3.7 draws on these findings and treats pacing as a separate decision from that choice. The support-preserving property of full randomisation becomes the channel-liveness principle described in Section 3.7.1, applied to each observation channel rather than to environment parameters. Despite the faster convergence Florensa et al. report, the adopted pacing decision proceeds with a fixed schedule regardless, since ablation arms governed by their own success would cross stage boundaries at different points. A difference at evaluation could then be attributed to the schedule as readily as to the head under test. Progression is instead fixed to a per-stage budget, accepting weaker pacing in exchange for a comparison in which the head is the only term that varies.

2.1.6 Evaluation Methodology in Deep Reinforcement Learning

Empirical comparisons between DRL methods are prone to specific, well-documented failure modes, and avoiding them requires meeting four conditions established across several studies of sound evaluation practice. Henderson et al. [50] demonstrated that outcomes in DRL are sensitive to the choice of random seed and hyperparameter configuration, to such an extent that two seeds of the same algorithm can differ by more than the performance gap between distinct algorithms. Single-seed comparisons are therefore unreliable, and multi-seed reporting with the variation shown is the minimum credible practice. Eimer et al. [51] extend this point to the tuning protocol, establishing that hyperparameters must be tuned for each algorithm rather than transplanted between them. The seeds employed for tuning must also be distinct from those used for evaluation, since a configuration tuned on the evaluation seeds is fitted to the very runs meant to judge it. The preceding requirements pertain to experimental protocol, while a further requirement concerns the statistical analysis of the outcomes obtained. Agarwal et al. [52] demonstrate that point estimates over small numbers of runs are statistically fragile, and recommend interval estimates derived through bootstrap resampling instead. Small samples are the norm rather than the exception,

since the reliability of an estimate improves as evaluation runs are added, yet computational cost restricts how many can be conducted and therefore keeps seed counts low in practice. Cobbe et al. [53] complete the discussion on generalisation, demonstrating that performance is overstated when training and testing occur under the same conditions. A credible assessment consequently requires evaluation under held-out conditions that the policy did not encounter during training.

All four requirements govern the experimental protocol that follows. Multi-seed reporting is ensured by a seed matrix specified in Section 3.8.2, and the spread across those seeds is examined in Section 4.7. Tuning imposes a tension between method-specific configuration and a controlled ablation, which is addressed explicitly when the ablation is designed in Section 3.8.1. Reporting over the resulting runs then rests on the bootstrap intervals recommended above, applied to the contrasts of this work in Section 4.1.5, and the held-out criterion establishes a generalisation boundary that is tested last of all in Section 4.8.1.

2.2 Literature Review

The prior work most closely related to the proposed architecture falls into three threads, one for each subsection below: uncertainty-aware RL for driving, evidential methods in RL, and learned autonomous parking. Each thread is structured according to the placement of uncertainty relative to the manoeuvre. All three threads are then brought together in Section 2.3, where the systems are compared side by side.

2.2.1 Uncertainty-Aware Reinforcement Learning for Driving

An uncertainty estimate can be added to the output of a driving policy, to its observation, or to its reward. Adding it at the output leaves the policy itself unchanged and gates the decisions it proposes. Hoel et al. [54] train an ensemble of value networks using randomised prior functions for tactical highway decisions. Epistemic confidence is derived from the spread of that ensemble, and the selected decision is replaced by a safe baseline action whenever its confidence falls too low. The spread only tracks epistemic uncertainty because of the randomised priors described in Section 2.1.2, which keep the members diverse in regions the trained agent has never visited. A state of that kind therefore produces a wide spread, which registers as low confidence and triggers the baseline action. The same spread instead narrows to zero once the members are trained on sufficient data, even for an outcome that remains genuinely variable, as each member predicts an expected return, and averaging discards the spread of the outcomes that produced it. The later ensemble quantile networks developed by the same Hoel et al. [55] address that limitation by replacing

each value network with one that predicts a distribution over returns instead of a single expected return. Aleatoric uncertainty is derived from the spread of that distribution, while epistemic uncertainty is obtained from disagreement between members. Any decision deemed unreliable on this basis is handed to a backup policy, which brakes to a stop if the remaining distance permits.

Despite the change of estimator between them, three characteristics are common to the value ensemble and its quantile successor alike. The quantity estimated is the value of candidate decisions rather than the state of the vehicle, and estimating it necessitates an ensemble, which imposes a computational cost of K forward passes at every decision. The estimate is then applied as a veto over the discrete manoeuvres proposed by the policy. The veto is activated at a confidence threshold selected after training as a runtime hyperparameter, and the policy itself never observes the uncertainty. Any decision left unvetoes is consequently executed identically, regardless of whether its uncertainty lies far below the threshold or just beneath it. Neither system grades that response at runtime, and the quantile successor does offer caution graded by the degree of uncertainty, though only by retraining with a risk-sensitive objective, so it cannot be adjusted at runtime. These findings establish that a value-based estimate can withhold an unsafe manoeuvre, yet both quantities are read off an ensemble of value predictions rather than emitted by the policy that produces the action.

Moving the estimate into the observation gives the policy what the value-based systems withhold from it, an input read before the action is chosen. The signal therefore shapes the manoeuvre as it is formed, instead of correcting one the policy has already committed to. Puthumanaillam et al. [56] condition navigation policies on a Task-Specific Uncertainty Map (TSUM), an abstraction that divides the workspace into regions and records the acceptable level of state-estimation uncertainty within each one. Their example is a surface vessel navigating an open lake under a single policy that acquires a costly Global Positioning System (GPS) fix near obstacles and relies on noisier sensing in open water. The policy reads the acceptable level at its location alongside a single scalar representing its current uncertainty. These two inputs enter a SAC learner, and the relevant property of that learner is the state-dependent action spread noted in Section 2.1.1. Adaptation to the context-dependent value of certainty is therefore divided between the map and the learner: the map retains the tolerance for each region, while the state-dependent spread converts it into the degree of caution applied by the vehicle. Kim et al. [57] retain the scalar form but remove the map, conditioning a policy solely on an epistemic uncertainty signal so that the vehicle drives differently in unfamiliar situations. Their motivating example is a driver slowing on an unfamiliar narrow road. The scalar alone has little effect in their ablation, so a mutual-information term linking uncertainty and action is added to the training

objective. The manoeuvre must consequently be made to vary with the uncertainty input, rather than uncertainty being treated as a quantity to minimise. Both signals describe the vehicle's own state, though a policy can be conditioned just as readily on how uncertain its surroundings are. Golchoubian et al. [58] showcase this by predicting the motion of nearby pedestrians and supplying a crowd-navigation agent with the confidence of each prediction, training the agent to lower its probability of colliding with them. A vehicle passing an unpredictable pedestrian consequently yields more room than one passing a pedestrian moving steadily, which comes closest among the surveyed systems to letting the amount of caution track the size of the estimate. The signal nevertheless reports the motion of other agents, so it is produced by a separate prediction module rather than the vehicle's own state estimator. These systems establish that a policy conditioned on an uncertainty input varies its caution continuously with that input, instead of driving unchanged until a threshold is crossed. Each of them nevertheless substitutes a proxy for the covariance the localiser already reports at every step, which remains unused.

A reward defined over the localiser's covariance consumes it without a proxy, making uncertainty the target of optimisation instead of a condition on it. Placed and Castellanos [59] reward an exploration agent using the D-optimality of the pose covariance maintained by the Simultaneous Localisation and Mapping (SLAM) filter, an optimal-design criterion that reduces the matrix to the scalar volume of the uncertainty ellipsoid it describes. The return is consequently highest wherever localisation uncertainty falls most rapidly, so a discrete-action deep Q-learner learns to steer back towards previously mapped landmarks. A repeat observation of a landmark already in the map then corrects the accumulated pose drift and tightens the filter's covariance. Seeking uncertainty to reduce it is the appropriate response for exploration, in which information is the objective. The parking task reverses this premise. Uncertainty arises exogenously from the satellite geometry, leaving the vehicle to manoeuvre under a level of uncertainty it has no means of lowering. A reward term penalising it would therefore charge the vehicle for a condition that no trajectory within a bay-sized workspace can improve.

The remaining placement sits above the manoeuvre altogether, where the signal becomes a guarantee about the policy. Zhou et al. [60] estimate the performance uncertainty of a driving policy from the sparsity of its training data and the residual error of the fitted network, then derive a lower bound on its return. Control is granted to the learned policy only where that bound exceeds the return of a rule-based planner, which retains control everywhere else. The uncertain quantity is therefore the overall reliability of the policy rather than the ego state. The estimate gates exploration during training and selects between the two controllers at runtime, yet remains absent from the observation on which the policy conditions. Across this thread, localisation-related

uncertainty becomes available to a driving policy only after an abstraction step has converted it into another form. None of the reviewed ego-motion driving policies therefore includes the state estimator’s reported covariance among its observation features.

2.2.2 Evidential Methods in Reinforcement Learning

Evidential techniques are particularly attractive when ensembles are prohibitively expensive because of the single-pass property established in Section 2.1.3, although their adoption within RL remains limited to a handful of systems. Most apply the NIG prior from Eq. (2.6) to the critic, causing the value of a state to be expressed as a distribution rather than a point estimate. The distribution serves three distinct purposes across the systems surveyed. The first is to maintain the learner’s adaptability as the dynamics of its environment change. Akgül et al. [61] apply precisely this construction to the value function of a PPO learner, the earliest evidential critic reported for an on-policy actor-critic. The resulting head is claimed to address non-stationary conditions through two properties, one on each side of the learner. On the critic side, the property concerns plasticity, with the spread widening whenever a change in dynamics makes the value target unfamiliar. The spread scales the update applied for a given error, so the critic retains the capacity to adapt long into training. On the actor side, the spread propagates into the advantage, with an optimistic bonus proportional to it added before the policy update. The method outperforms all on-policy baselines across continuous-control tasks perturbed at regular intervals. Both benefits nevertheless originate in a head placed on the value function and trained against the critic’s regression targets, so the advantage is the only route by which the estimate reaches the actor. The bonus accordingly lands during the update, once the actions of the batch have been sampled and executed. The sampling distribution remains the conventional Gaussian head described in Section 2.1.1, so the spread of the manoeuvre during execution is left to a variance that carries no per-state information.

Delivering the estimate to the manoeuvre is only the first requirement, as an estimate that does reach it remains uninformative while the aleatoric and epistemic moments of Eq. (2.7) are reported as a single quantity. Separating them constitutes the second purpose observed across the systems surveyed. Stutts et al. [62] combine evidential estimates of the action values of a quantile-regression Deep Q-Network (DQN) with quantile calibration based on conformal inference, evaluated on miniaturised Atari games over a discrete action space. Each half of that construction contributes a distinct mechanism. The evidential half has the network emit its parameters alongside the quantiles of the value distribution, so aleatoric and epistemic

uncertainty are read off separately through the two moments without sampling. The conformal half then widens those quantiles by a margin taken from the errors recorded on held-out data, which attains a nominal coverage level without assuming a form for the predictive distribution. Both scores and learning speed improve, so the separation is obtained at no cost to the performance of the agent. An equivalent separation is reached from an identical value-side placement by the one agent that Charpentier et al. [27] find to meet all four desiderata for uncertainty in a reinforcement-learning agent set out in Section 2.1.3. The agent is a deep Q-learner of the same value-based family, drawing its action values from the Posterior Network [63]. That model derives its evidence from a learned density over the training data rather than from a head of the NIG form given in Eq. (2.6). A Gaussian likelihood over each return carries the aleatoric term, whereas the evidence count scaling the epistemic term is set by a normalising-flow density over the encoded state rather than predicted by the network. That density distributes a fixed certainty budget over the latent space and is trained jointly with the encoder and the decoder, so the count reflects where the training states fell. A sparsely visited region of the state space therefore attracts a low density and a correspondingly high epistemic score, which registers the familiarity of a state and not the confidence of the pose held within it. Both systems obtain the separation over a discrete set of action values, so neither reaches the continuous steering and throttle commands that carry out the parking manoeuvre studied in the present work.

The third application places this type of estimate on a planning constraint rather than on a value. Ham and Ahn [64] apply EDL to the position of a detected obstacle, and a distributionally robust Model Predictive Control (MPC) planner consumes the resulting distribution to define an ambiguity set that grows and shrinks with perception confidence. The planner therefore remains efficient under the resulting constraint when the perception network is confident, while conservative margins are enforced when it is not. The evaluation isolates this effect in CARLA using an obstacle class absent from the detector’s training set. A deterministic constraint over the same detections results in collisions in all one hundred runs, whereas the evidential ambiguity set completes ninety-five runs without retraining the perception model. A response graded by confidence in the same way is sought from the evidential actor head developed in Section 3.5, where the spread of the action distribution widens as the evidence supporting the proposed action falls. The estimate is nevertheless derived from the position of another object rather than the ego pose, and it is produced by a planner downstream of any policy. The caution is therefore a property of the controller that overrides the manoeuvre rather than of the manoeuvre itself.

Beyond these three applications, one system moves the estimate off the learner entirely and onto the localisation stack that supplies its pose. Kalyanasundaram et al. [65] place a NIG head on a network that corrects the estimated velocity from

onboard motion sensing alone, so the correction supplied to an error-state EKF arrives with an uncertainty estimate obtained in the same forward pass. The estimate weights the correction as a virtual measurement, which reduces position drift through prolonged GNSS outages on real vehicle data at 40 Hz. Evidential estimation is therefore compatible with a localisation stack, and the quantity it is applied to is a design choice. The estimate is consumed by the filter upstream of any controller, leaving no learned policy to condition on it. In the architecture proposed in Section 3.5, the filter covariance is instead retained as the pose signal, and the evidential head is moved to the action selected under it.

The actor remains downstream of the estimate across all of these placements, and the evidential head is abandoned altogether in the one system that closes this distance. Kanazawa et al. [66] use an ensemble of actor-critic learners for continuous control, taking epistemic uncertainty from actor disagreement to drive exploration and aleatoric uncertainty to shape a risk-sensitive policy. The estimate is consequently available at the point of action, although reaching it costs the K passes established in Section 2.1.2 rather than the single pass motivating evidential methods in the first place. The decomposition is therefore recovered across the ensemble members rather than from the quantity emitted by the actor, which draws the manoeuvre from a conventional Gaussian head.

Setting the ensemble aside, none of the five systems carrying an evidential head places an evidential distribution over the action, as all share a constraint on the point at which such a head can be attached. Akgül et al., Stutts et al. and Charpentier et al. attach the head to the critic, where each prediction is regressed onto a bootstrapped return. Ham and Ahn instead attach it to a detected position and Kalyanasundaram et al. to a velocity, both regressed onto labelled ground truth. Each quantity supplies the observed target y required by the regulariser in Eq. (2.8). The action of a policy-gradient agent supplies no equivalent target because the manoeuvre emitted by a policy is never compared with a correct manoeuvre during training. Evidence over the action must therefore be accumulated through a route other than regression onto a target, and that route remains unexplored in the work surveyed.

2.2.3 Reinforcement Learning for Autonomous Parking

Production parking systems are organised as a modular pipeline, which fixes the points at which a statement of confidence can enter. Heimberger et al. [67] describe the layered architecture of a camera-based assistant, in which the outputs of embedded vision algorithms are integrated at a stage that subsequently informs the motion planning of the manoeuvre. The precision of that vision stage is fixed in advance as a specification: slot markings must be located to within thirty centimetres along the

optical axis and ten laterally. The planner thus receives a position presumed to meet that tolerance, never an estimate of the error actually incurred. The survey by Olmos Medina et al. [68] finds the same division across the field, where the perception stage is the only one to express confidence at runtime. A probabilistic occupancy grid assigns each cell an occupancy probability, and a detector reporting low confidence triggers a handover to a fallback. The control stage is instead defined by tracking precision rather than by any runtime uncertainty estimate of its own. Confidence in both stacks is hence an attribute of the camera and fusion stages, resolved upstream of the controller and not transmitted into the manoeuvre. The deployed systems therefore fall short of the point of action for the same reason as the evidential placements examined in Section 2.2.2.

A learned parking policy has no such stages to fix where confidence enters, collapsing planning and control into a single network. Three of these systems are examined below, and the manoeuvre itself is taken as the difficulty to be overcome in two of them, from Lazzaroni et al. [69] and Suleman et al. [70].

Lazzaroni et al. train a PPO agent to navigate to a randomly selected bay among sixty perpendicular parking spaces in a CARLA environment, the closest match among these systems to the task adopted in Section 3.2. The observation combines a LiDAR scan with the target-relative position and the velocity, acceleration and yaw rate of the vehicle. The agent then issues continuous throttle, steering and brake commands together with a reverse-gear signal. Difficulty is raised incrementally through a three-stage curriculum, initially omitting other vehicles and collision penalties, then introducing parked obstacles with a mild penalty, and finally making collisions terminal in the third stage. The agent attains a 97% success rate under that schedule. Pacing of this kind is one mechanism governing how readily the manoeuvre is acquired, while the reward function is another. Suleman et al. focus on the latter, comparing three reward formulations under matched hyperparameters on a task that fixes the target bay at a single location and restricts the agent to continuous steering. A milestone-augmented reward dispenses a proximity signal when the vehicle enters an intermediate region during its approach. This formulation outperforms both a sparse goal-only reward and a fully dense proximity reward under on-policy and off-policy optimisation alike.

Training in both studies occurs under nominal localisation, without inserting a state estimator between the simulator and the pose on which the policy operates. The observed performance is thus linked to the challenge of executing the manoeuvre and provides no insight into the difficulty of localising the vehicle during its execution. A staged training curriculum is designed in Section 3.7 to hold the localisation channel active from the first stage instead, so the manoeuvre is acquired under a pose that degrades as learning progresses.

Uncertainty goes unquantified by Lazzaroni et al. and Suleman et al. alike, whereas a judgement about it is reached by the remaining parking system, where it governs how the policy is fitted rather than how the vehicle is driven. Yang et al. [71] distil demonstrations from a classical planner-controller stack to formulate a goal-conditioned policy using end-to-end offline RL, incorporating a conservative value regulariser that pushes down the value of actions the demonstrations never took. This type of conservatism is an epistemic assessment of the dataset's coverage, resolved during the gradient updates. The deployed agent therefore inherits a value function already shaped away from the unsupported actions, and not a coverage score it can read at decision time. An evidential head is developed in Section 3.5 to obtain the same epistemic quantity at the output instead, scoring the observation currently being acted on rather than the coverage of a fixed dataset. The estimate is therefore recomputed at every decision and remains available during deployment.

In the learned policies of Lazzaroni, Suleman and Yang et al., the observation comprises the pose and the surrounding obstacles. Those inputs suffice because the simulator supplies the pose exactly, so none of the three carries an uncertainty estimate into evaluation, and none observes a covariance alongside the pose, as the policy specified in Section 3.3 does. That assumption of an exact pose carries a quantitative threshold, since the alert limit established in Section 2.1.4 places the lower tiers of the fix ladder beyond the accuracy a pose must hold to be treated as reliable. Fix loss is intermittent, so a manoeuvre proceeds within the bound for the most part and falls outside it on the occasions when the solution degrades. Olmos Medina et al. [68] indicate that such a deficit is typically addressed through probabilistic sensor fusion and perception-level confidence estimation. Neither element survives as a distinct stage in an architecture that maps raw input directly to a control action, and uncertainty quantification within such an architecture is described in the same survey as an open direction rather than established practice. The success rates reported by the three policies named above are therefore uninformative about parking under an error beyond the alert limit, of the kind that structures and large vehicles around a facility are liable to induce. Degradation of the pose estimate is accordingly treated as a controlled variable in Section 3.4.1.

2.3 The Research Gap

The two absences previewed at the start of this chapter are stated below, each bounded by the scope of work it is asserted against.¹

¹Both absence claims rest on searches of IEEE Xplore, Scopus, arXiv and Google Scholar, conducted in August 2026. The output side paired the evidential and normal-inverse-gamma vocabulary with policy-gradient and continuous-control terms. The input side likewise paired the localisation-uncertainty and

The first of the two concerns what a learned controller is given. No continuous-control policy for ego-motion has been given covariance-derived features from its own state estimator, such as the standard deviations along each axis, as observation inputs, whether for parking or for any other driving task surveyed. Two threads of the review converge on that point from different directions. The estimate in every placement analysed in Section 2.2.1 reaches the policy only through an abstraction of the covariance, while the parking agents of Section 2.2.3 receive the pose with no statement of confidence in it at all. Availability does not account for that absence, since the navigation-grade EKF of Section 2.1.4 produces the signal continuously at no additional inference cost. The covariance tracks the trustworthiness of the pose only if the filter’s noise models are tuned, and objective O1 accordingly requires that tracking to be confirmed before any policy is conditioned on it.

A corresponding absence on the output side governs what that controller emits. No published evidential RL system has attached the NIG distribution to the actor of a continuous-action policy-gradient agent. The restriction behind this absence was identified in Section 2.2.2, where every prior placement is anchored to the regression target presumed by the evidential loss, and no such target exists for an action. The only remaining alternative is the Gaussian head described in Section 2.1.1, which leaves per-state confidence inexpressible. Two questions are consequently left unexplored: whether an evidential action head can be trained without a ground-truth action target, and whether its uncertainty output carries a usable signal. Both are examined through objective O2, focusing on the NIG parameterisation and the single-pass property outlined in Section 2.1.3, with the pathologies catalogued alongside them constraining what such a head can be expected to deliver.

Both absences are visible at once in Table 2.2, where the surveyed systems are characterised along five axes. The first two concern the uncertainty estimate itself: the locus indicates where that estimate is placed relative to the manoeuvre, and the estimator is the mechanism that produces it. The remaining three bound what those systems can provide. The number of passes determines the per-decision cost, the action space determines the expressibility of a graded response, and the domain determines the proximity of the task to parking. A dash marks an axis that does not apply to a given system, either where no runtime uncertainty estimate is carried or where no action is emitted.

Across the rows, neither absence depends on the other, which is what makes their combination testable. An input-side mechanism can be integrated into a standard actor, an output-side mechanism can be trained independently of covariance observations, and both can be used together. This independence yields a factorial

pose-covariance vocabulary with observation-space and driving terms. No search establishes a universal negative, so both claims are limited to the work surveyed.

design in which each mechanism's contribution can be assessed separately. No cell of this design has been occupied by prior work under controlled localisation degradation, and parking is a context in which this omission has direct consequences, since an intermittent loss of the RTK fix carries the pose past the alert limit established in Section 2.1.4.

Objective O3 accordingly requires those contributions to be separated. The methodology for assessing this separation was established in Section 2.1.6, particularly the multi-seed reporting and interval estimates required by small seed counts. Reporting alone leaves the conditions under which the runs are produced unspecified, and the exposure driving them is controlled by the curriculum machinery described in Section 2.1.5. Objective O4 then requires evidence that any resulting benefit originates from uncertainty-conditioned behaviour rather than from the input alone. The evidence would take the form of a vehicle that drives measurably more conservatively as the reported uncertainty increases. No reward term refers to uncertainty, so caution of that kind would emerge on its own, whereas Kim et al. in Section 2.2.1 had to add one to the training objective before the manoeuvre varied at all.

Table 2.2 Prior work across the three threads of Section 2.2.

Work	Uncertainty locus	Estimator	Passes	Actions	Domain
Hoel et al. (2020) [54]	Output (decision veto)	Bootstrapped ensemble with randomised priors	K	Discrete	Tactical driving
Hoel et al. (2023) [55]	Output (decision veto)	Ensemble of quantile networks	K	Discrete	Tactical driving
Puthumanaillam et al. (2026) [56]	Observation (abstracted map)	Task-specific uncertainty map over the workspace	1	Continuous	Robot navigation
Kim et al. (2022) [57]	Observation (scalar signal)	Epistemic scalar with a mutual-information term	1	Continuous	Automated driving
Placed and Castellanos (2020) [59]	Objective (reward)	Particle-filter pose covariance, D-optimality	1	Discrete	Active SLAM
Zhou et al. (2023) [60]	Guarantee (policy-level fallback)	Bootstrapped ensemble of value estimates	K	Discrete	Automated driving
Akgül et al. (2025) [61]	Output (critic)	NIG on the state value	1	Continuous	Locomotion
Stutts et al. (2025) [62]	Output (action values)	NIG with conformal calibration	1	Discrete	MinAtar games
Charpentier et al. (2022) [27]	Output (predicted return)	Posterior Network with a normalising-flow density	1	Discrete	Classic control
Ham and Ahn (2026) [64]	Constraint (perception into planner)	Evidential perception uncertainty	1	Continuous	Automated driving
Kalyanasundaram et al. (2026) [65]	Input (filter measurement)	NIG on a velocity correction	1	-	Vehicle localisation
Kanazawa et al. (2022) [66]	Output (actor and critic)	Distributional actor-critic ensemble	K	Continuous	Continuous control
Lazzaroni et al. (2024) [69]	None	-	-	Continuous	Parking (CARLA)
Suleman et al. (2026) [70]	None	-	-	Continuous	Parking
Yang et al. (2025) [71]	Objective (offline conservatism)	Conservative value regularisation	1	Discretised	Parking (CARLA)

3 Methodology

The experimental system and the design decisions taken to answer the research objectives are documented in this chapter, in the order in which the system is constructed. The infrastructure and parking task are established in Sections 3.1 and 3.2, after which the policy interface and localisation pipeline are defined in Sections 3.3 and 3.4. The policy itself is developed and trained in Sections 3.5 to 3.7. The experimental isolation of the design, and its path into evaluation and deployment, are then specified in Sections 3.8 and 3.9.

3.1 System Overview and Infrastructure

The two-layer uncertainty architecture introduced in Section 1.4 is realised by a three-container simulation, localisation and training stack. Every signal crossing those containers is traceable to a named source, from the raw sensor streams to the rate-limited vehicle command.

The trajectory traced in Fig. 3.1 begins with the raw GNSS, IMU and LiDAR streams published by CARLA. Datasheet-derived noise is added to the GNSS and IMU measurements by dedicated relay nodes, which republish them with stamped covariances. The `robot_localization` EKF node [30, 31] fuses those two republished streams into a pose estimate with a live 3×3 covariance, and a state node serialises both outputs to a shared file. The LiDAR stream takes no part in that fusion, serving the obstacle-clearance features alone, so its degradation leaves the localisation uncertainty signal untouched. Ground-truth pose is routed separately into the reward calculation and never enters the observation, as it has no counterpart on a real vehicle (Appendix B.1). Once the serialised state is available in the training container, the observation is constructed by the Gymnasium environment and passed to the SB3 policy. The resulting action is then rate-limited by the actuator model and returned to CARLA as a vehicle command.

3.1.1 The Three-Container Stack

The experimental platform is organised into three Docker containers, each responsible for a distinct stage of the processing pipeline. First, the `carla-server` container runs CARLA 0.9.16 in headless mode with Graphics Processing Unit (GPU) passthrough. The resulting sensor streams are forwarded to `ros2-bridge`, which runs ROS 2 Jazzy and hosts the CARLA bridge, the `robot_localization` EKF, and the sensor-noise relay nodes. Finally, the filtered state estimate is passed to `training`, which is based on the

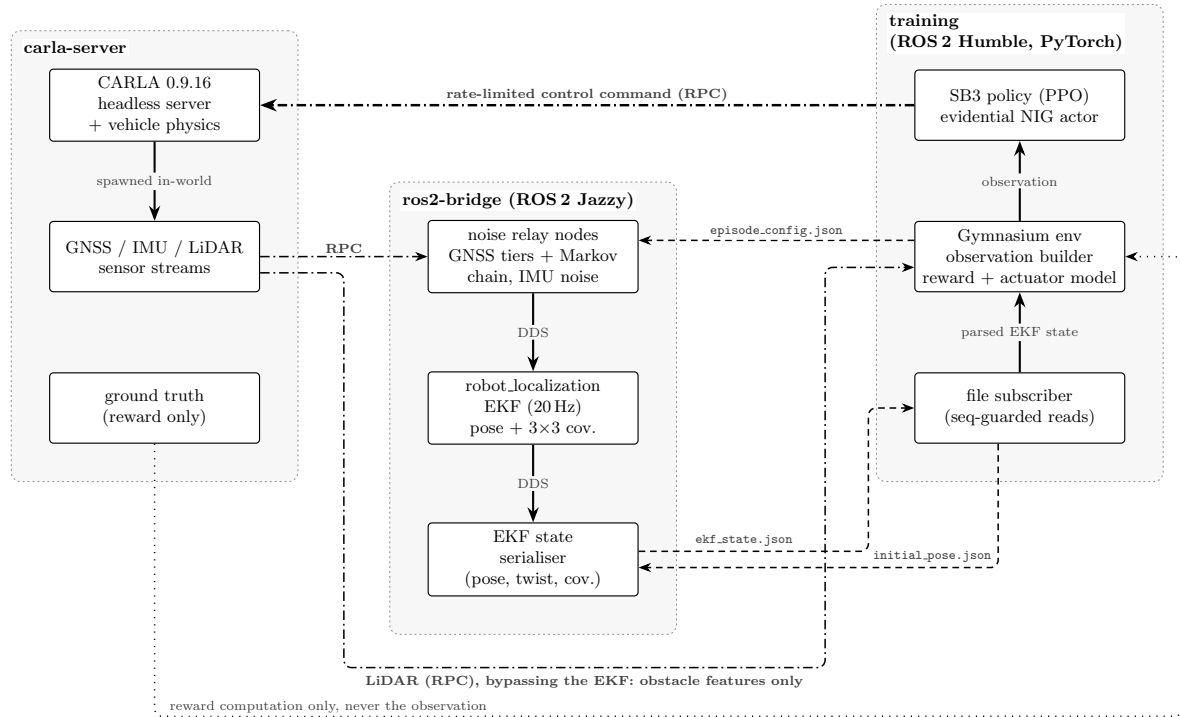


Figure 3.1 The three-container system and its data flow.

NVIDIA GPU Cloud (NGC) PyTorch 24.10 image and contains SB3, the evidential networks, and a ROS 2 Humble client library. The bridge runs passively, in that the sensors are spawned and the simulation clock is advanced by the training container alone, so simulation time has a single writer and an episode replays identically from the same seed.

This containerised architecture was adopted for three principal reasons. Reproducibility is the first: the Python dependencies are pinned to versions derived from the operational images, so the experiment can be replicated on any machine equipped with a container runtime and the NVIDIA Container Toolkit. Version isolation is the second, as CARLA, ROS 2 and the CUDA-enabled PyTorch stack depend on different Ubuntu releases. The bridge runs on Ubuntu 24.04, whereas the training stack sits on the 22.04 base supplied by the NGC PyTorch image, and Table 3.1 reports the complete version set for all three containers. Each component therefore runs within its required software environment only under containerisation. Operational robustness is the third, since the bridge container restarts automatically after a headless CARLA failure and re-establishes the connection without human intervention.

3.1.2 The Jazzy-Humble File Bridge

The version isolation described above creates a single discontinuity within the localisation pipeline, specifically at the boundary between the Jazzy bridge and the

Table 3.1 Software configuration of the experiment stack.

Component	Version	Container
CARLA simulator	0.9.16	carla-server
ROS 2 (bridge, EKF, relays)	Jazzy	ros2-bridge
ROS 2 client library	Humble	training
PyTorch base image	NGC 24.10	training
Stable-Baselines3	2.8.0	training
Gymnasium	1.2.3	training
Python	3.10 / 3.12	training / ros2-bridge

Humble training container. The relays, the EKF and the covariance extractor interact through standard ROS 2 topics within the bridge, drawn as the Data Distribution Service (DDS) links of Fig. 3.1. Jazzy and Humble serialise messages differently, so a subscription spanning both releases cannot be relied upon, and the state is instead transferred as three JavaScript Object Notation (JSON) files on a shared volume listed in Table 3.2.

Table 3.2 The three shared files that cross the container boundary.

File	Writer	Reader	Purpose
ekf_state.json	Covariance extractor (bridge)	Environment (training)	Latest EKF pose, twist and 3×3 covariance
initial_pose.json	Environment (training)	Covariance extractor (bridge)	Spawn pose; triggers an EKF state reset via /set_pose
episode_config.json	Environment (training)	GNSS noise relay (bridge)	Starting fix tier, spawn datum and drift override (Section 3.4.1)

Three characteristics of the file interface mitigate the absence of a live subscription. Durability is the first, as persisting each state to disk lets the most recent values recorded survive a bridge relaunch after a headless CARLA failure. Legibility follows from it, since those same records are held as plain text and are therefore open to logging, inspection and replay, which eases the diagnosis of faults spanning the two containers. Synchronisation is the third, as the two files written at a reset give the handshake a fixed point to wait on, so the first observation of an episode is withheld until the estimator has been re-initialised.

The three properties are obtained at no measurable cost in time. Each file holds a single sub-kilobyte record on a page-cache backed volume, so the exchange completes in memory rather than at disk latency. Under the 5 Hz decision rate adopted in Section 3.3.3, each decision is allowed 200 ms. Almost none of that budget is consumed by the JSON traffic between the containers, so nearly all of it remains available to advance the four simulation ticks that separate one decision from the next and to run the forward pass that yields an action.

Every file transmits data in one direction only, and the three are split between two cadences, one continuous and the other episodic. The continuous cadence is facilitated by the single bridge-to-training file, `ekf_state.json`, to which the covariance extractor writes the most recent pose, twist and 3×3 covariance at each EKF update for the environment to access at every decision step. The episodic cadence covers the two training-to-bridge files instead, each written during a reset and left unaltered in between. The environment records the sampled spawn pose in `initial_pose.json`, which the covariance extractor polls before issuing the corresponding `/set_pose` call that anchors the EKF state to the new episode. The environment additionally records the initial fix tier and the spawn datum to `episode_config.json`, which the noise relay accesses before the vehicle moves, together with an optional override that forces the fix quality to deteriorate without recovering, defined with the rest of the fix-state model in Section 3.4.1. Each run is therefore configured from its own draw rather than inheriting the relay's state from the previous run.

The environment and the bridge nodes access these files without a shared lock, so two mechanisms protect the exchange against race conditions. Each write operation is atomic, as the record is serialised to a temporary file and then renamed over the target using `os.replace`, which corresponds to the atomic POSIX rename system call. A partially written file is therefore invisible to every subscriber. Atomicity alone leaves a complete but stale record indistinguishable from a fresh one, so each file is additionally stamped with a monotonically increasing sequence number, alongside a modification timestamp that lets an unchanged record be skipped without reparsing. The subscriber records the most recent sequence observed at each episode reset and thereafter accepts only strictly higher values. The initial observation of a new episode therefore derives from a genuine post-reset EKF state.

3.1.3 Supporting Tooling and Verification

Three supporting tools were developed around the stack to accelerate development and make each design decision verifiable. A layout generator converts the geometry of each lot from a local-frame floor-plan module into a world-frame YAML description,

together with an accompanying bird's-eye plot. Every layout can therefore be defined and visually inspected before it is introduced into the simulator. At the opposite end of the workflow sits a detachable two-dimensional episode visualiser, which attaches to a training or evaluation run without altering it. The EKF estimate, the ground truth and the live covariance ellipse are displayed together as the episode is stepped. A drift between the estimate and the truth can thereby be read against the uncertainty claimed for it, so the response of the filter to each fault modelled in Section 3.4.1 can be confirmed as the intended one rather than assumed.

Between the two sits a unified lot inspector, which presents each configuration at the four progressively higher levels of fidelity set out in Table 3.3. The levels ascend from static geometry to a fully stepped environment, so an error can be attributed to the stage that introduced it, as a discrepancy absent at one level and present at the next implicates only what that level added. Faults in the geometry, the sensor placement and the observation pipeline are consequently separable before computationally expensive training begins.

Table 3.3 The four fidelity levels of the lot inspector.

Level	Adds	Verifies
1	The bare lot geometry	Bay placement and lot dimensions
2	The sensor mounts and their fields of view	Mounting poses and coverage
3	A live LiDAR, visualised against the simulated environment	Returns consistent with the modelled scene
4	A stepped training environment under a predetermined or manually selected action, with no policy loaded	The exact observation the policy would receive

All four levels are assessed visually, as judging whether a rendered scene is faithful demands an observer. The implementation behind it needs no observer, so an automated suite is relied upon to keep the code correct as it changes. The repository, laid out in Fig. D.1, holds 438 unit tests that execute within a container upon each push, under a continuous integration pipeline that also enforces linting, formatting and static type checks. Beyond the implementation, the properties on which the methodology depends are themselves encoded as tests, among them the channel-liveness principle of Section 3.7.1, asserted by checking that every observation channel varies from the first stage, and the arm definitions of Section 3.8, asserted by merging each configuration file onto the base and confirming that the result is the intended arm. A configuration change that would quietly compromise the experimental design is therefore surfaced as a build failure rather than left undetected.

The same intent governs how an experiment is launched, as every reported run was started by one of the `make` targets showcased in Table D.2, with explicit parameters. Every tuneable value is likewise declared in a YAML configuration file rather than left as a hidden code default.

3.2 The Parking Environment

The world is a tailored flat OpenDRIVE surface integrated into CARLA through its programmatic world generator, which restricts the manoeuvre to the horizontal plane and leaves the layout as the sole source of geometric variation. No camera sensor is affixed to the vehicle, so Unreal rendering is disabled throughout training. Two lots are defined through the layout generator of Section 3.1.3 and illustrated in Fig. 3.2, with every approach commencing at the cyan spawn marker and concluding at one of the blue bays within the shaded lot region, which the dashed out-of-bounds line encloses 5 m further out. Training takes place exclusively in `rectangle`, an axis-aligned 65×42.5 m lot with 47 targetable perpendicular bays arranged in five spatial clusters across three bay orientations. Its counterpart `irregular_a` is bounded by five sides, one corner cut away by a diagonal wall, and is reserved for the out-of-distribution evaluation in Chapter 4. The aisles of both lots are held to the 6.0 m minimum stipulated by published parking-design guidance for 90° bays [72]. The manoeuvre is therefore performed under the same spatial constraint as a compliant real facility.

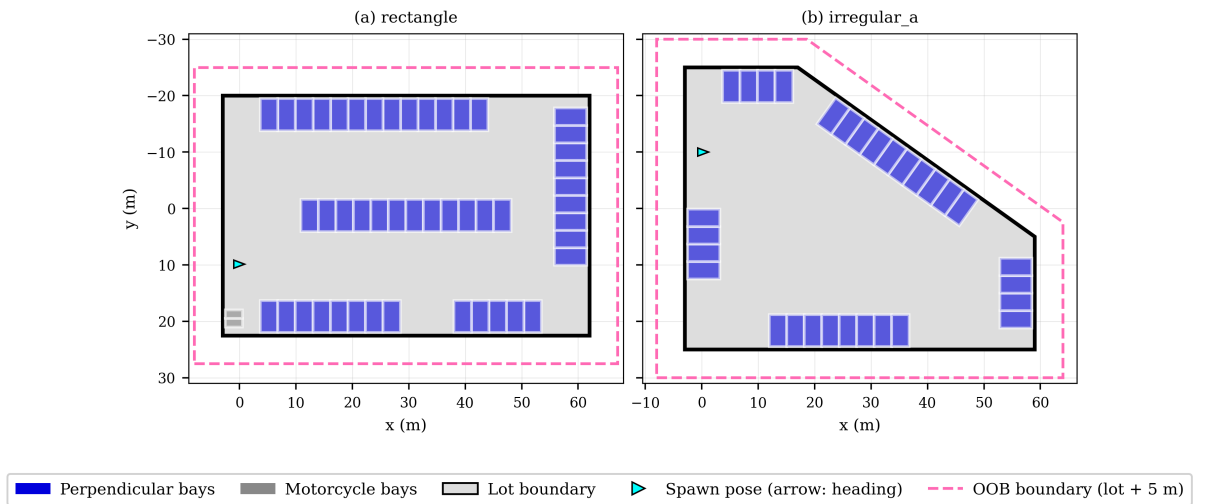


Figure 3.2 Bird's-eye plots of both lot layouts as generated, (a) `rectangle` and (b) `irregular_a`.

3.2.1 Scope Decisions

Beyond the layouts themselves, the reverse gear is disabled and the only obstacles in the lot are stationary parked vehicles. The constraint on the reverse gear stems from a single-antenna sensor configuration, specified later in Section 3.4. One GNSS antenna cannot measure absolute heading directly, so the angle is instead inferred from the COG of successive fixes and fused by the EKF as its sole yaw correction. The substitution is valid under forward motion, since COG reports the direction of travel rather than the orientation of the vehicle (Section 2.1.4). In reverse the two diverge by 180° , and a heading error of that size would propagate into the yaw standard deviation σ_ψ , one of the three covariance features given to the policy in Section 3.3. Restricting the vehicle to forward motion consequently preserves the integrity of the signal the study manipulates, at no cost to the task, since the success criterion of Section 3.6 requires no more than the lateral clearance achievable through a forward approach into a perpendicular bay.

Stationary parked vehicles are likewise sufficient as the sole obstacle population, since each neighbouring vehicle is simultaneously a LiDAR landmark from which the clearance features are computed and a collision hazard reachable under an accurate fix, through a mistimed or badly angled approach alone. Occupancy is thereby a genuine axis of difficulty, and failure in this task is not confined to what the localisation estimate reports.

3.2.2 Episode Mechanics

At the commencement of each episode, the ego vehicle is positioned at the lot's spawn pose. A target bay is subsequently selected uniformly from those available at the current stage of the curriculum yet to be presented in Section 3.7, four bays during the initial stages and all 47 in the final one. The occupancy bounds $o_{\min}$ and $o_{\max}$ are established by the same stage, and a single occupancy rate $u \sim \mathcal{U}(o_{\min}, o_{\max})$ is drawn from that range for the episode. Every bay other than the designated target is thereafter occupied with probability u by a vehicle blueprint drawn at random, and each bay is resolved by its own draw. Every parked vehicle is then rotated by 180° in half of all cases, so a target bay is approached with nose-in and nose-out neighbours alike.

Upon populating the lot in this manner, an episode that neither succeeds nor collides runs until a clock expiry of 1750 simulation ticks, capped at 437 policy decisions at the 5 Hz decision rate set out in Section 3.3. Termination before reaching that ceiling is governed by two mechanisms, the first applied at the lot boundary and enforced softly rather than through a collidable wall. The lot polygon is expanded outward by 5 m to form a run-off skirt. Every decision made with the vehicle centre

beyond that skirt incurs a reward of -0.5 , and the episode concludes once 10.0 of such cost has accumulated. Twenty decisions outside the skirt are therefore tolerated before the episode ends, so a single erratic deviation during a degraded fix is insufficient to terminate an otherwise salvageable rollout. The tolerance is most valuable during early exploration, as the outer perimeter is rarely approached under a converged policy. The reward function developed later in Section 3.6 draws the vehicle towards a bay within the lot instead, offering nothing for movement outwards.

A second mechanism addresses the opposite failure, an episode in which no progress is made. A rollout is deemed stalled once 50 consecutive decisions are taken outside the target bay below 0.1 m s^{-1} , the speed the success criterion of Section 3.6 requires, which amounts to ten seconds of standstill. The remainder of the episode is thereby reclaimed, since a motionless vehicle would otherwise be carried to the clock ceiling, flooding the buffer with repetitive zero-advantage frames. Deliberate waiting must be exempt from this rule, as pausing under a degraded fix is the behaviour that the uncertainty inputs of Section 3.3 are intended to elicit. The stall counter is consequently suspended whenever the live EKF position standard deviation exceeds 0.4 m, the boundary between the float and standalone fix bands subsequently defined in Section 3.4. The threshold of 50 decisions follows from those same bands, under which the longest expected degraded excursion lasts approximately seven seconds, comfortably within the ten permitted by the counter. A justified wait for recovery is therefore never cut short.

Truncation is likewise no cheaper than exhausting the clock, as a stalled episode is terminated under the same graded timeout penalty applied at clock expiry, as specified in Section 3.6.3. Discounting shrinks a penalty the later it is received, so the cost of stalling out of a failing episode exceeds that of seeing it through to the ceiling.

3.3 Observation and Action Spaces

The policy observes a 13-dimensional vector composed of four semantic blocks: vehicle state, localisation uncertainty, relative target pose and obstacle clearance. Table 3.4 defines every dimension and Fig. 3.3 shows the block structure. The covariance gate (indices 2-4) is the input-uncertainty factor of the 2×2 study, and disabling it yields the 10-dimensional observation of the covariance-blind arms. Those three dimensions carry the estimator-reported uncertainty of Section 2.1.4 into the policy, addressing the input-side absence identified in Section 2.3, where no continuous-control policy for ego-motion was found to receive covariance-derived features as observations.

Table 3.4 The 13-dimensional observation vector.

Index	Feature	Source	Units
0	speed (signed, body-frame)	EKF filtered twist	m s^{-1}
1	yaw rate $\dot{\psi}$	EKF filtered twist	rad s^{-1}
2–4	$\sigma_x, \sigma_y, \sigma_\psi$ (yaw)	EKF covariance diagonal	m, m, rad
5–7	$dx, dy, d\psi$ (target, ego frame)	EKF pose estimate	m, m, rad
8–9	left distance, left bearing	LiDAR (bearing $> +15^\circ$)	m, rad
10–11	right distance, right bearing	LiDAR (bearing $< -15^\circ$)	m, rad
12	forward distance	LiDAR ($ \text{bearing} \leq 15^\circ$)	m

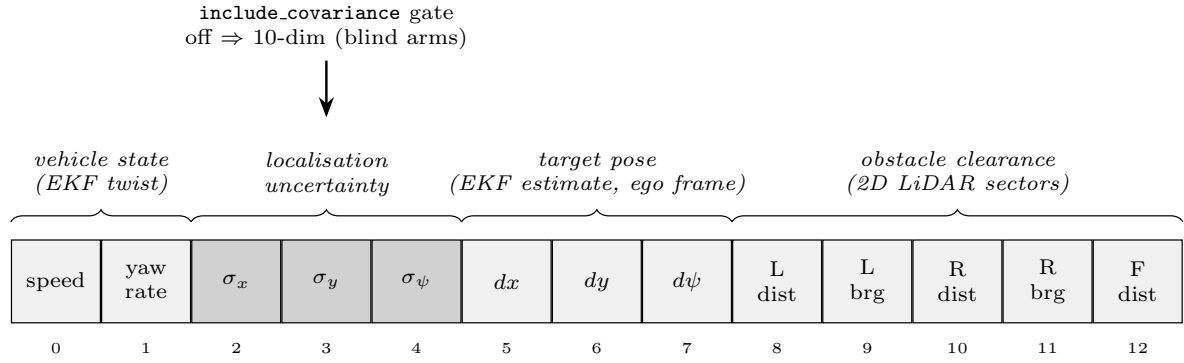


Figure 3.3 Block structure of the observation vector.

3.3.1 Justification of Each Block

The vehicle-state block encompasses the two components of the vehicle's motion, specifically its speed along the body axis and its yaw rate. Speed is reported as a signed body-frame quantity, and success is assessed against it directly. A rollout counts as a park only when the speed stays below 0.1 m s^{-1} for five consecutive decisions, one of the two conditions of the success criterion defined in Section 3.6. The second component carries no such threshold, as turning is expressed as yaw rate, the rate at which the vehicle is rotating under the steering command delivered on the previous decision. Both are derived from the filtered twist of the EKF, so the policy is trained against the Doppler and COG error characteristics described in Section 2.1.4.

Accompanying that motion is the uncertainty block, which consists of the three standard deviations along the diagonal of the EKF's $[x, y, \psi]$ covariance submatrix. The off-diagonal entries hold the correlations among those three states, and an earlier design of the observation supplied them to the policy accordingly, on the premise that the shape of the uncertainty is characterised by more than the variances alone. One

consideration stands against that premise. Each simulated fix carries a covariance with zero off-diagonal entries, as the degradation model introduced later in Section 3.4.1 scales the error on each axis independently and introduces no coupling between them. A change in the quality of the fix therefore leaves the correlations almost untouched, so no indication of that quality would reach a policy reading such an entry, while the three standard deviations already separate the localisation regimes to which the vehicle must respond. Supplying the correlations would therefore expand the observation without adding information, attenuating the three features where the uncertainty signal actually resides.

Those deviations describe the reliability of the third block, the target pose, which locates the assigned bay within the ego body frame through three quantities derived from the EKF pose estimate, a forward offset, a lateral offset and a heading discrepancy. As localisation deteriorates, so does the target pose in the observation, and this decline is simultaneously reflected in the covariance features, from which the appropriate degree of restraint can therefore be learned. The choice of frame serves a second purpose, as each lot has its own origin and its own bay positions, so world coordinates supplied as observations would bind a policy to the survey on which it was trained. An offset to the target carries no such dependence, since the same approach geometry yields the same observation irrespective of the frame in which the bay was surveyed. Absolute position is consequently absent from the observation altogether.

The target pose gives the direction to the bay without revealing what stands in the path towards it, so the vehicles parked in neighbouring bays are reported alongside it by the obstacle block. Its features come from the LiDAR scan, segmented at bearings of $\pm 15^\circ$ into a narrow forward cone and a sector to either side, each of the three reduced to its nearest return. The left and right sectors retain both its distance and its bearing, as an obstruction to the side must be located as well as detected, whereas the forward cone is narrow enough for the distance alone to serve that purpose. Two returns are excluded before any of this, those within 1 m, treated as the vehicle's own body, and those behind it, lying outside the 270° aperture of the front-mounted two-dimensional scanner [73].

3.3.2 Fixed Physical-Range Normalisation

The four blocks are quantified in metres, radians and their per-second derivatives, so a raw 40 m target offset is orders of magnitude larger than a heading standard deviation of only a few tenths of a radian, leaving the gradient dominated by scale alone. Some form of rescaling is therefore necessary, with the prevalent choice being a running normaliser that divides each dimension by statistics gathered throughout training. Such a normaliser was used in an earlier design of this environment, where an input was

amplified without limit at a boundary between the stages of the curriculum. The shift in range across such boundaries was sufficient to dislodge the accumulated statistics, which accounts for the failure. A fixed physical scale is therefore adopted for each dimension, as detailed in Table 3.5 and applied through the division and clip of Eq. (3.1):

$$\tilde{o}_i = \text{clip}\left(\frac{o_i}{s_i}, -5, 5\right) \quad (3.1)$$

where o_i represents the raw value of dimension i in the physical units of Table 3.5, s_i signifies the fixed scale by which it is divided, and $\tilde{o}_i$ is the normalised input supplied to the policy after the clip. A value at its scale, denoted by $o_i = s_i$, is thus presented to the network as $\tilde{o}_i = 1$, while any measurement of five scales or more saturates the clip. Every input is thereby bounded to ± 5 , irrespective of the magnitude of the raw measurement. Each s_i is set to the full physical range of its dimension, so the magnitude of a feature at the network input represents the fraction of that range it occupies rather than the unit in which it is measured. A dimension confined to a narrow part of that range is consequently compressed towards zero, at a cost in resolution that a running normaliser would have recovered. The scales function as structural constants of the environment rather than tuned parameters, remaining identical across every curriculum stage and layout, and that constancy is worth the resolution forgone. Adaptive scaling is consequently confined to the discounted returns maximised by the clipped surrogate objective of PPO, leaving the observations untouched.

Table 3.5 Fixed normalisation scales s_i by observation block.

Block	Dimension	Scale s_i	Rationale
Vehicle state	speed	5 m s^{-1}	forward parking-speed cap
	yaw rate	$\pi \text{ rad s}^{-1}$	wrapped to $[-\pi, \pi]$
Uncertainty	σ_x, σ_y	1 m	EKF position-std ceiling
	σ_ψ	0.5 rad	heading-std ceiling ($\sim 29^\circ$)
Target pose	dx, dy	40 m	offset over the $65 \times 42.5 \text{ m}$ lot
	$d\psi$	$\pi/2$	wrapped to $[-\pi/2, \pi/2]$ (180° symmetry)
Obstacle	distances	25 m	LiDAR maximum range
	bearings	$\pi/2$	forward-hemisphere bound

3.3.3 Action Space and Actuator Model

The simulator functions in synchronous mode at a fixed rate of 20 Hz, updating physics and sensors on every tick. An action is issued once every four ticks, producing a

decision rate of 5 Hz. A faster reaction provides no benefit at parking speeds, so the command persists across the three intervening ticks, throughout which the physics continues to integrate it. Deciding at every tick would require four times as many forward passes for the same duration of simulated driving. Consequently, a single call to the environment's step function represents one policy decision and one SB3 timestep spanning 0.2 s. Every budget and metric in this study is therefore expressed in decisions.

Each decision issues a continuous action over the three axes that control the vehicle, [steer, throttle, brake]. Every axis is emitted on $[-1, 1]$ by the tanh-squashed policy, which is constructed in Section 3.5. Steering uses that interval in full, since a negative command steers left and a positive one steers right. Throttle and brake carry magnitude only, so their negative halves are folded to zero and each pedal is driven on $[0, 1]$ by the positive half alone. The fold also withholds the reverse gear from the interface, under the single-antenna heading constraint of Section 3.2.1.

Requests from the policy are not delivered to CARLA directly, passing first through an actuator model. A steering wheel or pedal in a real vehicle moves at a finite slew rate, so the change attainable within one decision interval is bounded. A raw request is subject to no such bound, and full left lock may be demanded on one decision and full right on the next. Every axis is therefore confined to a band centred on its previous command:

$$a_t^{\text{cmd}} = \text{clip}(a_t, a_{t-1}^{\text{cmd}} - \Delta_{\text{max}}, a_{t-1}^{\text{cmd}} + \Delta_{\text{max}}) \quad (3.2)$$

where a_t is the raw action requested on that axis at decision t , a_{t-1}^{cmd} the command actually delivered on the previous decision, and Δ_{max} the per-axis rate limit. A request falling within that band is delivered unchanged, whereas one lying outside it is truncated to the nearest bound, so the delivered command a_t^{cmd} moves by at most Δ_{max} per decision.

Each Δ_{max} is selected to correspond to the rate achieved by a real actuator rather than being optimised for training, and Table 3.6 compiles the values adopted from the published measurements that anchor them. Steering is the most constraining of the three axes, restricted to 0.20 normalised units per decision, which over that interval translates to a road-wheel rotational speed of approximately 70° per second, below both the 140° per second ceiling of a driverless electric power-steering design [74] and the 115° per second MPC comfort envelope [75]. The pedals are limited less stringently, at 0.50 normalised units per decision, half the actuation range, reaching full application in 0.4 s, which remains slower than the sub-200 ms response of production drive-by-wire and electro-hydraulic hardware [76, 77]. Simultaneous demand for both pedals is unrestricted by the action space, so the brake is given

priority on safety grounds, with a delivered brake above 0.3 setting the throttle to zero immediately rather than following the throttle rate limit.

Table 3.6 Actuator rate limits.

Axis	Limit / decision	Physical rate
Steering	0.20	70°/s road wheel
Throttle	0.50	0 to full in 0.4 s
Brake	0.50	0 to full in 0.4 s
Priority	brake > 0.3	throttle cut to 0

A governor independent of the rate limits bounds the speed of the vehicle at 6 m s^{-1} and cuts the throttle above that threshold. The ceiling is intentionally positioned well above parking speed, as a limit set close to it would suppress the deceleration behaviour the evaluation is designed to detect.

3.4 Localisation Pipeline and Sensor Noise Models

The noise associated with the covariance features in Section 3.3 originates from published hardware specifications rather than being tuned for the specific task. The simulated suite is accordingly tailored to a named target platform, constructed around a VectorNav VN-100 IMU [78], a u-blox ZED-F9P RTK GNSS receiver [1] and a SICK TiM571 two-dimensional LiDAR [73]. Every value in Table 3.7 can be traced to one of those three datasheets except for the two marked †, each resting on a different substitute. No manufacturer figure exists for the ambiguity-float tier, as its accuracy varies both with atmospheric conditions and with baseline length. The tier is therefore assigned the decimetre regime that Reid et al. [34] report for float solutions over 30 000 km of highway driving, where the observed error distribution separates into centimetre, decimetre and metre regimes. The degraded tier rests on no measurement at all, being a stress condition rather than a documented operating mode, with its 5.0 m error selected to exceed the worst case the datasheet describes and to establish how the policy behaves once the fix is too poor for the manoeuvre to be attempted. The derivation of every row from its source specification, the two exceptions included, is given in Appendix A.

3.4.1 The Global Navigation Satellite System Fix-State Markov Chain

In a parking lot, the deterioration mechanisms described in Section 2.1.4 are influenced by the surrounding obstructions, among them the vehicles parked in neighbouring

Table 3.7 Sensor noise parameters and their provenance.

Sensor & quantity	Parameter	Value (1σ)	Provenance / derivation
ZED-F9P GNSS	rtk_fixed position	0.020 m	datasheet accuracy table [1]
	rtk_float position	0.360 m	measured decimetre float regime [34], no datasheet value [†]
	standalone position	1.802 m	datasheet accuracy table [1]
	degraded position	5.0 m	stress condition beyond the datasheet [†]
ZED-F9P velocity	Doppler speed (rtk_fixed)	0.05 m s^{-1}	datasheet velocity accuracy [1]
	per-tier scaling	$\times 1 / 18 / 90 / 250$	matched to the position-noise ladder
VN-100 IMU	gyro variance / sample	7.46×10^{-8}	noise density $0.0035^\circ/\text{s}/\sqrt{\text{Hz}}$ at 20 Hz [78]
	accel variance / sample	3.77×10^{-5}	noise density $0.14 \text{ mg}/\sqrt{\text{Hz}}$ at 20 Hz [78]
	bias (per episode)	gyro $5\text{--}7^\circ/\text{h}$ accel $< 0.04 \text{ mg}$	resampled per episode within datasheet in-run bias stability [78]
TiM571 LiDAR	systematic range bias	$\pm 60 \text{ mm}$	datasheet systematic error, per episode [73]
	statistical range noise	20 mm	datasheet statistical error, per point [73]

Variance units: gyro $(\text{rad/s})^2$, accel $(\text{m/s}^2)^2$, per sample.

bays. The degradation of fix quality is therefore determined by conditions external to the manoeuvre rather than by the vehicle's location.

The fix state is consequently represented as a discrete-time Markov chain over the four receiver tiers rather than as a constant sampled once per episode. Each tier introduces zero-mean Gaussian error at the per-axis standard deviation (1σ) listed in Table 3.7, obtained from the manufacturer's CEP figures under the circularly symmetric Gaussian assumption of Section 2.1.4. Those figures were measured under static open-sky conditions, which lack the reflectors of a parking lot. The radial value is therefore retained per axis rather than converted, which places every tier a factor of $\sqrt{2}$ above the strict interpretation, and the arithmetic behind each is set out in Appendix A.1. A transition structure is imposed across the four tiers of the discrete-state formulation adopted in Section 2.1.4, and is evaluated at every GNSS callback (20 Hz). The quality of the solution is thereby permitted to change within a single run, exposing the vehicle to deterioration and to recovery alike over the course of a manoeuvre.

Isolating a particular failure pattern requires a departure from that structure, so the chain can be modified for evaluation. The provision takes the form of a one-way

drift mode, beginning at `rtk_fixed` and deteriorating monotonically without any possibility of recovery. This mode is obtained by suppressing the recovery transitions through an override sent over the file bridge introduced in Section 3.1.2, and it serves as a stress condition in the evaluation of Section 4.1. A second setting reaches the relay by the same route, prompted by the units in which the injected error is quantified. The noise levels in Table 3.7 are given in metres, whereas the simulated receiver publishes latitude and longitude, so a displacement in metres corresponds to a different angular offset at every location on the globe. The mismatch is resolved using a local datum derived from the geolocation of the spawn point and re-latched at every reset.

The resulting configuration is depicted in Fig. 3.4, with the probabilities derived directly from the runtime settings. The tiers appear as nodes, annotated with the 1σ position noise each of them introduces, while the per-callback transition probabilities are shown on the connecting arcs, drawn in black for recovery towards `rtk_fixed` (upper) and in thin grey for degradation away from it (lower). An additional quantity appears beneath each node: the mean dwell for the three lost-fix tiers and the long-run occupancy for `rtk_fixed`, the latter derived from the transition matrix rather than configured.

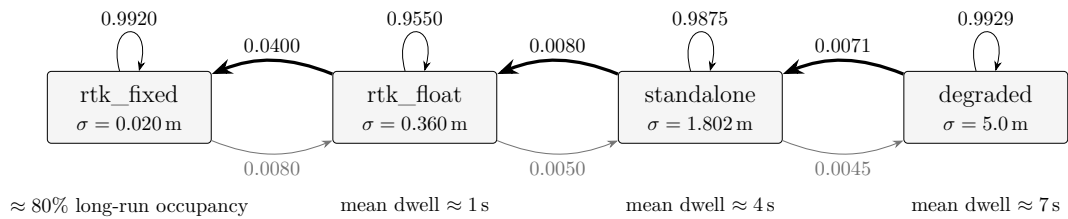


Figure 3.4 The four-state GNSS fix-state Markov chain.

The design is articulated through three attributes of the chain illustrated in Fig. 3.4. Transitions are permitted only between adjacent tiers, as a receiver cannot shift from centimetre to metre accuracy without passing through the float solution, the intermediate stage of carrier-phase ambiguity resolution detailed in Section 2.1.4. Movement through the tiers is then biased upward, as recovery is more likely than further deterioration at every level below `rtk_fixed`. The chain is therefore fixed-dominant, with around 71% of the long-run occupancy residing in `rtk_fixed` itself. The dwell times of the lost-fix tiers progressively increase down the hierarchy, from approximately one second in `rtk_float` to four in `standalone` and seven in `degraded`. Severity and duration consequently rise together, with the longest exposure occurring in the tier where awaiting recovery is the only appropriate response. None of these dwell times exceeds the ten-second stall-truncation threshold established in Section 3.2.2, so a justified pause is never penalised.

Every episode begins at an initial tier, from which the chain subsequently advances. That starting point is drawn from a fixed distribution placing probability 0.40 on `rtk_fixed` and 0.20 on each of `rtk_float`, `standalone` and `degraded`, and it reaches the relay through the file bridge described in Section 3.1.2. The split is intentional, as a substantial proportion of episodes is seeded into a diminished fix from the outset rather than allowed to deteriorate over the course of the run. Competent action is therefore required on a fix that was already poor at the onset of the episode, without any preceding deterioration to indicate it.

The upward bias was adopted only after an earlier chain had proved unsuitable. Its matrix was free-diffusing, retaining approximately 30 % of its long-run time in `rtk_fixed` and recovering over intervals measured in minutes. A degraded episode was consequently unwinnable, as the fix would not return before the episode had ended. Under the chain adopted in its place, waiting for recovery is both physically plausible and learnable within a single episode. That wait is the behavioural mechanism the evaluation is designed to detect.

Beyond the fix state, the noise model extends to the rest of the sensor suite. The chain governs the magnitude of the GNSS position error and nothing else, so five secondary effects are established for the error characteristics it leaves unmodelled.

The first effect concerns the direction of the positional error. Each tier in Table 3.7 carries a single per-axis figure, which reproduces the circular idealisation that Section 2.1.4 sets against the elliptical error of a real fix. Direction is restored by drawing an east/north ratio of at most 1.5 and applying it to the tier's 1σ value, multiplying on a randomly chosen axis and dividing on the other. The eastward and northward components thereby diverge, while their geometric mean remains the tabulated figure. Satellite geometry orients the ellipse [32] and holds throughout a manoeuvre before shifting between one arrival and the next, so the ratio is drawn once per episode to match. It reaches the two diagonal components of the reported covariance, and not the position alone, since altering the position by itself would leave the eastward and northward uncertainties indistinguishable and the anisotropy would never arrive at the covariance features observed by the policy.

Four further effects correspond to independent failure modes distributed across the receiver, the IMU and the LiDAR. Two originate in the receiver, one affecting the reported position and the other the heading. The positional mode consists of cycle slips and transient satellite obstructions [32], reproduced by discarding each fix with a probability of 0.02, after which the EKF estimate advances solely on the prediction step until the next fix arrives. The COG heading is instead conditioned on speed through a variance obtained by analytic error propagation that scales as the Doppler noise divided by speed. The variance is published alongside the COG measurement and used as the measurement noise R_k of the yaw channel in the Kalman gain of Eq. (2.11),

which downweights a heading obtained at low speed. That same reasoning suppresses the heading outright in two cases. The first is a displacement between consecutive fixes too small to define a direction, and the second an IMU reading of standstill. The second of those is handled on the inertial side by a Zero-Velocity Update (ZUPT), the standard inertial-navigation correction applied over a detected stationary interval to bound drift [32], during which the gyro and planar accelerations are clamped to zero. Sensor noise from a motionless vehicle is thereby kept from being integrated as motion.

The remaining two effects introduce no failure at all, each randomising instead a characteristic that a physical unit would hold fixed. The IMU bias and scale-factor inaccuracies are resampled for every episode within the bounds of the datasheet, so no single calibration becomes ingrained during training. The same principle governs the LiDAR noise, applied in polar coordinates as a systematic bias drawn once per episode alongside statistical noise drawn for each point. The parameters governing both the IMU and the LiDAR are those listed in Table 3.7.

3.4.2 The Extended Kalman Filter

The `robot_localization` EKF node described in Section 2.1.4 performs localisation at 20 Hz in planar mode. Three aspects of its configuration bear on the study, namely the inputs that are fused, those deliberately withheld, and the tuning of the process noise.

The receiver observables account for two of the three inputs to the filter. The first input is a single fused quantity arriving together from the receiver, combining the flat-earth projected position with the Doppler-derived longitudinal velocity, while the gated COG heading forms a second input that corrects the yaw. The IMU yaw rate is the third, driving the prediction step. Linear acceleration is intentionally left unintegrated, as the velocity states are already constrained by the 20 Hz velocity measurement, so an accelerometer channel would introduce integration drift without any benefit at low speeds. The LiDAR is withheld on different grounds, serving obstacle clearance alone, so the covariance features carry no trace of the obstacle channel.

On the evidence of Kovacs et al. [29] reviewed in Section 2.1.4, the process-noise covariance Q is treated as a tuning variable rather than left at its default values, configured so that the posterior grows appreciably whenever the corrections weaken. A degraded fix carries large measurement noise R , which the gain defined in Eq. (2.11) renders small. Each update consequently removes less uncertainty than the preceding prediction added, causing the reported covariance to climb for as long as the fix remains poor. The steepness of that climb is set by the active diagonal of Q , which holds one spectral density in variance per second for every state. Those densities are integrated over the timestep to give the Q_k of the prediction step in Eq. (2.10). The yaw rate is set to 1×10^{-3} and anchors the diagonal, as the only fused IMU channel and

the dominant source of growth between fixes. The two remaining entries are sized against it by the measurement each depends on, 1×10^{-2} for the position (x, y) and 1×10^{-4} for the yaw, and together the three govern the speed at which a lost fix becomes apparent in the covariance features. The full diagonal and the reasoning behind each value are given in Appendix A.4.

A representative trace of σ_x across a single training episode is illustrated in Fig. 3.5, with the fix-state tiers shaded in the background, presenting the empirical sawtooth anticipated in Section 2.1.4. The reported σ_x closely follows those tiers, increasing as the solution deteriorates into standalone and degraded, then holding for as long as the vehicle remains within them. The decline of the uncertainty is far steeper than its rise, as a reaccepted clean fix restores a small R and with it a large gain, so a handful of updates recover the confidence that many predictions had eroded.

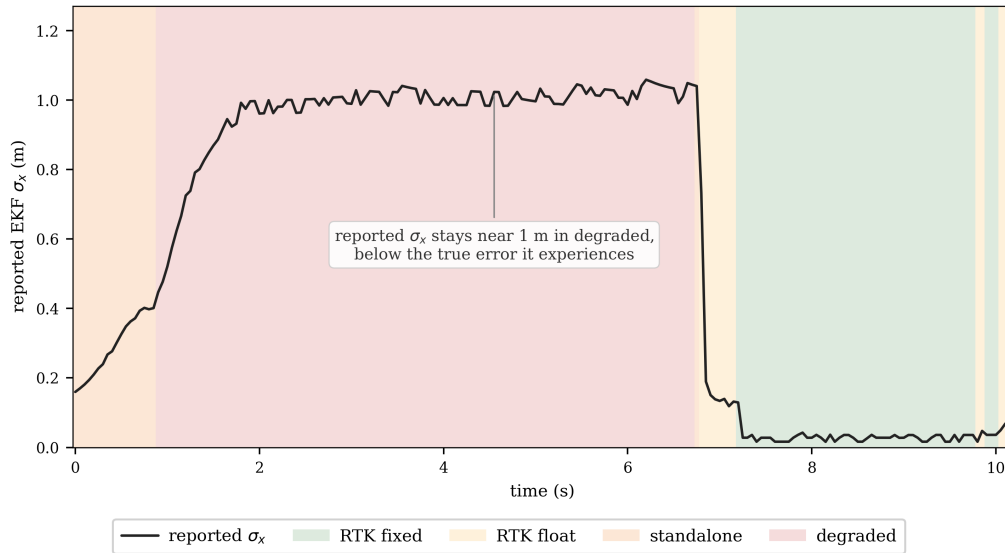


Figure 3.5 Reported EKF σ_x over one training episode.

The limit of the reported σ_x as an uncertainty signal is visible in the same trace. During the dwell in the degraded tier, where the injected position noise reaches 5 m (Table 3.7), the reported σ_x remains at approximately 1 m and so understates the true localisation error several times over. The discrepancy arises from how the quantity is formed. The measurements received during the dwell are too noisy to constrain the estimate, so the posterior is inflated at the bounded rate determined by the tuned Q and not measured against the error the vehicle is actually incurring. That bound puts the full injected error beyond reach within a single dwell, so the reported standard deviation indicates the fix state and the direction in which it is moving without estimating the instantaneous error.

Any localisation approach for this task has to produce an uncertainty that varies within an episode and is not tied to one vehicle, and two alternatives were

implemented and rejected before those requirements became clear. The first involved scan-matching against perimeter cones using Cartographer [79], which was attractive because match quality decreases as parked vehicles occlude the cones, coupling uncertainty to occupancy by construction rather than by accident. The coupling was also its defect, as occupancy remains fixed throughout an episode, leaving a rollout either well-localised or poorly-localised from beginning to end. The reported uncertainty would consequently never rise and fall within a single episode, rendering the appropriate response to a given level unlearnable. The second alternative, wheel odometry fused with the other sensors, was briefly attempted before being reverted. In principle, it would have sufficed on this lot, but the signal would thereby be tied to a particular drivetrain, restricting the setting to platforms that expose clean wheel encoders.

Neither objection applies to RTK GNSS with IMU fusion. Its degradation ladder is documented by the manufacturer, so the injected noise is defensible rather than tuned, and the fix state varies throughout an episode rather than being established at the outset. The choice constrains the experiment but not the method, as the policy conditions on a faithful covariance rather than on the sensor producing it. Any localisation pipeline reporting a well-founded covariance would therefore be equally applicable, and the receiver modelled in Table 3.7 is one instance of that input rather than a prerequisite for it.

3.5 The Evidential Actor

The covariance in Section 3.4 serves as an observation for the policy, measuring the uncertainty in the estimated pose without imparting any confidence in the selected action. The evidential NIG actor head provides that confidence, producing an action distribution and a decomposition of its uncertainty from a single forward pass.

The head uses the DER construction from Section 2.1.3 without any structural alterations, differing solely in the regression target. The original formulation was designed for a supervised label y , while the Gaussian-NIG hierarchy in Eq. (2.6) is instead applied to a sampled action a along a single axis of the continuous action space defined in Section 3.3.3, with the substitute for the absent label established in Section 3.5.2. Four parameters (γ, ν, α and β) are generated for each action dimension, and the virtual-observation interpretation from Section 2.1.3 transfers with them, so ν reflects the evidence supporting a proposed action rather than a predicted label. The two uncertainty channels subsequently follow from the NIG moments in Eq. (2.7):

$$\sigma_{\text{alea}}^2 = \frac{\beta}{\alpha - 1} \quad \sigma_{\text{epi}}^2 = \frac{\beta}{\nu(\alpha - 1)} = \frac{\sigma_{\text{alea}}^2}{\nu} \quad (3.3)$$

A single NIG head therefore yields σ_{epi}^2 by dividing σ_{alea}^2 by ν , tying the two channels together through the evidence count alone, and the consequences of that coupling are examined in Section 4.3.2. Neither estimate is claimed as an exact Bayesian posterior, both being treated throughout as the heuristic single-pass signal argued for in Section 2.1.3.

3.5.1 The Normal-Inverse-Gamma Actor Head

The two uncertainty channels are meaningful only while the four parameters remain well defined, and three properties of the head must hold for that condition to be met. A constraint on the outputs must bound the region the parameters can occupy, an initialisation must fix their starting point within it, and a sampling rule must convert them into an action.

The constraint on the outputs arises from the HUA described in Section 2.1.3, the region surrounding $\alpha = 1$ where the evidential gradient vanishes and training stalls. The consequence of entering that region depends on the setting. In a supervised regressor, only accuracy is affected, whereas in an actor the cost is considerably greater. According to Eq. (3.3), the aleatoric variance is the action sampling variance, which becomes unbounded as $\alpha \rightarrow 1$, and an unbounded sampling variance signifies policy failure.

This head employs the generic softplus-with-offsets constraint from Section 2.1.3, configured so that the HUA lies beyond the reachable range, with an offset applied to every activation rather than to α alone:

$$\nu = \text{softplus}(o_\nu) + 10^{-6} \quad \alpha = \text{softplus}(o_\alpha) + 1.5 \quad \beta = \text{softplus}(o_\beta) + 10^{-6} \quad (3.4)$$

The offsets on ν and β are negligible floors that ensure the values remain strictly positive, with each activation clamped at 100 at the opposite end as a numerical safeguard. The offset on α is the critical choice, set to 1.5 rather than the usual 1.0. The denominator $\alpha - 1$ of Eq. (3.3) is thereby held at 0.5 or above by construction, so the pathological region is structurally unreachable and not merely regularised against.

The offsets in Eq. (3.4) bound the region in which the head can settle while leaving its starting point unconstrained, and that initial state presents a hazard of its own. Conventional weight initialisation drives the raw outputs towards zero, and a softplus input near zero sends both ν and β to their 10^{-6} floors. Since ν sits in the denominator of the epistemic variance (Eq. (3.3)), that estimate diverges from the first step and takes the gradient updates with it. A deliberately selected initial state is therefore required instead of the default, and it is established in three parts.

Control over the outputs is first transferred from the weights to the biases by scaling all head weights down by a factor of 0.01, so each output is dominated by its bias term at the first step. Those biases are subsequently assigned to the targets in Table 3.8, one group of which determining the action means for the driving task. The throttle bias is positive, whereas the brake bias is firmly negative, so the freshly initialised policy makes the vehicle creep forward rather than engaging both pedals simultaneously. The initial evidence is set by the remaining group: ν starts below one, at approximately 0.313, so according to Eq. (3.3), the starting prior is epistemic-dominant, the regime intended for a novelty-reading gate in unfamiliar states. These targets survive integration only if they are re-applied, as the SB3 orthogonal-initialisation pass zeroes all biases after the head is built and would otherwise erase them. The NIG bias targets are consequently restored once that pass has finished.

Table 3.8 Bias-driven NIG initialisation targets.

Parameter	Bias	Activation	Value at init	Interpretation
γ steer	0.0	linear	0.0	straight ahead
γ throttle	+0.5	linear	+0.5	pedal default-on during approach
γ brake	−1.0	linear	−1.0	pedal default-off (folded to zero)
ν	−1.0	softplus + 10^{-6}	≈ 0.313	sub-one evidence: epistemic-dominant prior
α	0.9	softplus + 1.5	≈ 2.741	well-defined finite variance, clear of the HUA
β	0.0	softplus + 10^{-6}	≈ 0.693	moderate scale

A head constrained by the softplus offsets set out in Eq. (3.4) and initialised to the bias targets of Table 3.8 yields its actions through a tanh-squashed Gaussian. An unbounded latent variable z , the action before squashing, is sampled from the Gaussian likelihood of Eq. (2.6), centred on the location γ and scaled by the square root of the aleatoric estimate from Eq. (2.7). Its epistemic counterpart is excluded from that scale, as it quantifies uncertainty over the mean γ rather than any further noise on the action itself. The latent is drawn under a clamp on the sampling variance, motivated in Section 3.5.2, and is then transformed through tanh into the bounded action a , with the throttle and brake axes folded to their non-negative halves as described in Section 3.3.3:

$$a = \tanh(z) \quad z \sim \mathcal{N}\left(\gamma, \text{clamp}\left(\frac{\beta}{\alpha-1}, \sigma_{\min}^2, 1.0\right)\right) \quad (3.5)$$

3.5.2 Adapting Evidential Learning to Reinforcement Learning

Positioning an evidential head on an actor instead of the supervised regressor from Section 2.1.3 introduces two contrasting pathologies. The first is a divergence, under

which the regulariser increases indefinitely and the uncertainty estimates rise with it. The second is the inverse failure, a collapse under which the predicted variance approaches zero and exploration halts. Both can be traced to the same origin, the absence of a fixed regression target under policy sampling, and each is identified and addressed in turn below.

The divergence stems from the loss itself, which requires modification before it can be applied to RL. The missing target established in Section 2.2.2 bears directly on the evidence regulariser in Eq. (2.8), since the sampled action must substitute for the label under its residual form. The residual $|a - \gamma|$ then represents the policy’s inherent exploration noise rather than a prediction error, so the regulariser rewards evidence reduction across exploratory actions by penalising the very variability on which exploration relies.

The effect of that penalty became apparent during an early training run, one conducted with the unmodified loss and without any of the constraints of Section 3.5.1. Its training traces, which record the uncertainty estimates and task metrics at each step, show the regulariser loss increasing indefinitely, driving the aleatoric estimate from approximately 2 to above 11 000. Exploratory actions were suppressed under that penalty, so the policy entropy collapsed alongside it and success remained at zero throughout. Two faults combine to produce that divergence, the first being a residual-weighted regulariser poorly specified under policy sampling. That condition is absent from every supervised placement surveyed in Section 2.2.2, and the penalty consequently grows with the very exploration it suppresses. Compounding that instability, the NIG parameters are left unconstrained, so their growth under the penalty is subject to no upper limit.

The final design addresses each fault in turn. The runaway parameters are contained by the output clamping and initialisation protocol of Section 3.5.1, which restricts the NIG to a bounded operating range, and the poorly defined penalty is eliminated by replacing the residual-weighted term with the prior-anchoring regulariser below.

$$\mathcal{L}_{\text{reg}} = \mathbb{E} \left[\left(\frac{\alpha}{\alpha_0} - 1 \right)^2 + \left(\frac{\beta}{\beta_0} - 1 \right)^2 \right] \quad \alpha_0 = 2.741 \quad \beta_0 = 0.693 \quad (3.6)$$

This raw-ratio quadratic does not require a target, as each factor compares a live parameter against a fixed reference and not against a data label, with the expectation $\mathbb{E}$ taken over a batch of collected experience across which both parameters vary from state to state. The reference is the initialisation prior of Section 3.5.1, so the loss falls to zero precisely at $\alpha = \alpha_0$ and $\beta = \beta_0$, the post-softplus outputs of the initial biases in Table 3.8.

The head is therefore drawn back towards the controlled configuration established by those biases, although it is never fixed to it, since the restoring term

counteracts the PPO gradient instead of overriding it. The parameters ultimately settle at the balance between those two forces, so they track the task while the extent of their deviation stays constrained. Normalising by α_0 and β_0 before squaring keeps the penalty independent of scale, as the restoring gradient represents the relative deviation from the prior rather than its absolute magnitude. The strength of that gradient follows from the quadratic form, rising linearly with the relative distance, so a parameter far from the prior is drawn back more firmly than one already close to it. Both parameters are thereby anchored with comparable force, even though α and β differ by a factor of four in magnitude. Those references remain fixed for the duration of training, and the resistance the anchor offers is therefore undiminished by the competence the policy acquires, where a residual-weighted term would have relaxed as the predictions improved.

The scope of the anchor is deliberately restricted to those two parameters, as its extension to ν would bind the epistemic evidence to its initial value and undermine the purpose of the head, under which the epistemic uncertainty rises in unfamiliar states and falls as familiarity grows. A further safeguard governs the anchor itself, in that the regulariser coefficient λ_{reg} of Eq. (2.9) increases gradually from zero to 0.02 over the first 50 000 decisions of each stage, ensuring that the averages stabilise under PPO before the anchor takes effect.

A restoring term of that strength leaves β substantial freedom, so once training was otherwise stable a second pathology emerged, in which the sampling variance collapsed towards zero. The mechanism behind it stems from the sampling standard deviation in Eq. (3.5), a quantity derived from the loss in which β , unlike α , is floored only at 10^{-6} and can therefore approach zero. As some episodes begin to yield returns, β can be reduced profitably by the policy gradient, diminishing the sampling noise with it and shifting the policy from exploration towards exploitation. The conventional Gaussian PPO head of Section 2.1.1 is not exposed in the same way. Its log standard deviation is a free parameter, independent of the input and beyond the reach of the reward gradient, whereas the evidential head derives that quantity from β and α . The collapse is therefore a cost of the substitution, to be addressed within the evidential design itself.

The severity of that cost was measured on a run following the divergent one described earlier, carrying the anchor and the constraints adopted in response to it, in which the shrinkage in β ran its full course. Across its traces the aleatoric estimate fell from 0.40 to 0.05 and the epistemic estimate from 0.32 to 0.04, reducing the sampling variance by approximately an order of magnitude. Driving performance declined alongside that narrowing spread, the success rate falling from a peak of 0.74 to 0.52 and the mean final position error rising from approximately 3 to more than 9 metres.

The decline represents early convergence towards the suboptimal deterministic policy the entropy term of Eq. (2.4) exists to prevent.

A spread this narrow confines the vehicle to manoeuvres that have already earned rewards, so alternatives are sampled only rarely and a better approach to the bay is never found. These traces document the variance-minimisation bias of Section 2.1.3 in its actor-side form, where the evidential objective drives the sampling distribution towards a degenerate density. No prior instance of this failure has been reported, and the placements surveyed in Section 2.2.2 account for its absence. Each of them attaches the evidential head to a critic, a detected position or a velocity correction, none of which sets the spread from which an action is drawn, so a variance driven towards zero costs them nothing beyond an overconfident report.

Eliminating the drift requires three mitigations, the first being an aleatoric floor. The sampling variance in Eq. (3.5) is constrained to a minimum of $\sigma_{\min}^2 = 0.04$, ensuring that the sampling standard deviation cannot fall below 0.2 regardless of the value attained by β . The exploration noise is thereby bounded from below by construction, which the policy gradient cannot overcome however profitable it finds the shrinkage. Such a bound is the operational form of the deprecation of the variance-minimising regularisation term documented in Section 2.1.3, applied to the sampling spread rather than the objective. Its reach is nonetheless confined to the sampling trajectory, never extending to the value reaching the uncertainty consumers, which would distort the signal under study.

Holding the sampling spread above 0.2 conceals the shrinkage in β without opposing it, so a second mitigation is directed at that shrinkage, raising the anchor coefficient of Eq. (3.6) from 0.005 to 0.02. The increase is deliberately restrained, as a stronger anchor would hold α and β near the prior and flatten the epistemic channel along with them, inflating the reported uncertainty beyond what the data warrants. The value adopted is accordingly the smallest that counters the shrinkage while leaving the channel responsive.

Neither the bound nor the anchor incentivises exploration, which remains the function of the entropy term $\mathcal{H}$ of the PPO objective in Eq. (2.4). The manner in which that term is computed is the concern of the third mitigation, as no closed-form entropy exists for the tanh-squashed distribution of Eq. (3.5). The SB3 treatment of such a distribution reports no entropy at all, leaving the negative log-probabilities of the squashed samples as the fallback estimate. Saturating the squashed action raises that estimate at less cost than widening the standard deviation, so the bonus would reward saturation in place of the exploration it was intended to sustain. The head therefore overrides the fallback, supplying the differential entropy of the pre-squash Gaussian in

its place:

$$\mathcal{H}[\pi_\theta](s_t) = \sum_{i=1}^3 \frac{1}{2} \ln(2\pi e \sigma_i^2) \quad (3.7)$$

The index i runs over the steer, throttle and brake axes, and each σ_i is the square root of the clamped variance from which Eq. (3.5) draws that axis. Widening those standard deviations is now the only route to a higher bonus, so the floor and the bonus meet on the same quantity, the first holding it at 0.2 from below and the second rewarding every value above.

3.5.3 Integration

The gated observation described in Section 3.3 enters the actor-critic architecture depicted in Fig. 3.6, which carries it through a shared trunk to the evidential actor at the output, all of it trained under a modified PPO iteration.

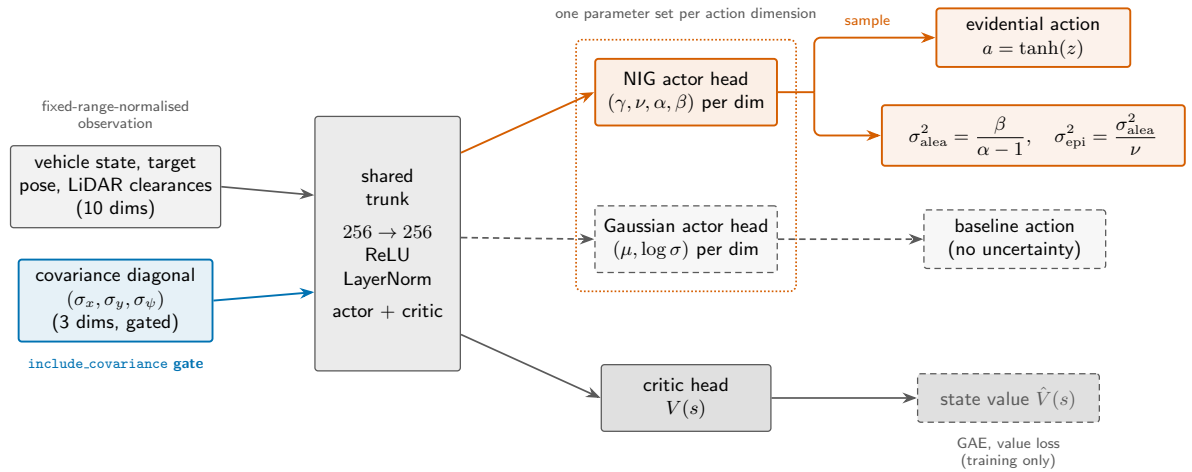


Figure 3.6 The actor-critic architecture.

The gated observation is processed by a shared trunk comprising two 256-unit layers, each applying a Rectified Linear Unit (ReLU) activation followed by layer normalisation to guard against the loss of plasticity described in Section 2.1.5.

The actor branch feeds those features into one of two interchangeable heads, selected by the arm under training. The NIG head of Section 3.5.1, drawn solid, supplies the sampled action with the aleatoric and epistemic estimates of Eq. (3.3) attached, and the standard Gaussian head, drawn dashed, emits only $(\mu, \log \sigma)$ and so leaves the action unaccompanied by any uncertainty estimate. No such swap applies to the critic branch, which stays the same across all arms and provides the scalar state-value estimate $\hat{V}(s)$ on which the GAE advantages and the value loss depend. That estimate is consumed within the update rather than issued as a command, thereby shaping the policy solely through the gradient and never reaching the vehicle.

The interchangeable pair constitutes the output-uncertainty axis of the ablation to be designed in Section 3.8. Swapping one head for the other changes nothing upstream of the trunk, leaving the input axis free to vary without an architectural confound between the two.

Training proceeds under SB3, where the head is accommodated by three replacements to the stock implementation, one for the action distribution, one for the policy class and one for the PPO algorithm itself. Those replacements confine the evidential changes to the sampling distribution (Eq. (3.5)), the entropy (Eq. (3.7)) and the anchoring regulariser (Eq. (3.6)). The remainder of the optimisation pipeline therefore retains its stock form, namely rollout collection, GAE and the clipped surrogate. That optimiser is consequently identical across all four arms, joining the architecture as a factor held constant between them.

None of those three stock components carries an uncertainty estimate out of the head, so a read-out is added for each of its two consumers. At inference the estimate serves the safety wrapper, a component Section 3.9 covers in full, reaching it through a dedicated interface that returns the action and the complete uncertainty dictionary in a single call. Every command is consequently gated on the uncertainty from the same forward pass that generated it. During training the estimate instead serves the diagnosis of the head itself, for which the aleatoric and epistemic means, the regulariser loss, and the entropy are logged to TensorBoard at every update, the record from which the divergence and collapse of Section 3.5.2 were identified.

Algorithm 1 Evidential PPO training iteration

- 1: collect n_{steps} decisions, sampling a via Eq. (3.5) from the normalised observation (Eq. (3.1))
 - 2: compute returns and advantages $\hat{A}$ with GAE
 - 3: **for** each of n_{epochs} epochs over minibatches **do**
 - 4: evaluate $(\gamma, \nu, \alpha, \beta)$, values, log probabilities and entropy
 - 5: $\mathcal{L} \leftarrow -L^{\text{CLIP}} + c_1 L^V - c_2 \mathcal{H} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$
 - 6: clip gradients and take an Adam step, halting the epochs early on the KL target
 - 7: **end for**
 - 8: advance the learning rate, entropy coefficient and λ_{reg} per the stage schedule
-

The update departs from stock PPO at four separate points, and a single training iteration is presented in Algorithm 1 to locate each of them. The floored sampling rule of Eq. (3.5) acts during collection, so the bound on exploration noise applies before any gradient is formed, whereas advantage estimation and the clipped update that follow are left unmodified, optimising the PPO objective of Eq. (2.4). The epoch loop carries the remaining two, evaluating the pre-squash entropy derived in Eq. (3.7) in place of the fallback estimate and adding the prior-anchoring regulariser introduced in Eq. (3.6) to the loss alongside it, so both safeguards against collapse act on the same update.

The strength of that anchor is set by the closing step, where λ_{reg} follows the stage schedule that raises it from zero once the averages have stabilised.

3.6 Reward Design

The advantage estimates incorporated into the PPO update of Section 3.5 are derived from a scalar reward, structured so that its components characterise the parking task. Five components contribute to it, four dense and one sparse, each set out with its magnitude and role in Table 3.9. Those five are summed at every decision, so the vehicle receives their aggregate rather than any one of them in isolation:

$$r_t = r_{\text{prog}} + r_{\text{end}} + r_{\text{clear}} + r_{\text{oob}} + r_{\text{term}} \quad (3.8)$$

Three of the dense terms shape the approach itself, in that r_{prog} rewards progress towards the bay, r_{end} rewards a clean endgame, and r_{clear} penalises encroachment on a neighbouring vehicle. The fourth, r_{oob} , charges for crossing the out-of-bounds line defined in Section 3.2.2, and that charge accumulates across the whole episode rather than clearing on re-entry. The remaining term r_{term} is the sole sparse one, the payment issued once the episode ends.

None of the five draws on the localisation covariance or the two uncertainty estimates, so the signal is an outcome-only one, scoring the parking manoeuvre alone. Any uncertainty-conditioned behaviour measured later is emergent under such a reward rather than incentivised.

3.6.1 Progress: the Bay-Frame Shaping Potential

The progress term r_{prog} is formulated from a shaping potential expressed in the frame of the target bay, taking the parked pose as its origin and the bay's depth direction as its axis. Measuring the ego pose against that axis separates the manoeuvre into three components a potential can weight independently. These are the along-track distance down the axis, the cross-track distance across it, and the symmetry-wrapped heading error between the two headings, represented by d_{along} , d_{cross} and θ_{err} respectively. Combining the three yields the potential ϕ as:

$$\phi = -(w_{\text{along}} |d_{\text{along}}| + w_{\text{cross}} |d_{\text{cross}}| + w_{\text{head}} \theta_{\text{err}}) \quad (w_{\text{along}}, w_{\text{cross}}, w_{\text{head}}) = (1, 2, 3) \quad (3.9)$$

This value reaches zero only at the perfectly parked pose and remains negative everywhere else. Its monotonicity in each coordinate means that every error reduction in depth, offset or heading raises it, so the direction of an approach governs its reward

as much as the ground it covers. An earlier design measured radial distance to the bay alone and was discarded for its indifference to direction, rewarding a fast, crooked arrival at the bay mouth as generously as a slow, aligned one.

The coefficients w_{along} , w_{cross} and w_{head} determine the extent to which each coordinate contributes to ϕ , and they cannot be compared directly because the two distances are measured in metres and the heading error in radians. The heaviest coefficient is assigned to the heading error, although at the magnitudes presented by the task a 20° error and a metre of depth incur roughly equal penalties, with a metre of lateral offset incurring double the penalty of either. Those penalties set the priority among the three corrections, so reaching the bay centreline pays most, while straightening onto the bay axis and closing the remaining depth into the bay pay comparably less. Advancing into the bay is therefore never the best-paid correction available, so the vehicle aligns outside it rather than approaching from an angle.

A ceiling nonetheless applies to w_{cross} and w_{head} , though not to w_{along} , since closing depth earns its reward immediately while the other two corrections do not. In an S-shaped correction the vehicle turns away from the bay axis before it gains any ground towards the centreline, so the heading penalty is incurred ahead of the cross-track reward. Raising either weight beyond the value in Eq. (3.9) would deepen that intervening loss until a forward-only vehicle stalls in aligned-but-offset poses.

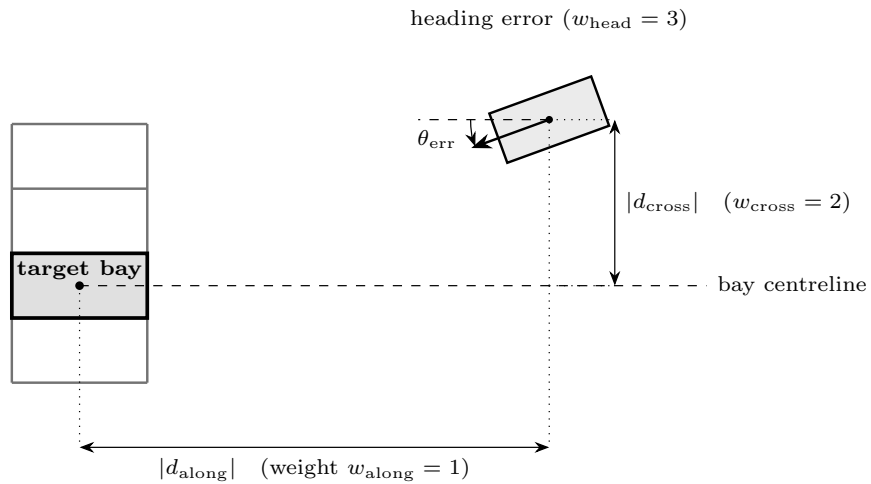


Figure 3.7 The bay-frame coordinates of the shaping potential.

The three coordinates and the axis against which they are measured are illustrated in Fig. 3.7. Their combination characterises the quality of a pose, and a difference converts that characterisation into a reward, so r_{prog} is the change in ϕ across a single decision, normalised per episode and scaled:

$$r_{\text{prog}} = \frac{\phi_t - \phi_{t-1}}{\max(|\phi_{\text{start}}|, 5)} \times R_{\text{target}} \quad R_{\text{target}} = 39 \quad (3.10)$$

This is the potential-based formulation for which optimal-policy invariance is established in Section 2.1.1. The calculation employs $\phi_t - \phi_{t-1}$ instead of discounting ϕ_t , so successive terms cancel exactly and the sum telescopes to the change from the initial pose. The cyclic accumulation identified in Section 2.1.1 is closed off as a result, since a stationary vehicle earns nothing, a loiter sums to zero, and a full, clean approach earns the multiplier irrespective of the route taken. A full approach therefore earns R_{target} against the +50 paid for success, a terminal figure given later in Table 3.9, so approaching a bay perfectly remains less valuable than parking in it.

Holding the episode total at R_{target} is the function of the denominator in Eq. (3.10). An unconstrained total would scale with the commanded distance, so a far assignment would pay more dense reward than a near one purely for having further to cover, and for the most distant targets would surpass the terminal payment itself.

3.6.2 Endgame Bonus and Clearance Nudge

The progress term is the primary driver of the reward, but it is complemented by two additional terms that influence the agent's behaviour in the final stages of an episode. The first is an endgame bonus, awarded for holding a pose near the parked target and graded by the offset, heading and depth remaining against it:

$$r_{\text{end}} = (0.008 + 0.006 \cdot s) \cdot f_{\text{line}} \cdot f_{\text{align}} \cdot f_{\text{depth}} \quad (3.11)$$

The factors f_{line} , f_{align} and f_{depth} rescale the three bay-frame coordinates onto $[0, 1]$, each reaching unity at the parked value and decaying to zero over a cutoff of its own, 2 m of offset for f_{line} , 45° of heading for f_{align} and 6 m of depth for f_{depth} . Multiplying them rather than summing means a deficiency in any one suppresses the bonus, so a vehicle that stops short and crooked collects almost nothing.

The two leading coefficients determine the rate at which the bonus accrues, a base rate of 0.008 while the vehicle is moving and a further 0.006 scaled by the slowness factor s , which reaches unity only at a held stop. Even that settled rate aggregates across an episode to less than the discounted terminal payment, so completing the park is worth more than hovering beside it.

The second of the two dense terms is the clearance nudge r_{clear} , a penalty for nearing an adjacent parked vehicle, and the distance it acts on is the closest clearance d taken across the LiDAR sectors described in Section 3.3:

$$d = \min(d_{\text{fwd}}, \min(d_{\text{left}}, d_{\text{right}}) + 0.5) \quad (3.12)$$

The side returns are discounted by 0.5 m before the minimum to ensure that the forward cone dominates, since a return directly ahead of a moving vehicle is the real

collision risk. That clearance then drives a ramp governed by a gate on the pose:

$$r_{\text{clear}} = -w_{\text{clear}} \underbrace{\min\left(\max\left(\frac{0.8 - d}{0.5}, 0\right), 1\right)}_{\text{ramp}} \underbrace{(1 - f_{\text{line}} f_{\text{align}})}_{\text{gate}} \quad (3.13)$$

The gate reuses the corridor factors from Eq. (3.11) and falls to zero as the vehicle squares onto the centreline, while the 0.8 m onset lies below the approximately 0.98 m side gap left beside an occupied neighbour. Parking next to another vehicle is therefore not penalised, and the penalty falls only on a vehicle that drifts toward a neighbour while still manoeuvring.

3.6.3 Terminal Payments and the Success Criterion

Unlike the dense terms, which are paid at each decision throughout an episode, the terminal payment r_{term} is issued once at its end, fixed for success and collision but graded by the final error when the budget expires:

$$r_{\text{term}} = \begin{cases} +50 & \text{success} \\ -25 & \text{ego-fault collision} \\ \max\left(-\frac{1.5 e_{\text{pos}} + 2.5 e_{\text{yaw}}}{\max(|\phi_{\text{start}}|, 5)} \times R_{\text{target}}, -24\right) & \text{timeout} \end{cases} \quad (3.14)$$

The timeout branch assesses the final pose, assigning the orientation error e_{yaw} a greater weight than the position error e_{pos} . A vehicle ending square with the bay but slightly out of position is therefore charged less than one ending closer but badly angled. It incorporates the normaliser from Eq. (3.10), which places the penalty on the dense progress scale and prevents a distant bay from being penalised for its distance. That scaled penalty is bounded below at -24 , one point above the -25 charged for an ego-fault collision, so ending a failing episode in a collision is never cheaper than letting the clock expire. Payment at the opposite end of that scale, the $+50$ for success, requires the vehicle to be both inside the bay and stationary. Every corner of the ego bounding box must lie within the target bay polygon, and the speed must remain below 0.1 m s^{-1} , with both holding for five consecutive decisions so that a fast drive-through cannot register as a park.

The polygon is subject to a dimensional tolerance, the margin, applied before the corners are tested against it. That tolerance is the sole quantity within the criterion that varies. A negative value expands the polygon outwards and permits a looser park, while one closer to zero contracts it and requires a tighter fit. The margin begins loosely in the early curriculum stages and is stepped down as training advances, on a

schedule Section 3.7 defines in full. It ends at the tightest setting, which is the one the evaluation in Chapter 4 applies.

Table 3.9 The additive reward of Eq. (3.8).

Term	Type	Magnitude	Role
Progress r_{prog}	dense	sums to R_{target} per episode	potential difference of Eq. (3.9), normalised so every bay offers the same dense pool
Endgame bonus r_{end}	dense	below discounted terminal	aligned, near-depth, stopped factors that settle the vehicle in
Clearance nudge r_{clear}	dense	small ramped penalty	discourages closing on a parked neighbour, off once square
Out-of-bounds r_{oob}	dense	-0.5 per decision outside	the accumulating drift-out cost of Section 3.2.2
Success	terminal	$+50$	bounding box inside the bay, held stopped
Ego-fault collision	terminal	-25	striking a vehicle
Timeout	terminal	floored at -24	graded on final pose error (Eq. (3.14)), above the ego-fault floor

3.7 Training Protocol: the Curriculum

The reward in Section 3.6 is sparse in its crucial component, as the $+50$ terminal payment is awarded only after the successful completion of an entire manoeuvre under drifting localisation. Such a payment would rarely be reached by a policy initialised from scratch on the full task across all 47 bays, so the difficulty is organised into stages.

3.7.1 The Channel-Liveness Principle

The staging is constrained throughout by the liveness of the observation channels, each of which is required to vary from the first stage rather than being introduced once the geometry has been mastered. Every channel therefore remains active from the outset, while each subsequent stage expands only the extent or severity of one difficulty axis. The support-preserving property carried over from fixed domain randomisation in Section 2.1.5 is thus combined with the gradual widening of a curriculum, so no channel remains degenerate long enough for its input weights to decay. The principle is recorded for each channel in Table 3.10, together with the stage

at which its range is subsequently opened up. Those transitions are scheduled manually rather than gated on measured performance.

Table 3.10 The channel-liveness principle.

Channel	Kept live from stage 1 by	Ramped later by
$\sigma_x, \sigma_y, \sigma_\psi$	the always-on GNSS fix-state Markov chain (mid-episode drift)	not ramped: the chain is stage-invariant
$dx, dy, d\psi$	four bays spanning all three bay orientations	more bays, reaching all 47
LiDAR clearance	parked neighbours at up to 10 % occupancy	denser occupancy, reaching 80 %

The liveness requirement follows from the shortcomings of an initial prototype for the curriculum, built on the more intuitive assumption that geometry and noise constitute separate skills to be taught in sequence. Training accordingly began with one bay in an empty lot, with the fix-state chain from Section 3.4.1 pinned to its best tier throughout a clean opening phase. Those three simplifications were intended to ease the task, but every one of them flattened a group of observation dimensions instead. Holding the chain at one tier fixes the covariance it reports, pinning those three dimensions to the floor of their range, and a single bay does the same to the target pose and an empty lot to the five clearance readings. Only two of the thirteen dimensions consequently varied, too few to compel a target-conditioned controller, so a lone trajectory was fitted and the weights on the remaining eleven decayed towards zero, leaving them unusable by the time variation reached them.

This is the dormant-neurone and plasticity-loss failure described in Section 2.1.5, under which an input that has fallen out of use is not readily re-engaged once it starts carrying information again. The sequential ordering was abandoned as a result, and the liveness requirement put in its place grades the task without silencing any channel to do so.

The same collapse arose through a second fault, left untouched by the liveness requirement and remedied separately. Once every channel was live, a revised prototype still normalised observations with the running statistics identified as a liability in Section 3.3.2, and those statistics crossed each stage boundary with the weights. A dimension held narrow during one stage consequently crossed into the next carrying the variance measured under that setting, and its values were divided by a figure too small for their new range. The bay set expands at each of the first two transitions, widening the range of the target offset more than any other dimension. Offsets drawn from a larger set of bays, divided by a variance gathered over a smaller one, fall outside the clip bounds of Eq. (3.1), so the policy received a saturated constant

on that dimension precisely as it began to carry information. The remedy adopted in Section 3.3.2 removes the learned statistics altogether, normalising against fixed physical scales so that no measurement taken within one stage is carried into another.

The principal implication of the liveness requirement is that the GNSS degradation process is not part of the curriculum at all. The single Markov chain described in Section 3.4.1 operates unchanged throughout, without gating or scaling by stage, so localisation uncertainty is present from the first training decision.

3.7.2 The Six Stages

The six stages are defined in Table 3.11. Budgets are counted in policy decisions, and the learning rate decays linearly within each stage rather than across the complete run. The occupancy column sets only the upper bound $o_{\max}$ of the per-episode draw defined in Section 3.2, with $o_{\min}$ held at zero throughout, so every density from an empty lot up to that stage’s maximum remains reachable.

Table 3.11 The six curriculum stages as committed.

Stage	Bays	Occ. max	Margin	Budget	LR	Primary axis
1	4	0.1	−0.75	1.5×10^6	$3.0 \rightarrow 0.5 \times 10^{-4}$	bootstrap manoeuvre
2	15	0.1	−0.55	1.5×10^6	$1.5 \rightarrow 0.5 \times 10^{-4}$	bay variety
3	47	0.1	−0.40	2.0×10^6	$1.5 \rightarrow 0.4 \times 10^{-4}$	bays to full
4	47	0.1	−0.25	1.5×10^6	$1.2 \rightarrow 0.4 \times 10^{-4}$	margin to strict
5	47	0.5	−0.25	2.0×10^6	$1.2 \rightarrow 0.3 \times 10^{-4}$	occupancy
6	47	0.8	−0.25	1.5×10^6	$1.0 \rightarrow 0.2 \times 10^{-4}$	occupancy to final

The order of those columns is not arbitrary, as each stage widens an axis that becomes tractable only once the skill beneath it has been consolidated in the preceding stage. The hardest skill is concentrated in stage one, involving parking at all three bay orientations with both turn directions while every other axis remains forgiving. The loosest acceptance margin, light occupancy, and the dense per-bay-normalised shaping from Section 3.6 combine to ensure that the +50 terminal payment is received often enough to bootstrap PPO from scratch. Once that manoeuvre is reliable at four bays, it is progressively generalised across the lot in stages two and three, with the bay set expanding through supersets from 4 to 15 and finally all 47 targets. The acceptance margin is tightened to the strict evaluation criterion in stage four, only after that generalisation is complete and while conditions are otherwise clean. Precision is therefore acquired before any obstruction is present in the lot. Occupancy is increased

to the operating point last, during stages five and six, since a crowded lot obstructs a manoeuvre that can by then be executed precisely at any bay.

That ordering fixes what each stage contains, while a separate choice governs when one gives way to the next. Progression follows the difficulty setting rather than the outcome achieved within a stage, the fixed pacing argued for in Section 2.1.5. The acceptance margin, bay set and occupancy ceiling therefore advance to the values in Table 3.11 once each stage's decision budget is spent, and the run is never stopped early or held back for missing a success rate. That uniformity is what keeps the stage boundaries identical across the arms, a comparison undertaken in Section 3.8.

A schedule that advances on budget alone is only as sound as the training state it hands to the next stage. The learning rate is the parameter most at risk across a boundary, since the finding in Section 2.1.1 places its value among the most influential in on-policy training while leaving its decay schedule far less consequential. Three choices therefore determine what crosses that boundary, the first being that every stage runs its own linear learning-rate and entropy curves over its designated budget, so no schedule state is inherited from the one before it. Restarting those curves still leaves the learning rate needing a starting value, which is set to roughly three times the final value of the preceding stage rather than to the from-scratch rate, nudging a nearly converged policy instead of destabilising it. Each stage's configuration is restricted to schedule parameters alone, with the learning rate, entropy and difficulty settings permitted under the allowlist of Appendix C.1 and anything that would change the network excluded from it. No stage can therefore alter the network, and the weights are loaded into an unchanged one at every resume.

Those three choices apply to every arm, and the schedules they carry differ in a single respect between the two head types. The two per-policy blocks, one for the standard-head arms and one for the evidential arms, are initialised identically except for a raised entropy floor in the evidential case and, in the later stages, a higher starting value. The asymmetry follows from where that entropy pressure lands. On the evidential head, the action standard deviation is the square root of the aleatoric estimate (Eq. (3.5)), so the pressure acts directly on that estimate. The standard head's log standard deviation, by contrast, is decoupled from the reward gradient. This single difference is the planned response to the collapse tendency described in Section 3.5.2, applied uniformly across both evidential arms so that the ablation's covariance axis remains controlled.

3.7.3 The Learnability of Waiting

The exploration those schedules protect has to reach the behaviour the uncertainty inputs exist to elicit, which is a pause under a degraded fix. No reward term pays for

that behaviour, yet four choices taken independently across this chapter combine to make it learnable. Target-pose features are derived from the EKF estimate, while success is assessed using ground truth unavailable to the policy (Section 3.3.1). The perceived target is therefore corrupted under a poor fix at the same moment that the covariance features indicate the degradation. A vehicle that commits to the manoeuvre in that state misses the bay and earns no terminal payment under the outcome-only reward of Section 3.6. The tier is nonetheless restored within seconds by the upward-biased chain in Section 3.4.1, while the wait is tolerated in the meantime by the stall counter in Section 3.2.2. Waiting is therefore reachable without ever being paid for.

3.8 Ablation Design and Hyperparameters

Uncertainty enters the agent through two pathways: the covariance observation on the input side and the evidential head on the output side. The contribution of each is then evaluated by crossing them in a 2×2 factorial ablation. The shared trunk and the stock SB3 optimiser from Section 3.5 hold learning constant, while the stages advance according to a fixed budget rather than measured performance (Section 3.7). The four arms of Table 3.12, each a complete training configuration, consequently differ from one another only in the actor head and in whether the three covariance features from Section 3.3 are admitted through the gate.

Table 3.12 The 2×2 ablation arms.

Arm	Covariance obs.	Actor head	Obs. dim
vanilla_ppo	no	standard Gaussian	10
input_uncertainty	yes	standard Gaussian	13
output_uncertainty	no	evidential NIG	10
full_method	yes	evidential NIG	13

Each axis carries two settings, the mechanism under test and the component it replaces. The evidential head is therefore compared against the standard Gaussian head, and the covariance observation against its own absence. A further setting on either axis would divide a contribution this design is meant to attribute whole, so the finer-grained arms that would apportion it are left to the future work of Section 5.2.

Every arm draws its hyperparameters from Table 3.13, which summarises the committed configuration files reproduced in full in Appendix C. The discount of 0.997 yields an effective horizon of approximately 333 decisions, around 66 seconds, long enough that the terminal payment remains visible from the spawn pose throughout the

approach. The entropy coefficient decays within each stage to a floor, which is raised for the evidential arms because the bonus is the second safeguard against variance collapse on that head, alongside the stage schedule of Section 3.7.2. Its floor is therefore a safety parameter rather than a tuning mechanism. Neither that coefficient nor the learning rate holds the tabulated value during training, as both are listed as starting points that the per-stage schedules in Table 3.11 supersede.

Table 3.13 Committed hyperparameters.

Parameter	Value	Rationale
rollout length n_{steps}	2048	~ 4.7 episodes per update at ~ 437 decisions each
batch size / epochs	256 / 5	standard PPO regime for this rollout size
discount γ	0.997	effective horizon $1/(1 - \gamma) \approx 333$ decisions
GAE λ	0.95	standard bias-variance point
clip range	0.2	PPO trust region; primary stability control
target KL	0.04	early-stops update epochs; backup to the clip
value coefficient	0.5	equal-footing value regression
gradient-norm clip	0.5	guards the shared trunk
learning rate (base)	$3 \times 10^{-4} \rightarrow 1 \times 10^{-5}$	linear decay; stage schedules override (Table 3.11)
entropy coefficient (base)	$0.02 \rightarrow 0.006$	decaying pressure with a floor; evidential floor higher (Section 3.7.2)
log-std init (standard head)	-0.6	initial exploration $\sigma \approx 0.55$, matching the evidential prior scale
λ_{reg}	$0 \rightarrow 0.02$ over 5×10^4	prior anchor of Eq. (3.6); brittle, raised cautiously (Section 3.5.2)
aleatoric floor	0.04	sampling-variance floor of Eq. (3.5); exploration guard

3.8.1 Fixed Rather Than Tuned Hyperparameters

No Hyperparameter Optimisation (HPO) was conducted for any of the experiments, the values in Table 3.13 being predetermined defaults applied identically across every arm and seed. That decision resolves a conflict between two requirements. Established practice, recorded in Section 2.1.6, is to optimise each method under comparison separately, as the values that suit one method rarely suit another [51]. In a 2×2 ablation, by contrast, the actor head and the covariance gate are isolated by holding every other variable constant. The configuration would become a fifth variable under a separate search for each arm, and any measured difference would then be partly attributable to the four searches converging unevenly. Priority is therefore given to the controlled comparison, since the contribution rests on the ablation.

That priority leaves a residual weakness that merits explicit mention, as a set selected once may suit the optimiser dynamics of one arm better than another, and no claim is made that it suits all four equally. No such preference is introduced by construction, however, since every value is derived from the task or from established PPO practice rather than from any arm’s measured performance, and the complete set was committed before the ablation began, placing it beyond the influence of the results. A search would also have consumed GPU time on the scale of a training chain, and that time was allocated to the seed matrix of Section 3.8.2 instead. An advantage arising from the suite rather than from the architecture would be expected to vary between seeds, so averaging over three reduces its influence on the reported contrasts. That reduction falls short of elimination, and the prospect that the committed values suit one arm better than another is carried forward as a limitation in Section 4.8.3.

3.8.2 Seeds and Compute

The seed matrix trains every arm as a complete six-stage chain under each of the three evaluation seeds 42, 123, and 7, giving twelve in total. Each spans ten million policy decisions and approximately 23 to 33 hours of wall clock time at the observed rate of 85 to 120 decisions per second for the stack in Table 3.1. Reproducing the full seed matrix therefore requires around two weeks of continuous GPU time. The per-stage budgets were established using the first seed alone and then reused unchanged for seeds 123 and 7, with training traces inspected at each stage and investigated only upon divergence. No per-seed tuning could consequently leak into the comparison.

3.8.3 Training Trajectories

Every chain runs under the curriculum of Section 3.7 and the shared hyperparameter suite of Table 3.13. The resulting training trajectories are plotted in Fig. 3.8, averaged across the three seeds and marked at the stage boundaries. Success and collisions are recorded there against the training distribution, not under the evaluation protocol later defined in Section 4.1.3, so the two panels show how each configuration converges rather than how well it performs.

Success separates the full method from the rest, which leads throughout and reaches approximately 50 % by the end of the final stage, against around 27 % for the input- and output-uncertainty arms and approximately 23 % for vanilla PPO. The three lower arms therefore finish within six percentage points of one another, so neither the covariance observation nor the evidential head lifts an arm far above vanilla PPO on its own. The gain appears only where the two are combined, as an interaction between the axes.

Collisions separate the arms by head instead. The rise in occupancy at stage five lifts every arm from below 10 % to a peak above 30 %, which is the intended consequence of admitting parked-vehicle obstacles rather than a loss of competence, and success recovers over stage six. The two evidential arms then settle near 20 % while vanilla PPO and the input-uncertainty arm remain around 30 %. Both arms carrying the evidential head end the curriculum in the lower band, which is consistent with the state-dependent sampling variance of Eq. (3.5) narrowing the action distribution as a crowded bay is approached, an adjustment unavailable to the standard head.

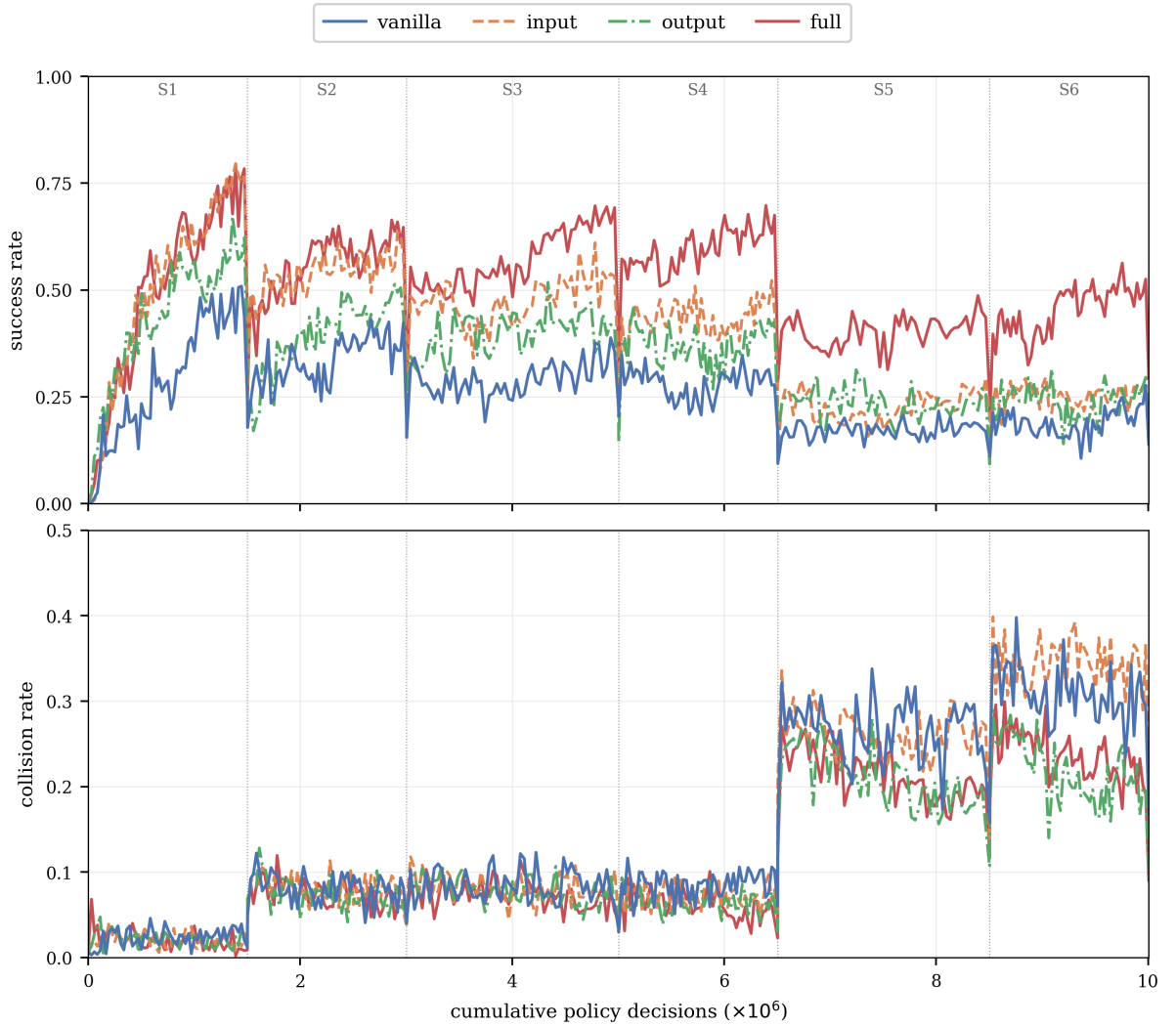


Figure 3.8 Training success rate (top) and collision rate (bottom).

3.9 Safety Wrapper and Deployment Path

A runtime safety layer is established around the environment during deployment but not during training, enabling unrestricted learning while supervision is applied only where actions reach the environment. The rule was designed to distinguish between the two uncertainty channels, stopping on a high epistemic reading as an indication of an unfamiliar state while treating a high aleatoric reading as intrinsic task noise, for which deceleration is the appropriate response rather than a handover. Separation of this kind requires the channels to operate independently, which they do not on a single-head NIG actor for the reasons provided in Section 4.3.2. Separate thresholds therefore do not differentiate between the two conditions they were intended to identify. The aleatoric and epistemic estimates defined in Eq. (3.3) nonetheless remain variances of a single predictive distribution, and they sum to the total variance given by Eq. (2.5), which therefore increases whenever either channel does. The wrapper is therefore triggered on that sum rather than on the two channels separately, which also spans a wider range than either alone and so admits a threshold positioned more precisely.

Once the total reaches 1.2, the commanded action is overridden: steering is zeroed and full braking is applied, bringing the vehicle to a stop. The handover is then flagged and the episode ends at that point, so no post-intervention behaviour is attributed to the policy. The threshold remains consistent across every stage, ensuring that all arms are subject to one identical rule.

Arbitration between a learned policy and a fallback based on a runtime signal places this wrapper within the guarantee-and-fallback family of Section 2.2.1, although the two differ in the signal that activates the fallback. The surveyed systems condition a stop on a verdict concerning the reliability of the policy, whereas the total predictive uncertainty is emitted by the actor itself and heightened by the state it encounters.

A rule based on magnitude alone provides no information about duration, so a momentary excursion above 1.2 and a sustained one both stop the vehicle. A persistence variant is consequently examined alongside the magnitude rule, requiring the total to remain above the same 1.2 for ten consecutive decisions, approximately 2 s, while any reading at or below the threshold resets the count. The delay rests on the expectation that a sustained rise signifies a genuine loss of confidence, whereas a brief one reflects noise, reserving the stop for cases that warrant a handover. Neither configuration is assumed superior, so both are replayed against the outcome reached by each episode, with Section 4.8.2 reporting what the wrapper can and cannot discriminate.

3.10 Summary

The development of the two-layer architecture outlined in Section 1.4 is complete, and three of its characteristics bear on the evidence presented in the next chapter. The first concerns provenance, as every underlying noise level is sourced from the datasheets of the specified target hardware rather than chosen for convenience. Departures from that baseline form the second characteristic, with the degradation modelled as an upward-biased GNSS fix-state Markov chain, so the uncertainty encountered by a policy arises from a defined process rather than being introduced arbitrarily. The third concerns the evidential configuration, under which a regulariser divergence and a variance collapse arise. Both are diagnosed in Section 3.5.2 alongside the fixes that resolve them, and the head does not converge on any seed until those fixes are applied.

Training and evaluation use a single code path, a unified configuration source, and consistent observation scaling. Under these controls, the two uncertainty channels remain the sole variables across the four arms of Section 3.8. The system built to those specifications is assessed in the next chapter against the objectives of Section 1.3.

4 Evaluation and Results

The framework established in Chapter 3 is evaluated in this chapter against the four objectives of Section 1.3, using the trained policies generated by the factorial ablation of Section 3.8. The groundwork is laid in Section 4.1, covering the experimental procedure, the policies compared, and the criteria for interpreting each subsequent number.

Both ends of the policy interface are examined before any task-performance result rests on them. The integrity of the EKF covariance entering the policy is evaluated in Section 4.2 by comparing the reported uncertainty with the pose error actually incurred. The estimates leaving the evidential head are then analysed in Section 4.3, which assesses what they distinguish and what they leave unpredicted.

Once the interface has been verified at both ends, task performance is detailed in Section 4.4, where the contrast central to the research question is reported. The response of each policy as GNSS quality declines is subsequently analysed in Section 4.5, and the behaviour producing the performance difference is characterised in Section 4.6. The persistence of the effect under retraining with new seeds is examined in Section 4.7, after which Section 4.8 sets out the three limitations that qualify the findings, each accompanied by its underlying mechanism. Every objective is ultimately traced to its evidence in Section 4.9.

4.1 Experimental Setup

No established benchmark is available for this evaluation to adopt, as no prior system combines closed-loop parking control with controlled localisation degradation (Section 2.3). The absence extends to the available environments, since the public parking environment closest to this task supplies neither the state estimator this work observes nor an axis along which its sensing can be degraded [80]. The protocol is therefore assembled from the practices established in reinforcement learning and in parking control, the two fields this task spans. Reporting practice is drawn from the former, following the multi-seed requirement established in Section 2.1.6, under which every headline result is pooled across three training seeds rather than taken from a single run. Measurement adopts the task success rate reported as the headline outcome by Lazzaroni et al. [69] and the final position error reported alongside it in end-to-end benchmarks [68], the two metrics conventionally used in parking and low-speed control work. Both are supplemented by a five-class outcome taxonomy and by per-step behavioural proxies, each defined in Section 4.1.4.

4.1.1 Arms and Controlled Contrasts

The four arms of Table 3.12 each serve as the controlled comparator for their adjacent counterparts on either axis, a role that no external trained system could provide.

Any two arms adjacent on a single axis differ by one factor, producing four contrasts that the chapter uses as required. Along the covariance axis, the full method against the output-uncertainty arm shows the effect of the covariance observation on the evidential head, while the same effect on the standard Gaussian head appears in the input-uncertainty arm against vanilla PPO. Along the head axis, the full method against the input-uncertainty arm isolates the evidential head where the covariance observation is available, and the output-uncertainty arm against vanilla PPO does so where it is withheld. A pair on either axis is analysed together rather than separately, since a factorial design determines whether the effect of one factor depends on the setting of the other.

4.1.2 Evaluation Conditions

Each arm is evaluated under the same four conditions, with each condition isolating a specific property of the deployment environment that the results are intended to characterise. The conditions are defined in Table 4.1 and grouped into a reference pair, a generalisation probe, and a within-episode drift case. Each varies one factor against the deployment anchor, so any performance difference can be attributed to the factor varied. Closed-loop control in simulation is not a supervised prediction task, so no held-out labelled test set can be constructed, and a fixed, seeded battery of conditions is used in its place.

Table 4.1 The four evaluation conditions.

Group	Condition	Factor varied	Purpose
Reference pair	anchor_deployment	none (occupancy 0.5, training noise process live)	in-distribution reference
	anchor_empty	occupancy $\rightarrow$ 0.0	no-obstacle control
Generalisation probe	ood_irregular_rtk_fixed	unseen irregular layout, clean GNSS	geometry as the only OOD factor
Within-episode drift	gnss_degrade_one_way	mid-episode one-way drift to degraded	ecologically valid degradation

The reference pair provides the baseline, holding conditions as close to the settings under which the policy was trained as the deployment target allows. The

anchor condition (anchor_deployment) departs from those settings only in bay occupancy, which is held at the deployment value of 0.5 rather than resampled during each episode as it is under the curriculum described in Section 3.7. Fixing it prevents a GNSS-tier effect from being confounded with how busy a particular episode’s lot happened to be. Localisation remains unchanged, so an RTK fix-state tier is drawn for each episode and remains free to drift through the Markov transition defined in Section 3.4.1. The condition is therefore reported in its own right, representing the performance that the instrumented lot would obtain, while the three remaining conditions are each evaluated against it.

The second member of the pair (anchor_empty) is identical to the anchor except that its occupancy is reduced to 0.0. No parked vehicles remain to return a LiDAR echo or present a collision hazard, leaving the filter estimate as the only signal by which a bay can be reached. Any performance difference between the two conditions consequently measures the extent to which task performance depends on the presence of parked vehicles.

The remaining two conditions carry the evaluation beyond what the anchor represents, one through the lot geometry and the other through localisation quality. The generalisation probe (ood_irregular_rtk_fixed) moves the task to the held-out lot established in Section 3.2. Localisation remains clean throughout the episode rather than following the Markov chain, leaving the unfamiliar geometry as the sole OOD factor. The geometry is therefore reported alone in Section 4.8.1, establishing the cost of an unfamiliar layout on its own terms, since degrading the fix at the same time would leave the layout and the drift inseparable.

A guaranteed transition from confident to degraded localisation is provided by the second condition (gnss_degrade_one_way). An anchor episode follows the Markov chain wherever it leads, so a degradation may or may not occur within it, whereas Section 4.8.2 later replays the handover threshold and requires a degradation that is certain. Localisation therefore starts at RTK fixed and proceeds one way down the ladder without recovery.

4.1.3 Protocol

Every arm is assessed over 200 episodes under all four conditions, and the whole battery is repeated for the three training seeds fixed in Section 3.8.2, so any condition-arm combination rests on $n = 600$ episodes. The policy operates deterministically throughout at the fixed 5 Hz decision rate adopted in Section 3.3.3, where the 0.2 s a decision spans is established as the unit in which every budget is expressed. Any quantity reported in decisions is converted on that basis, and the protocol in its entirety is reproducible using the guide in Appendix D.

The safety mechanism described in Section 3.9 remains disabled throughout every episode. An activation would replace the selected action with a full stop and truncate the run, corrupting the behavioural signal and withholding the outcome that the episode would otherwise have reached. Each episode is consequently permitted to conclude naturally, and the implemented rule is assessed in Section 4.8.2 by replaying its threshold over these records rather than intervening within them.

4.1.4 Metrics

Each numerical value in this chapter is constructed from four types of measurements, comprising a pair of primary results for each episode, one categorical label recording how the episode ended, four per-step behavioural proxies, and two grouping quantities that sort episodes into comparable bands rather than reporting an outcome.

The first of those types, the primary pair, establishes both the outcome of an episode and the margin attending it. Success is recorded categorically, awarded once the vehicle comes to rest fully inside the target bay on the criteria fixed in Section 3.6.3, evaluated at the tightest bay margin of the curriculum. A binary label of that kind records nothing of how an unsuccessful episode ended, so it is paired with the final position error. That second quantity measures the Euclidean distance in metres from the ego to the target bay centre when the episode terminates, and a failed attempt is thereby recorded according to how close it came to the target. Episodes ending tens of metres away pull the mean of this quantity several times above the median, which makes the mean the informative measure where the composition of outcomes is itself the subject, and the median where the concern is how precisely the vehicle parks. Both are therefore reported, and the caption of every results table in this chapter states which of the two it carries. Aggregating the binary outcome over a set of episodes then yields a success rate, and one such set of episodes is compared against another by the difference between their rates. Such a difference is expressed in percentage points (pp), denoting an absolute change in the proportion of episodes parked rather than a relative one. Thirty points added to a base rate of 20% therefore gives 50%, whereas a relative increase of thirty percent would raise that base to only 26%.

Neither of that pair records how an episode came to end, which is the work of the outcome label. Each episode is assigned exactly one from five mutually exclusive categories: success, collision, out-of-bounds, near-miss, and stuck. The first three are determined by the event that concludes the episode, success being the criterion defined above, collision a registered contact, and out-of-bounds the ego leaving the drivable area. The last two cover episodes where no such event occurs and the run reaches the step limit, distinguished between them by the final position error, as a near-miss ends within a specified distance of the target bay while a stuck episode ends

beyond it. The finer split matters because a bare success-or-failure label would score an attempt that ran out of time short of the bay and one that halted far from it as the same failure.

A hierarchy resolves the cases in which more than one label could apply, with the physical outcome taking precedence over any other event at the same step. To take one example, a collision rate reported in this chapter is the proportion of episodes labelled that way once the hierarchy has been applied, not simply a count of contact events.

The outcomes and labels above characterise an episode by its conclusion, whereas the conduct of the vehicle within it is characterised by four per-step behavioural proxies, on which Section 4.6 later builds its account of how the vehicle was driven. The first two record the commands issued to the vehicle, read from the vector delivered to the simulator once the rate limits of Section 3.3.3 have been applied. The brake command is the brake axis of the delivered vector, on $[0, 1]$, whereas action jerk draws on every axis of it, being the Euclidean norm of the difference between the full steering, throttle and brake vectors of successive decisions, so a large value marks a decision that departed sharply from the one before it. Jerk in this sense describes the commanded signal rather than the resulting motion, and is distinct from mechanical jerk, the rate at which the vehicle's acceleration changes.

The remaining two record the resulting motion instead of the commands producing it, and each imposes a restriction that keeps the measurement interpretable. Speed while moving is the vehicle's speed recorded only during steps in which it is actually moving, defined as speeds at or above the same 0.1 m s^{-1} threshold that the success criterion of Section 3.6.3 treats as stationary. Omitting steps below that threshold prevents the stationary hold at the end of a completed park, and any deliberate wait, from pulling the average towards zero. A similar restriction applies to the the magnitude of the EKF yaw rate, which remains unsigned so that opposing turns do not cancel one another across an episode. All four are per-step quantities averaged within a run, so every attempt contributes one value per proxy, and together the four describe how the vehicle behaved during that attempt.

Every measurement to this point describes a single episode, whereas the last two, realised difficulty and reported uncertainty, instead group episodes for comparison and are never reported as outcomes. Each condenses an episode into a single value used to sort it into a band, the term this chapter assigns to a group of complete episodes. Realised difficulty is the mean per-step distance between the EKF position estimate and the simulator's ground-truth pose across the episode. It is a ground-truth measure that groups episodes according to how difficult the localisation problem actually was, and it is on that basis that Section 4.5 later bands the episodes. Reported uncertainty is instead derived from the filter's own output, as the

episode-mean reported position standard deviation. It therefore groups episodes by the signal available at the policy's input at the time, which is the grouping the behavioural comparison further on relies on (Section 4.6). The two cannot substitute for one another, since the reported posterior inflates only at the bounded rate its process noise allows and so understates the true error, as established in Section 3.4. Both are divided into bands of equal population, so that the same number of episodes supports every value reported from a band.

A separate term is reserved for the per-step groupings used in Section 4.2, which are called bins rather than bands, since they sort individual decision steps rather than episodes and characterise the EKF alone rather than any policy.

4.1.5 Reporting Statistics

The metrics defined above convey the claims of this chapter through three statistics, each addressing a distinct question about the numbers they produce. An interval estimate indicates how precisely a contrast is known, a rank correlation measures how strongly two quantities track one another, and the area under the Receiver Operating Characteristic (ROC) curve assesses how effectively a scalar signal separates two outcomes.

The pooled episodes of an arm are a finite sample of its performance. A success rate calculated from them therefore approximates the rate the arm would attain over every episode it could run, rather than establishing it exactly. Quantifying that variability by running further episode pools is prohibitively expensive, so the existing episodes are instead resampled with replacement and the contrast recomputed for each resample. The spread of those recomputed values indicates how far the estimate could shift if the evaluation were repeated, while the 2.5th and 97.5th percentiles of their distribution provide the stated Confidence Interval (CI) on the small-sample grounds established in Section 2.1.6. The interval is read by its width, with a narrow one signifying a precise estimate and a wide one an imprecise estimate, while an interval containing zero signifies an effect that cannot be distinguished from sampling variability.

An interval quantifies a single contrast, while the association between two quantities requires a different statistic, called upon twice further on, where Section 4.2 tests the covariance against the true pose error and Section 4.6 grades the driving against the covariance. The Spearman rank correlation [81] fulfils this role by replacing each metric with its rank within its own sample and measuring the degree of agreement between the two rank sets. A value of $+1$ signifies a quantity that consistently rises with the other, 0 indicates no reliable direction, and -1 represents one that falls as the other rises. The coefficient therefore establishes the dependability of a direction rather

than the magnitude of an effect. Ranking discards the measured values once the order is fixed, so the coefficient is settled by that order alone. Neither a handful of unusually large values nor a difference in the units of the two quantities can therefore influence the result, which is the property for which Spearman proposed the measure.

The correlations above relate two measured quantities to one another, whereas a third statistic sets a scalar signal against a binary outcome and measures how far the two classes are distinguished by its value. Separation of this kind is what Section 4.3.1 later measures, scoring the failure-prediction gate on how far its signal ranks failing episodes above successful ones. Separation is quantified by the Area Under the Curve (AUC), the probability that a randomly selected member of the first class receives a higher signal value than a randomly selected member of the second [82]. The measurement rests on a threshold above which the signal counts as a detection, and each setting of that threshold captures some proportion of the first class while incorrectly capturing some proportion of the second. Plotting those two proportions against one another across the full range of settings forms the ROC curve, of which the AUC is the area. A value of 1.0 therefore signifies complete separation, 0.5 a signal carrying no information about the outcome, and anything below 0.5 a signal ordering the two classes in the opposite direction.

4.2 The Precondition: Covariance Honesty

Each claim made in this chapter assumes that the standard deviations of the EKF provide a meaningful conditioning signal, such that higher reported uncertainty predicts greater true error. A policy conditioned on a signal that fails this test would instead be responding to noise, rendering any resulting behavioural outcome uninterpretable. This prerequisite is therefore established before any policy comparison. To determine this, the standard deviation reported at each decision step is paired with the true error of the estimate it describes, and the covariance ranks that error to the extent that the two rise together. The EKF implemented in Section 3.4 is identical across the four arms, so what is measured is the localisation pipeline rather than any policy. A single arm is therefore sufficient to characterise it, and every decision step analysed in this section is drawn from the vanilla PPO runs.

The relationship is illustrated in Fig. 4.1 and quantified by a rank correlation and a binned mean error. Assessing the covariance requires completed parking episodes, a criterion the unseen irregular layout does not meet, since no arm records a single successful park (Section 4.8.1). Its exclusion leaves three conditions, pooled into the single population from which every figure reported in this section is drawn. The two panels cover position and heading and use the same series of decision steps, plotted at

two granularities. The blue dots are the individual measurements, one per step, each positioned by the standard deviation reported by the filter against the true error incurred at that step. The joined red markers aggregate those same points, one per bin. To construct them the steps are ordered by reported standard deviation, divided into five equal-count bins, and the mean error within each is calculated. A marker therefore lies at the mean absolute error of its whole bin.

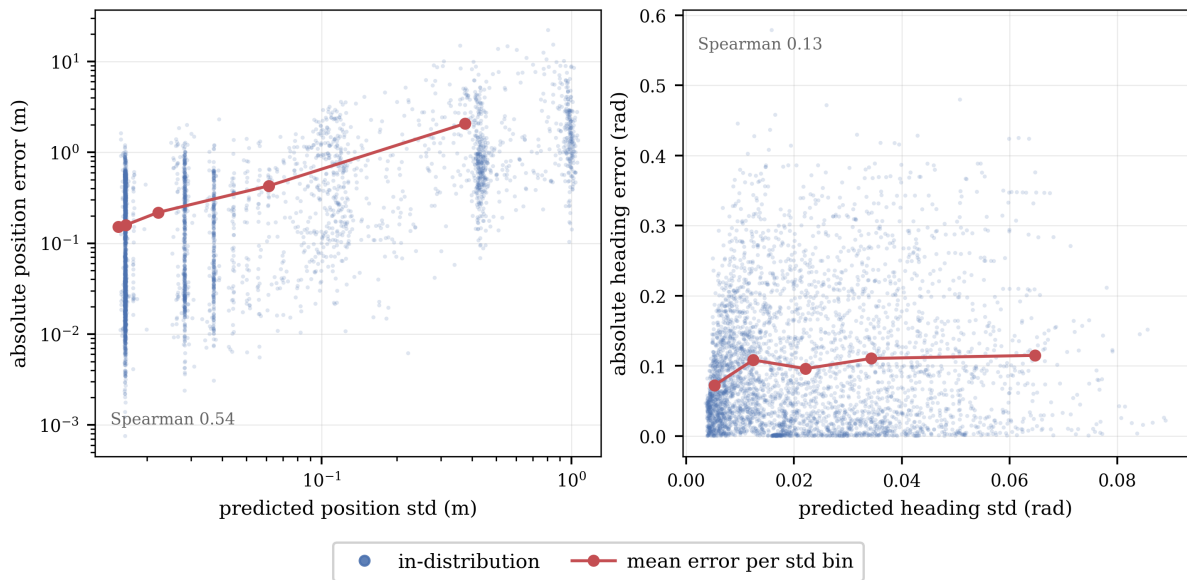


Figure 4.1 Reported EKF standard deviation against true error, pooled over the three completed-episode conditions.

Interpretation of either panel depends on its scaling, which differs between the two. Both axes of the position panel use a logarithmic scale, since around 48% of the steps cluster within a few centimetres of the RTK-fixed floor of the reported standard deviation, while the small minority of steps taken under a degraded fix report values beyond 1 m. Almost two decades separate the two extremes, and a linear axis would compress that right-skewed spread into a single column at the floor. The heading panel exhibits no such crowding because its reported standard deviation spans less than one decade, so both axes remain linear. In the position panel the binned markers climb from left to right, and the correlation coefficient and bin means measure how steep the climb is. The heading markers instead rise over the sweep as a whole but not at every step, an irregularity that the weaker correlation for that channel reflects.

Quantifying the rise in the position channel begins with the ordering shared by the reported position standard deviation and the true position error. Their Spearman rank correlation (Section 4.1.5) is 0.54 over that population, against a value of one for a perfect ordering. Much of the distance from that ceiling is attributable to ties in the data, since a large share of the steps run on a stable RTK fix, for which the filter reports nearly the same value every time. A ranking cannot separate steps that share a value,

so those pairs contribute no agreement to the coefficient despite the estimates being accurate to within a few centimetres. Recomputed above that floor, where the reported figure genuinely moves, the coefficient rises to 0.63, which places much of the shortfall with the ties. The agreement between reported uncertainty and true error is therefore one of population ordering rather than of the individual step, which is the level at which a conditioning signal is consumed. One ordering held across the pooled steps can be produced by conditions that differ in both reported uncertainty and true error, without the two rising together inside any single condition, so the coefficient is recomputed separately for each. The correlation stays positive throughout, at 0.41 for the one-way drift, 0.42 for the obstacle-free control and 0.63 for the deployment anchor, though two of the three fall below the pooled 0.54, which places part of that figure with the contrast between conditions rather than the ordering within one.

The bins address a question the coefficient leaves open, since a rank correlation gives the direction of the relationship without showing whether the error grows enough to justify slowing the vehicle. The mean absolute position error rises at every step of the sweep across five equal-count uncertainty bins, taking the values 0.15, 0.16, 0.22, 0.43 and 2.07 m. The bin containing the steps with the largest reported standard deviation therefore averages roughly thirteen times the error of the bin containing those with the smallest, a gap wide enough to warrant action rather than one detectable only through a statistic.

Two limits govern how far that climb can be read. The first concerns how the bins were formed, as holding equal numbers of steps rather than equal ranges of reported uncertainty makes their widths differ by orders of magnitude. The two lowest bins sit at the RTK-fixed floor of the reported standard deviation because part of every condition runs on a stable solution. A large mass of steps therefore accumulates at a single floor value, which equal-count binning must split. The second bin is narrower than 0.0001 m, whereas the widest admits 0.98 m. The second limit concerns magnitude, as the bounded inflation rate of the posterior caps how high the reported figure can climb within a single dwell. The consequence is the understatement of the true error identified in Section 3.4.2, where the tuned inflation rate is shown to indicate the fix state without tracking the error incurred under it.

The heading panel permits no such interpretation, since that channel does not rank its error as the position channel does. Its reported standard deviation has a pooled Spearman correlation of 0.13 with the true heading error. The estimate itself remains sound, with a median heading error of 0.07 rad and a 99th percentile of 0.38 rad. Yaw is propagated from the IMU yaw rate at 20 Hz and corrected by the gated COG estimate, as specified for the pipeline in Section 3.4. Even as the GNSS tier degrades and position error grows to metre scale, this dead reckoning keeps the heading corrected independently of GNSS, so its error remains small. The 99th

percentile of heading error is roughly five times its median, compared with a factor of 49 for position. High position error therefore occurs without a corresponding rise in heading error, which remains uniformly small from step to step. A rank correlation needs variation in the quantity being ranked, so the low coefficient reflects that absence rather than a failure of the filter. The weak heading correlation nonetheless leaves every result in this chapter intact, since the response graded in Section 4.6.1 is measured against the position standard deviation and no claim conditions on the heading figure. A policy responding to the position standard deviation is therefore responding to a signal that ranks its own error, and the behavioural contrasts that follow can be read as responses to uncertainty rather than to noise.

4.3 The Evidential Actor's Uncertainty Channels

The evidential head, presented in Section 3.5, reports a second uncertainty, quantifying the policy's confidence in its own actions, which warrants examination now that the covariance it receives as an observation has been shown to rank the error it describes. Under the NIG parameterisation a single forward pass should produce both per-action epistemic and aleatoric uncertainty (Table 2.1), and their sum is the total predictive uncertainty defined in Section 3.9. The total is evaluated against each episode's outcome to determine whether it carries any warning, and the evidence gathered along the way also bears on whether what remains is simply a heteroscedastic Gaussian.

4.3.1 Failure Prediction from the Head's Reported Uncertainty

The claim under test is that the uncertainty reported by the head warns of failure, which holds only if failing episodes can be separated from successful ones by that signal. The head's total predictive uncertainty is therefore the quantity on trial, formed by summing the epistemic and aleatoric estimates of Eq. (3.3) at each decision step. Scored beside it is the EKF standard deviation for the two position axes, averaged into a single planar figure. That covariance enters as a reference measure rather than a candidate replacement, since it reports on the state estimate alone and stays silent on the collisions and stalls the manoeuvre itself produces, which remain reachable under an accurate fix (Section 3.2.1). Only a warning attached to the chosen action is answerable for both, and each signal is reduced to the maximum value it reaches over an episode.

The scores come from episodes run under the protocol of Section 4.1.3, in which no threshold is active and every run concludes naturally. A definitive outcome label therefore attaches to each episode, and the four outcome categories short of success defined in Section 4.1.4 are collapsed into a single failure class. Separating

failures from successes requires both to be present, which restricts the analysis to the three conditions of Section 4.1.2 where the GNSS tier varies within the episode. A candidate threshold is swept over the scores, and each setting of it yields the two outcomes a live gate would produce, a catch on a failing episode the rule would have stopped and a needless abort on a successful park it would have halted. Plotting one rate against the other gives Fig. 4.2, a curve per arm and signal, and each curve is summarised by its AUC (Section 4.1.5) so the reading rests on the whole sweep rather than one operating point [82].

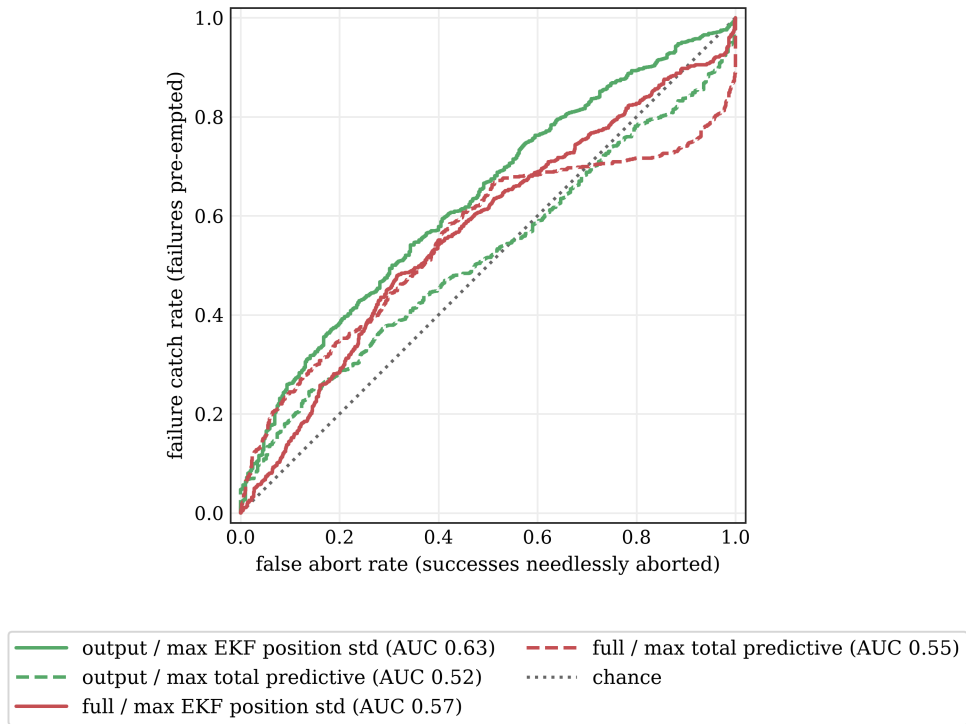


Figure 4.2 Pooled failure-prediction ROC curves over the three varying-GNSS conditions.

Aggregated across the three seeds, the gate on the head's total predictive uncertainty is close to random on both evidential arms, yielding an AUC of 0.55 for the full method and 0.52 for the covariance-blind evidential arm. The control gate on the EKF position standard deviation scores the same episodes higher on both arms, at 0.57 for the full method and 0.63 for the covariance-blind evidential arm, the highest value recorded. All four values exceed the 0.5 baseline, establishing that a per-step signal retains some discriminative power over these episodes and that the prediction task is therefore attainable in principle. None exceeds it by a margin sufficient to separate the two outcomes, however, so the uncertainty the head reports on its own actions affords no warning of failure and the claim under test is unsupported.

A per-condition analysis of the same scores sharpens that reading. In the obstacle-free control, the maximum score of the full method distinguishes between unsuccessful and successful episodes with an AUC of 0.88, compared to 0.44 in both

obstacle-bearing conditions. Manoeuvring near stationary cars raises the head's reported uncertainty in successful and unsuccessful episodes alike, so the episode peak levels off against that ceiling whenever obstacles are present. The pooled result therefore indicates saturation rather than an absent signal.

4.3.2 The Coupling of the Two Channels

The evidence parameter ν accounts for the near-chance gate AUC the head's total predictive uncertainty records, failing for the same reason as the divergence and the collapse addressed in Section 3.5.2, all three tracing to the absence of a fixed regression target under policy sampling. The epistemic estimate is the aleatoric one divided by ν , per Eq. (3.3), so the two channels separate only when ν varies across states, and nothing supervises ν towards doing so.

Separation therefore demands a state-dependent ν , as epistemic uncertainty is the channel expected in Section 2.1.2 to rise with novelty and fall with familiarity. The head was constructed to accommodate precisely that behaviour. The initialisation protocol of Section 3.5.1 establishes ν below unity, the regime in which the epistemic estimate exceeds the aleatoric one, while the anchor of Eq. (3.6) is imposed on the remaining parameters alone, leaving ν unconstrained as training proceeds. The parameter nonetheless converges to a state-invariant value under evaluation, as an unconstrained parameter acquires state dependence only under a gradient that rewards it, and the actor-side objective provides none.

Such a gradient arises under the two placements surveyed in Section 2.2.2, supervised regression against a labelled target and value-side RL against the TD error, and neither is available to a head positioned on the actor, where a sampled action does not serve as its own label and no action-value is regressed from which a TD error could be formed. Either would resolve the collapse at the cost of abandoning the actor-side placement set out for examination in the research gap of Section 2.3, and both are returned to as future work in Section 5.2. The deficiency is therefore one of placement and not of evidential methods in general, though it leaves a single signal reported under two names as the most the present head can offer, which meets none of the four desiderata for uncertainty in a reinforcement-learning agent established in Section 2.1.3, since each is defined over an epistemic estimate that moves independently of the aleatoric one.

4.3.3 The Head Against a Heteroscedastic Baseline

The collapse of the epistemic channel leaves the evidential head close to a heteroscedastic Gaussian head, emitting a per-state variance over the action

distribution. The conventional SB3 PPO head of Section 2.1.1 structurally lacks that property, so the operating mechanism of the evidential head is best described as heteroscedastic. The two differ in what each can represent out of distribution: the NIG head has an evidence mechanism under which uncertainty can rise in new states, whereas the Gaussian standard deviation is a reward- and entropy-adjusted exploration parameter. The distinction remains a property of the method rather than a measured outcome, since no arm in the factorial isolates the NIG parameterisation from state-conditioned variance. Holding the standard head fixed is what isolates the contribution of the replacement (Section 2.1.1), so the control needed to separate the two mechanisms falls outside the factorial and is proposed as future work in Section 5.2. State-conditioned variance nonetheless survives the collapse, and the covariance contrast of Section 4.4 rests on it, so what the collapse costs is confined to the independent failure warning tested in Section 4.3.1.

4.4 Task Performance Across Conditions

Both uncertainty channels are now defined: the covariance reported by the filter ranks the error of its own estimate, while the evidential head emits a per-state variance over the action distribution. The question examined below is what their combination is worth on the task itself, and the pooled results give the first view of it. Figure 4.3 shows the pooled success rate and mean final position error for the four arms under the four conditions, with Table 4.2 giving the success rates numerically and Table 4.4 the mean error alongside the median. The full method achieves a success rate ranging from 45.8% to 64.7% across the three trainable conditions, while the other three arms yield results between 19.0% and 33.3%. The unseen irregular layout is the exception: every arm registers a 0% success rate on that layout, a finding examined among the limitations in Section 4.8.

Table 4.2 Pooled success rate by condition and arm ($n = 600$ episodes per arm and condition).

Condition	Vanilla	Input	Output	Full
anchor_deployment	23.8	25.2	19.0	45.8
anchor_empty	33.3	30.7	28.5	64.7
gnss_degrade_one_way	27.3	28.5	20.0	52.5
ood_irregular	0.0	0.0	0.0	0.0

A margin read off the pooled figures compares the full method against the remaining three at once, so it attributes the gain to the pairing rather than to the covariance input or the evidential head separately. Apportioning it between the two

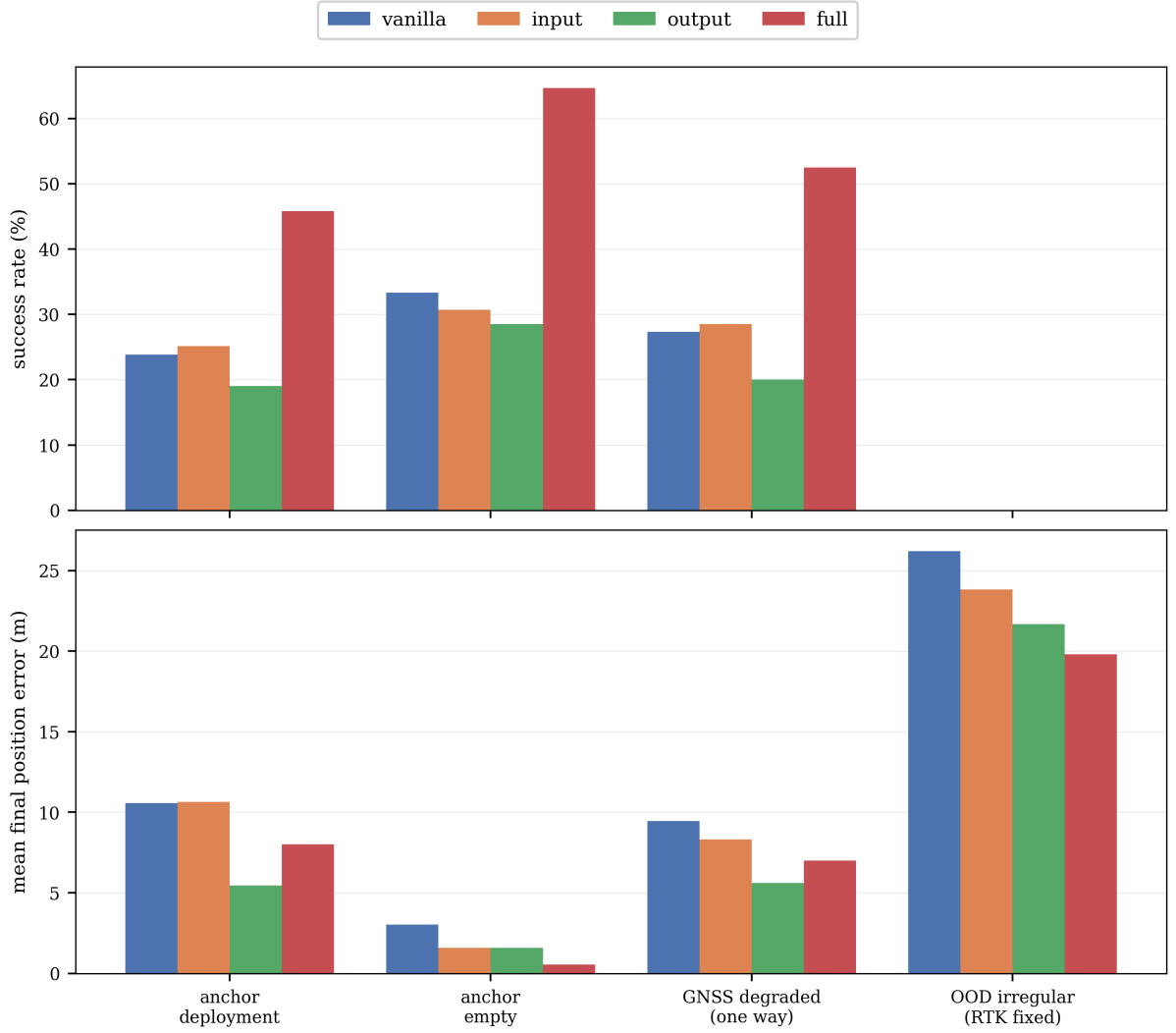


Figure 4.3 Pooled task performance ($n = 600$ episodes per bar).

requires the single-factor contrasts for which the factorial was constructed. The first of these, reported in Section 4.4.1, measures the effect of supplying the covariance input to the evidential head, and the second repeats that comparison on a standard Gaussian head in Section 4.4.2, thereby establishing how far the gain depends on the architecture receiving the input.

4.4.1 The Covariance Contrast

The contrast examined below sets the evidential head trained with the covariance against the same head trained without it, under an identical curriculum, reward and set of seeds. Any difference in outcome is thereby attributable to that input alone. The contrasts are reported with a bootstrap CI, and an interval clear of zero indicates an effect sustained across the resampled episodes.

The covariance raises parking success by 26.8 pp at the deployment anchor (95% CI [21.7, 32.0]), and by 26.8 to 36.2 pp across all trainable conditions. Those figures are drawn from Table 4.3, which reports every contrast under the success rate and the final position error defined in Section 4.1.4, listing the evidential head on the left and the standard Gaussian one on the right. Exposing the localisation covariance is therefore the sole modification that more than doubles the success rate of the evidential arm across all trainable conditions, which constitutes the primary quantitative finding of the evaluation.

The magnitude of the controlled effect is best assessed beside the end-to-end comparison. Vanilla PPO and the full method differ in both the covariance input and the actor head, so their success-rate gap confounds the two changes. That gap spans 22.0 to 31.3 pp across the same three trainable conditions. Two beneficial changes acting together would be expected to outperform the covariance alone, yet this range falls below the single-factor contrast at both ends. The output-uncertainty arm therefore performs worse than vanilla PPO, with the evidential head in isolation proving a liability rather than a benefit. The covariance is the sole indication available to that head of a deteriorating position estimate, and in its absence the vehicle is uniformly cautious rather than cautious in proportion to the circumstances, an account elaborated in Section 4.6.1. The same covariance input supplied to a standard Gaussian head yields no benefit either, its three success-rate contrasts on the right of Table 4.3 spanning -2.7 to $+1.3$ pp. The variance of that head is a single learned parameter that the observation never reaches, under the standard PPO head described in Section 2.1.1. The outcome therefore applies to the control specifically rather than to Gaussian policies in general, and a Gaussian head conditioning that variance on the observation might put the covariance to use, a possibility weighed against the evidential head in Section 4.3.3.

Table 4.3 Covariance contrasts in success rate (treatment minus control) with 95% bootstrap confidence intervals, pooled over $n = 600$ episodes per arm and condition.

Condition	Evidential head		Standard head	
	$\Delta\text{success (pp)}$	95% CI	$\Delta\text{success (pp)}$	95% CI
anchor_deployment	+26.8	[21.7, 32.0]	+1.3	[−3.5, 6.2]
anchor_empty	+36.2	[30.8, 41.3]	−2.7	[−8.0, 2.7]
gnss_degrade_one_way	+32.5	[27.3, 37.7]	+1.2	[−3.8, 6.3]
ood_irregular	0.0	[0.0, 0.0]	0.0	[0.0, 0.0]

An arm parking more often should finish closer to the bay on average, yet the mean and median columns of Table 4.4 disagree on that point at the deployment anchor and under the one-way drift alike. At the anchor the full method records a

higher mean than the output-uncertainty arm, 8.02 m against 5.45 m, yet a lower median, 0.76 m against 1.23 m. The parked vehicles obstruct the approach, so a failed attempt halts metres from the bay rather than beside it, and an arm that parks more often is left with failures further out that dominate its mean. The reversal is confined to that statistic, since the full method holds the lowest median under all three trainable conditions, the ordering that a higher success rate would predict.

Table 4.4 Pooled final position error over all episodes, successful and failed alike ($n = 600$ episodes per arm and condition).

Condition	Mean (m)				Median (m)			
	van.	in.	out.	full	van.	in.	out.	full
anchor_deployment	10.58	10.65	5.45	8.02	2.83	2.22	1.23	0.76
anchor_empty	3.02	1.58	1.59	0.54	0.99	0.80	0.93	0.41
gnss_degrade_one_way	9.45	8.33	5.61	6.99	1.70	1.46	1.18	0.64
ood_irregular	26.22	23.84	21.69	19.82	20.49	19.62	19.31	19.60

4.4.2 The Interaction: The Covariance Depends on the Evidential Head

The standard-head columns of Table 4.3 show the same covariance input providing no benefit once the evidential head is removed. Its three success-rate contrasts under the trainable conditions span -2.7 to $+1.3$ pp, none on an interval that clears zero, so the covariance dimensions reach the mean of a state-independent Gaussian and go no further. The evidential head instead makes the action variance a function of the input, and the same three dimensions produce the gain recorded alongside it.

A claim resting on the two sets of columns together is stronger than one resting on either. The standard-head pair provides no evidence for a covariance observation beneficial in isolation, the effect demonstrated being instead that of an input which helps where the architecture can represent uncertainty-conditioned behaviour. The central claim of Chapter 1 thereby rests on a measured interaction rather than a presumption.

4.5 Graceful Degradation

Graceful degradation denotes the vehicle operating more conservatively as localisation uncertainty rises over the course of a run. Observing such behaviour requires an episode in which the fix state deteriorates mid-manoeuve, which the live Markov chain of Section 3.4.1 supplies. The transitions of that chain determine how long an episode remains poorly located, so the realised difficulty, the first of the two measures

set out in Section 4.1.4, is a property of the episode itself rather than of the condition from which it was drawn.

The 2400 anchor episodes, 600 for each arm, are arranged according to that realised difficulty. The sequence is then divided into thirds to form the three bands of Fig. 4.4. Grouping by true error rather than by the filter’s own account of it establishes the difficulty of each band independently of the estimator, which matters for an outcome comparison because the reported posterior systematically understates the true error, for the reason established in Section 3.4. A division into equal thirds determines the population of each band but not the range of position error it spans, which the median error establishes. It stands at 0.16 m in the low band, rises to 0.39 m in the mid band and reaches 1.38 m in the high band, where the most severe episodes extend to 5.4 m. Approximately 200 episodes per arm fall in each, and the rate plotted is the proportion of those episodes that end in a successful park.

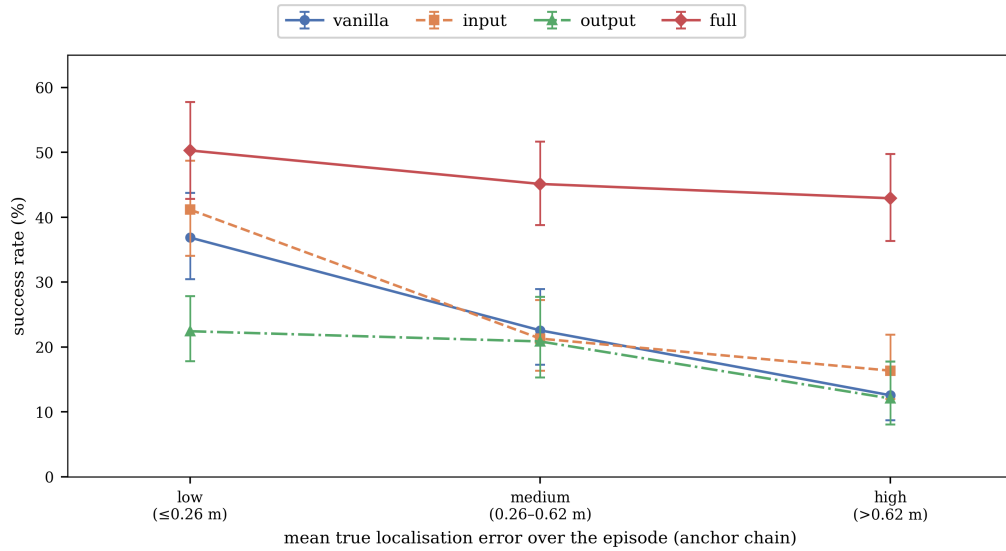


Figure 4.4 Pooled success rate under the live anchor GNSS chain (~ 800 episodes per band).

Across that range, the full method’s success rate remains almost level, falling by 7.4 pp from 50.3% to 42.9%. The other three arms instead lose roughly half their success or more over the same bands: vanilla PPO falls from 36.9% to 12.6%, the covariance-fed standard head from 41.2% to 16.4%, and the covariance-blind evidential arm from 22.4% to 12.1%. The full method’s CI therefore clears the CI of every other arm in the mid and high bands, while in the low band it still overlaps the CI of vanilla PPO and of the covariance-fed standard head.

The full method alone holds its performance as localisation error increases. The two single-factor arms, one carrying the covariance without the evidential head and the other the head without the covariance, both regress to the level of vanilla PPO in the high band. Robustness to a deteriorating estimate is thus a property of the pairing,

which is the same interaction Section 4.4.2 reports on the pooled outcomes. The mechanism behind that robustness remains open, as an outcome records whether an episode ended in a park and nothing of the conduct that led there. The driving itself is accordingly examined decision by decision in Section 4.6.

4.6 The Timing and Effort of the Response

An arm that conditions on localisation uncertainty should produce more conservative driving when the filter reports a poor estimate. The episodes are therefore grouped by reported uncertainty, the second of the two measures set out in Section 4.1.4. The covariance is the sole indication of localisation quality available to the policy, establishing it as the appropriate benchmark against which a response is assessed. That response is quantified through the four per-step behavioural proxies defined alongside the measure.

A condition can support such a comparison only if its reported uncertainty varies, since a covariance that remains constant provides no basis against which behaviour can be graded. The irregular layout of Section 4.8.1 does not meet that requirement, because it holds RTK fixed throughout the episode, and is therefore excluded. The three conditions in which the GNSS tier changes mid-episode are consequently pooled, giving $n = 1800$ episodes per arm.

The magnitude of the behavioural response is read from Fig. 4.5, where each of the 1800 episodes of an arm is summarised by five averages taken across its steps. One of them is the standard deviation the filter reports, and the other four are its behavioural proxies, brake command, speed while moving, yaw rate and action jerk, as discussed in Section 4.1.4. Those episodes are arranged by their reported uncertainty and divided into five bands of equal population, the division being made within each condition separately to keep the mix of conditions constant from one band to the next. Dividing the pooled 1800 instead would concentrate the obstacle-free episodes in the highest band, where the absence of parked vehicles elevates the mean speed on its own, as the band figures reported later in Section 4.6.1 show. Every proxy is then averaged a second time over the episodes each band contains, and those values are the markers plotted, weighting every attempt equally regardless of its duration. An average of this kind is raised as readily by a proxy that moves moderately in every episode of a band as by one that moves sharply in only a few, so the consistency of the response is established separately in Table 4.5. Its upper block correlates each proxy with the reported uncertainty through the rank statistic discussed in Section 4.1.5, yielding one value per proxy and arm, whereas its lower block averages the operating quantities those correlations leave unrecorded.

Pairing the two blocks matters because a coefficient conveys no information about the magnitude of the response. An arm may grade its driving by the reported uncertainty more sharply than a covariance-blind control and gain nothing in success for it, a case examined in Section 4.6.2. The analysis therefore begins by comparing the full method with the output-uncertainty arm, since that pair differs only in the covariance input and thus isolates its effect on the proposed architecture.

4.6.1 The Effort and Consistency of the Response

A vehicle driven more conservatively as uncertainty rises should brake harder and manoeuvre more gently, which across the bands means an increase in the brake command and a decrease in the speed, the yaw rate and the action jerk. Those trends are read against bands plotted at their midpoints on a logarithmic axis, since the reported standard deviation carries the skew established in Section 4.2.

Every one of those expectations is met by the full method. Its mean brake command, delivered on $[0, 1]$ by the action interface of Section 3.3.3, increases by a factor of 2.7 across the sweep, from 0.125 in the lowest band to 0.333 in the highest, rising at every band. The mean speed in motion falls over the same range from 3.06 m s^{-1} to 2.63 m s^{-1} , so the vehicle is both slowing and braking as the estimate deteriorates. Its manoeuvring softens alongside that response, as the mean absolute yaw rate more than halves, from 0.186 rad s^{-1} to 0.091 rad s^{-1} , ending below every other arm, while the mean action jerk falls from 0.297 to 0.227. The vehicle consequently approaches the bay more slowly and turns half as much once its own position can no longer be pinpointed.

The output-uncertainty arm shows almost none of that response over the same sweep. Three of its four proxies move in the conservative direction: the brake command rises across the bands from 0.054 to 0.166, the yaw rate declines from 0.134 rad s^{-1} to 0.103 rad s^{-1} , and the jerk falls from 0.246 to 0.214. Speed alone shows no such ordering, with its means confined between 2.30 m s^{-1} and 2.57 m s^{-1} and oscillating between bands rather than declining as the filter deteriorates.

Braking, yaw rate and jerk therefore follow the trend expected of a conservative policy, yet none of the three reaches the magnitude at which the full method operates. The yaw rate and jerk of the output-uncertainty arm begin at the lowest levels recorded for any arm and move very little across the sweep, with the widest gap appearing in the yaw rate. Across the lowest and highest bands, the full method's yaw rate falls by 0.095 rad s^{-1} , whereas that of the output-uncertainty arm falls by 0.031 rad s^{-1} . The full method therefore traverses three times the range from a starting level 1.4 times as high and finishes below the point at which the output-uncertainty arm even began.

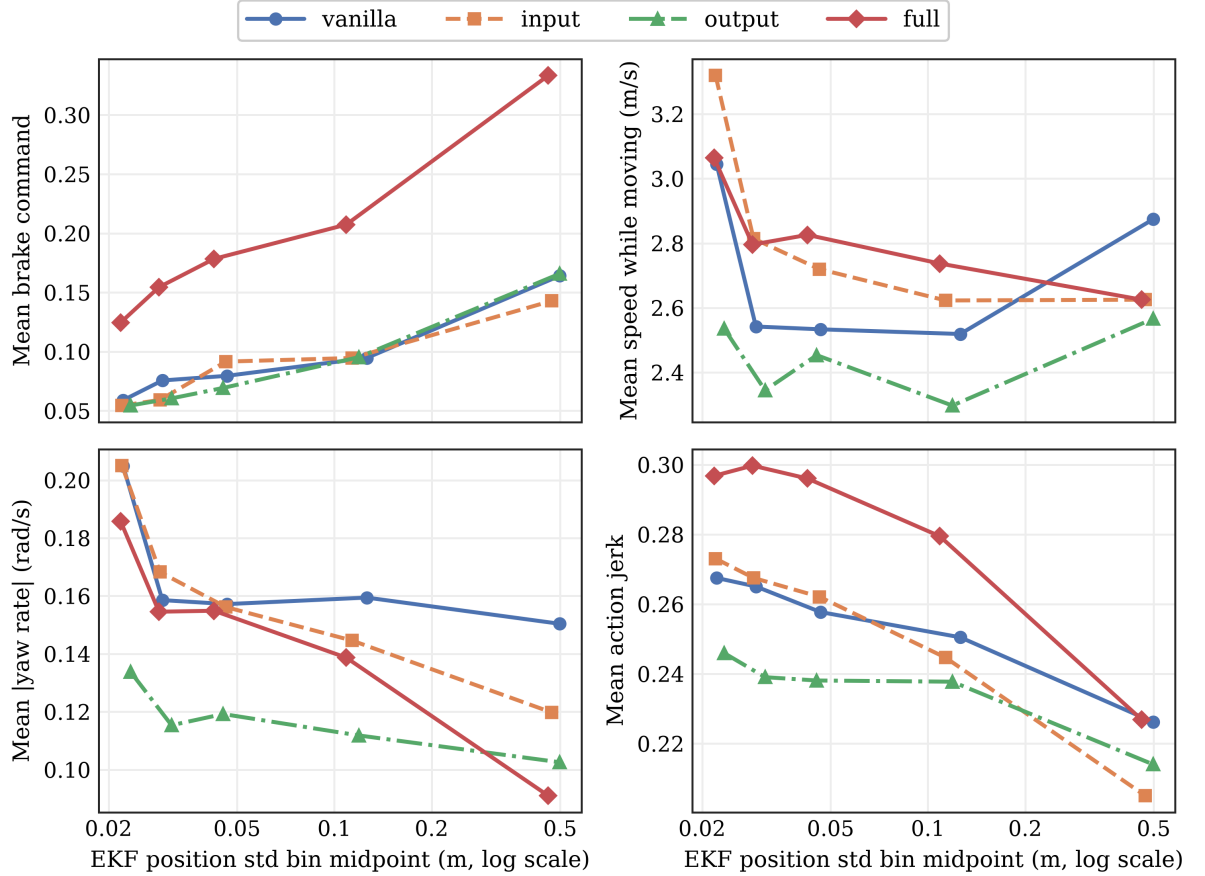


Figure 4.5 Per-episode behavioural proxies, pooled over the three conditions in which the GNSS tier changes mid-episode.

The weakness of this response is a consequence of the ablation itself. Covariance is withheld from the output-uncertainty arm leaving high-uncertainty and clean episodes represented by the same set of ten features, so nothing distinguishes them at the point where a slower approach must be selected. Unable to anticipate which episodes warrant caution, the arm hedges across all of them, holding the lowest mean speed of any arm in Table 4.5 at 2.44 m s^{-1} . The safety of that hedge is not repaid at the bay, where the vehicle comes to rest a median 1.09 m from the centre against 0.54 m under the full method.

The response the full method shows across the bands counts for little if it appears in only some episodes, so the rank correlations described in Section 4.1.5, reported in the upper block of Table 4.5, provide the second half of the evidence. The sign a coefficient must carry to count as conservative differs by proxy. A positive one on the brake command indicates firmer stopping effort as the filter deteriorates, while a negative one on the remaining three indicates the slowing, straightening and smoothing that accompany it. Each coefficient is computed within a condition and averaged across the three. The full method carries the conservative sign throughout, with $+0.406$ on braking, -0.127 on speed, -0.446 on yaw rate and -0.432 on jerk. The

Table 4.5 Uncertainty-conditioning analysis pooled over the three conditions in which the GNSS tier changes mid-episode ($n = 1800$ episodes per arm).

Metric	van.	in.	out.	full
<i>Rank correlation (Spearman vs episode-mean position std, within condition)</i>				
Brake command	+0.265	+0.133	+0.223	+0.406
Speed while moving	−0.039	−0.209	+0.010	−0.127
Absolute yaw rate	−0.207	−0.353	−0.187	−0.446
Action jerk	−0.257	−0.437	−0.165	−0.432
<i>Operating level (pooled)</i>				
Mean speed while moving (m s^{-1})	2.70	2.82	2.44	2.81
Mean brake command	0.09	0.09	0.09	0.20
Median final error (m)	1.46	1.10	1.09	0.54
Successful parking (%)	28.2	28.1	22.5	54.3

ordering it imposes is therefore consistent across every axis and is not the product of one proxy alone.

The coefficients of the output-uncertainty arm fall below those of the full method on all four proxies, most markedly on speed, where its +0.010 carries the opposite sign against the full method’s −0.127. The gaps on the other three are narrower, the difference between the two arms’ coefficients ranging from 0.14 to 0.27 across the four proxies. Differences of that size are comparable to the output-uncertainty coefficients themselves. The benefit conferred by the covariance is therefore distributed across all four proxies rather than concentrated in any one of them.

4.6.2 The Limits of Timing Alone

Under the full method the vehicle’s behaviour is both graded by uncertainty and parked successfully more often than under any other arm, and the two effects cannot be separated in an architecture that carries the covariance input and evidential head together. Establishing the value of grading alone therefore falls to the input-uncertainty arm described in Section 4.1.1, which retains the covariance without the evidential head. The standard head described in Section 2.1.1 lets the covariance reach the commanded action and not the variance around it, and its success rate is accordingly level with vanilla PPO in Section 4.4. The evidence available from the arm is therefore confined to the grading of the driving rather than the outcome of the episode.

Three of the four proxies confirm that expectation. In Fig. 4.5, the input-uncertainty arm opens the sweep level with vanilla PPO in yaw rate and ahead of every other arm in speed (3.32 m s^{-1}). Its yaw rate and jerk decline consistently from those values, reaching 0.120 rad s^{-1} and 0.205 , whereas speed descends unevenly, flattening at the end of the sweep before finishing at 2.63 m s^{-1} . Those trends yield -0.209 on speed against vanilla PPO's -0.039 , -0.353 against -0.207 on yaw rate and -0.437 against -0.257 on jerk, the strongest speed correlation of any arm and the second strongest on the other two.

None of that ordering reaches the outcome. The input-uncertainty arm parks on 28.1% of episodes and vanilla PPO on 28.2%, both well short of the full method's 54.3% (Table 4.5). Speed, yaw rate and jerk are therefore graded by uncertainty without an evidential head, with the covariance reaching them through the commanded mean alone.

Braking is what partly accounts for that result, since all three baselines fall well short on it. Vanilla PPO reaches $+0.265$, the output-uncertainty arm $+0.223$ and the input-uncertainty arm $+0.133$, the lowest of the four, whereas the full method stands clear of them all at $+0.406$. The operating levels give the same picture, with the input-uncertainty arm producing a mean brake command of 0.089 against 0.095 for vanilla PPO, 0.089 for the output-uncertainty arm and 0.200 for the full method. The two readings together describe an arm braking more firmly on uncertain episodes than on clean ones, yet too gently throughout for that ordering to register in the parking outcome.

Vanilla PPO receives no uncertainty signal at all, and its speed across the bands shows what follows from that. The value fluctuates between 2.52 m s^{-1} and 2.87 m s^{-1} over the upper four bands without settling into a decline, so the vehicle is as likely to be travelling quickly on a poorly localised episode as on a well localised one. A coefficient of -0.039 records that absence of ordering, and the vehicle comes to rest a median 1.46 m from the bay centre, the worst of the four arms by roughly a third of a metre. Nothing in the observation of this arm separates a well localised episode from a poorly localised one, and its standard head holds the action variance fixed whatever the state, so neither route to a graded response is open to it.

Setting the mean speed of the two extremes side by side across the bands shows what the signal provides. Both begin the sweep at 3.06 m s^{-1} , and from there they move in opposite directions. The full method sheds speed as the estimate deteriorates, finishing at 2.63 m s^{-1} , whereas vanilla PPO runs slower than it across the lower four bands and then accelerates in the highest to 2.87 m s^{-1} . The two therefore cross, and the uninformed arm carries most speed precisely where the estimate is least trustworthy.

A crossing of that kind establishes the timing of a cautious response without establishing how cautiously the vehicle drives, a distinction the two blocks of Table 4.5 preserve, the correlations addressing the former and the operating levels the latter. The correlations establish the covariance as the prerequisite for that timing. The output-uncertainty arm is equipped with the evidential head and nonetheless grades speed, yaw rate and jerk no more sharply than vanilla PPO, which is equipped with neither component, whereas the input-uncertainty arm grades all three on a standard head. The head therefore contributes nothing to the timing once the input is withheld.

The operating levels divide the arms on a different principle. Three axes are graded by uncertainty under the input-uncertainty arm, yet the vehicle brakes no harder than under the arm supplied with no uncertainty input at all, so timing on its own leaves the outcome unaltered. The effort is supplied by the evidential head, and the two components are combined only in the full method, which constitutes the mechanism behind the benefits reported in Section 4.4.

4.6.3 The Origin of the Behaviour

The uncertainty-conditioned behaviour reported above was never specified in the objective against which the policy was trained. The reward formulation of Section 3.6 contains no uncertainty component, certainty incentive or covariance-weighted penalty, and the standard deviations enter only through the observation. The vehicle nevertheless adjusts all four proxies with the reported position uncertainty, establishing the behaviour as emergent. That adjustment earns the terminal bonus of Section 3.6.3 more often, so it was learned for the outcome it produces rather than for any intrinsic merit. Its discovery depended in turn on the curriculum of Section 3.7, which kept the uncertainty channel active from the first stage and thereby left the strategy within reach throughout training.

Slowing while the estimate is poor and resuming once it recovers requires an action distribution able to widen with the reported uncertainty, a capacity available to the evidential head and to no other arm. The account of the mechanism is thereby complete, and it accords with the finding of Section 4.4.2 that the covariance raises success on that head alone. Its achievement remains bounded, with the gain being one of accuracy rather than safety, and the collision rates behind that distinction are reported in Section 4.8.3.

4.7 Robustness to Seed Selection

Training each arm under three seeds, as established in Section 4.1, produces a second body of evidence that the pooled means leave unread. Every CI reported so far

resamples episodes while the three policies trained for that arm remain fixed, following the resampling procedure described in Section 4.1.5. Such an interval therefore bounds the precision of the arm’s estimate without revealing the variability produced by the same training procedure across seeds. Separating the three runs recovers that second spread.

At $n = 3$ seeds, the strongest claim available is unanimity, which the per-seed record provides for the covariance contrast measured in Section 4.4.1. Comparing each full-method policy with the output-uncertainty policy trained from the same seed measures every difference within a matched pair. The three seeds combined with the three conditions in which any arm achieves a successful park give nine such pairs. The contrast is positive in all nine, with margins ranging from +16.0 to +59.0 pp (Table 4.6). Discarding the seed matching leaves the advantage intact, since the weakest full-method policy surpasses the strongest output-uncertainty policy in every condition, by +14.0 pp at the narrowest. No ordering of the six policies therefore leaves the advantage attributable to a fortunate draw, which is the failure mode the multi-seed protocol is designed to rule out.

Table 4.6 Per-seed success ranges for the evidential-head pair and the matched-seed covariance contrast ($n = 3$ seeds; 200 episodes per seed).

Condition	Success [min, max] (%)		Matched-seed Δ (pp)	
	full	output	min	max
anchor_deployment	[39.5, 51.0]	[17.0, 22.0]	+21.5	+34.0
anchor_empty	[48.0, 78.5]	[19.5, 34.0]	+16.0	+59.0
gnss_degrade_one_way	[40.0, 62.5]	[16.5, 23.0]	+17.0	+46.0

The unanimity of the matched-seed contrast does not imply narrow per-seed ranges within the individual arms. The output-uncertainty arm is the most consistent of the four, spanning 5.0 to 14.5 pp depending on the condition, whereas the other three span 11.5 to 30.5 pp (Fig. 4.6). The full method ranges from 48.0% to 78.5% in the obstacle-free control, whereas vanilla PPO spans 12.5% to 41.5% in the one-way drift. A single run of the latter arm would place it either just behind the full method or far below it, depending on the seed drawn. Spreads of this magnitude further limit what the per-seed record can settle. The full method’s range overlaps those of both standard-head arms on the one-way drift, over the interval from 40.0% to 41.5%, as shown by the overlapping whiskers in Fig. 4.6 and the per-seed scores in Appendix E. The overlap is a property of individual runs, not of the pooled estimate on which the comparison rests. Every finding in this chapter is reported on that basis, as a pooled average over all episodes from all three seeds (Section 4.1) rather than as the result of any single run.

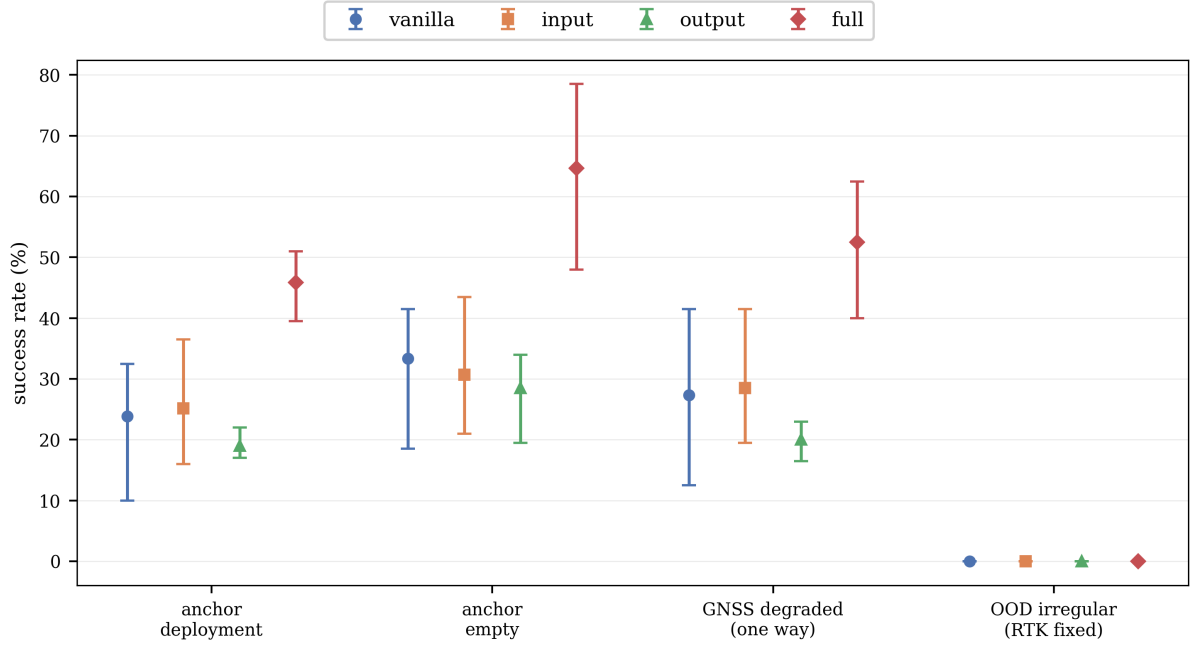


Figure 4.6 Pooled success rate, with whiskers spanning the per-seed minimum and maximum.

The standard Gaussian head yields no comparable consistency. The contrast between the input-uncertainty arm and vanilla PPO varies in sign across seeds in all three trainable conditions, spanning -22.0 to $+29.0$ pp on the one-way drift. The covariance observation therefore raises the success rate on one seed and lowers it on another. The per-seed record thus reproduces the pooled inertness reported in Section 4.4.2. It is consistent with the asymmetry lying in the head rather than in the seeds, since the identical covariance input is decisive for the evidential head while remaining directionless for the standard one.

The seed evidence thus strengthens the architectural claim rather than simply passing a robustness check. The covariance observation is of value only in a head able to represent what it encodes, and that dependence is reproduced on every trained seed.

4.8 Limitations

Three limitations bound the results reported above. The mechanism behind every one is set out below, together with a precise statement of what it does and does not invalidate.

4.8.1 No Successful Park on Out-of-Distribution Geometry

No arm records a success on the unseen irregular layout, all four sitting at a 0% success rate across 600 pooled episodes apiece (Table 4.2). Each phase of the curriculum described in Section 3.7 is conducted on the rectangular lot, while the irregular lot is reserved for assessment, which is the held-out measure of generalisation required in Section 2.1.6. The training thus produces a policy linked to a single geometry.

A complete failure on unseen geometry raises the possibility that no transferable capability was acquired and that the policy merely memorised the lot on which it was trained. Evidence against that interpretation is found on the trained lot itself. The observation defined in Section 3.3 is egocentric and lacks absolute position, so no representation of the entire lot can be retained. The target at the deployment anchor is drawn from 47 bays, of which the early phases described in Section 3.7 use only 15 before the remainder become eligible. Memorisation would leave that heavily practised group the more reliable of the two, yet it attains a success rate of 0.49 against 0.44 for the bays introduced later.

A skill that falters in unfamiliar geometry can fail at more than one stage, and a success rate of zero provides no indication of which. The vehicle may reach the bay and execute the manoeuvre poorly, or it may never arrive at all. Under the full method at the deployment anchor, the median final distance from the target is 0.7 m and 73% of episodes end within 5 m, whereas across the remaining 27% that median is 24.2 m. Those proportions reverse on the irregular layout, where the median over all episodes extends to 19.8 m and fewer than 8% end within 10 m, at step counts and speeds matching the anchor. A failed manoeuvre would leave the vehicle a few metres from a bay it had reached, yet hardly any episode ends at such a distance. The closest approach to any bay is also confined to the clusters nearest the spawn. The failure therefore lies in traversing the lot rather than in the final manoeuvre. All four arms fail in this manner, so the comparison among them remains unaffected.

The training set contained a single layout, so the result reported above cannot speak to the case where a distribution of layouts is used instead, which might yield a policy that does transfer. The layout generator that produced this probe could supply that varied training set, an experiment deferred to Section 5.2 because it calls for a fresh training campaign on every arm. The contribution of this work is therefore uncertainty-conditioned control on the trained geometry class, not zero-shot generalisation to unseen layouts.

4.8.2 Total Uncertainty Does Not Separate Outcomes

The safety layer specified in Section 3.9 brings the vehicle to a stop and flags a handover once the total predictive uncertainty crosses a fixed threshold. Its activations follow that signal alone and are therefore blind to the outcome an episode is heading for. All rates in this subsection are derived from replaying the threshold over the one-way drift episodes run under the protocol established in Section 4.1.3, where the rule was inactive and a degradation is guaranteed rather than left to the Markov chain (Section 4.1.2). Each episode consequently carries both the moment the rule would have fired and the outcome it actually reached, a combination that a live intervention disrupts by terminating the run.

The total predictive uncertainty rises over the course of an episode under every assessed condition, including those without degradation, and it rises for the same reason in each. As the vehicle slows and brakes harder, the signal tracks the signature of a slow, heavily braked final approach to the bay rather than any degradation. The one-way drift adds a degradation-driven rise on top of that approach signature, as the localisation estimate moves from confident to uncertain within a single episode.

A useful handover should fire on the episodes heading for failure and spare the rest. The threshold is instead crossed in 56.2% of the episodes that ended in a successful park, marginally more often than the 50.5% reached on those that did not. The two rates stand 5.7 points apart, so the rule surrenders roughly one successful park for every failure it averts, which is the 0.44 gate AUC this condition records in Section 4.3.1 restated as a firing rate. Moving the threshold cannot repair this, since the evidence parameter ν holds at a constant rather than tracking familiarity (Section 4.3.2). Every threshold on either channel, or on their sum, consequently fires on the same episodes in the same order.

A rule that waits for the threshold to be exceeded on several consecutive decisions, rather than firing on the first, widens that separation without resolving it. The 600 one-way drift episodes of the full method were replayed at the threshold fixed in Section 3.9, now demanding ten successive decisions above it, approximately 2 s, before a handover is issued. The firing rate falls from 56.2% to 10.5% on the episodes that would have parked successfully and from 50.5% to 26.7% on those that would have failed. The larger fall is the desirable one only if a surrendered park costs more than a collision, which it does not, so the delay improves the rule on the side that matters less.

Selectivity must therefore come from a signal that separates the two outcomes, or from a criterion reading more than just the uncertainty level. The task phase [56] is a prime candidate, since the approach and the final manoeuvre tolerate markedly different amounts of it, and a tolerance that varies with context has already been

shown to be learnable when supplied explicitly. Both routes are set out as future work in Section 5.2.

4.8.3 Scope Limitations

Every result in this chapter was obtained in simulation. The real-vehicle pipeline is implemented, unit-tested and structurally identical to the simulation path, and it covers the safety wrapper specified in Section 3.9 along with the calibration and pre-deployment sequence set out in Appendix B. None of it has ever been executed on hardware. The correspondence between the two paths is nonetheless substantial, since the observation is built by the same code from the same message types on both, as recorded in the interface listing of Appendix B.2, so the trained weights would run on the vehicle without retraining.

A second bound arises from the hyperparameter suite. One configuration is held fixed across all four arms rather than tuned for each, since a separate search per arm would add a fifth variable to the ablation and confound the two under test, as argued in Section 3.8.1. The arms are therefore ranked at a single operating point rather than at the best one available to each, and any arm poorly served by the fixed values is understated as a result. Averaging over three seeds reduces that effect without removing it.

A third bound concerns what the uncertainty-conditioned behaviour measured in Section 4.6 achieves. The gain is one of accuracy rather than of safety. Conditioning on the covariance raises success and lands the vehicle closer to the bay centre, yet the collision rate does not fall with it, standing at 19.9% for the full method against 17.3% for the output-uncertainty arm. The behaviour is therefore established as a route to more precise parking and not as a demonstration of safer driving.

4.9 Summary

Each objective of Section 1.3 is now traced to its evidence.

A localisation pipeline that honestly predicts its own error, objective O1, is demonstrated by the calibration analysis of Section 4.2. Across the three conditions in which the GNSS tier changes during the episode, the reported position standard deviation ranks the actual error at a Spearman correlation of 0.54. The mean error meanwhile rises from 0.15 to 2.07 m across five equally populated uncertainty bins, roughly a thirteenfold increase between the outer bins.

Training to convergence, the stability half of objective O2, was obtained on every seed once the variance-collapse pathology of Section 3.5 was mitigated (Fig. 3.8). The separability half is answered negatively by the channel-coupling analysis presented

in Section 4.3.2, where the evidence parameter ν settles at a constant across states, leaving the epistemic estimate a fixed multiple of the aleatoric estimate. The head therefore returns one signal under two names.

The 2×2 ablation reported in Section 4.4, aimed at the gap identified in Section 2.3, crosses a single factor pair, as objective O3 requires. Supplying the EKF covariance to the evidential actor raises parking success across all trainable conditions, with a gain of +26.8 pp on the deployment anchor and up to +36.2 pp on the empty layout. Those figures carry 95% bootstrap intervals of [21.7, 32.0] and [30.8, 41.3], computed by the resampling procedure of Section 4.1.5. The advantage further persists through the seed draw. All nine matched-seed comparisons between the full method and the covariance-blind evidential arm favour the full method, at a margin of +16.0 pp in the worst case, on the per-seed record given in Section 4.7. In the absence of the evidential head, the benefit is lost. The same covariance input on a standard Gaussian head shifts success by between -2.7 and $+1.3$ pp, a range that includes zero. The contribution is consequently a synergy rather than beneficial with either component alone: the covariance input raises success only for an actor that can adjust its action distribution from state to state, the heteroscedastic property isolated in Section 4.3.3.

The synergy is visible in the driving itself, which objective O4 requires, and the behavioural evidence is set out in Section 4.6. Under the full method the vehicle slows, brakes harder, turns less sharply and changes its commands less abruptly as the reported uncertainty rises. The operating level is more conservative throughout as well, with an average brake command of 0.20, while all three baselines hover around 0.09. No part of the reward pays for any of it, so the behaviour was never bought in its own right and was learned instead because it leads to more successful parking.

A failed transfer and a coarse signal bound the scope of the four claims above without weakening their basis. No arm succeeds on the withheld irregular layout, all four recording 0% success on it, which ties the competence acquired to the geometry it was trained on. On the conditions where that competence does hold, the uncertainty signal conditioning it is too coarse to gate on. Thresholding the head’s total predictive uncertainty over them separates the episodes heading for failure from those heading for success at a rate barely above chance. The near-constant ν of O2 is the reason, since both channels and their sum then carry a single ordering. Both are reported with their mechanisms among the limitations set out in Section 4.8.

5 Conclusions and Future Work

The preceding chapters examined how much localisation uncertainty matters to a continuous-control parking strategy, supplied to the policy as an observation and accompanied by a head reporting the confidence attaching to the actions selected. The evidence in Chapter 4 answers the research question posed in Section 1.1 affirmatively, within the two bounds recorded in the evaluation summary given in Section 4.9, namely that the head’s epistemic and aleatoric channels do not separate and that no arm parks successfully on the unseen layout. Both are findings in their own right. Each of the four objectives set out in Section 1.3 is assessed against the reported results in Section 5.1, alongside the five contributions claimed. The research that follows from those results is set out in Section 5.2, ordered by how much of the existing system each item reuses.

5.1 Conclusions

5.1.1 The Objectives Revisited

The covariance computed by the pipeline of Section 3.4 is informative about that pipeline’s own error, which satisfies objective O1. The calibration coefficient of Section 4.2 licenses an ordering rather than a magnitude, so a larger reported uncertainty marks a genuinely worse estimate, and it is on that ordering alone that the policy acts. One constraint qualifies the conclusion. Completed parking episodes are required for the coefficient to be scored, and none are obtained on the unseen irregular layout by any of the four arms (Section 4.8.1), so the ordering is established over the three conditions run on the trained lot and not over unseen geometry.

A trustworthy input is only half of the architecture, and objective O2 is met on the output side in the matter of stability alone. The convergence recorded in Section 4.9 was obtained under the prior-anchoring regulariser of Section 3.5, where the collapsed head is repaired against fixed references rather than against a label. An evidential action head can therefore be trained without a ground-truth action target, the first of the two questions raised in Section 2.3, since the target presumed by the evidential loss is required by the regulariser alone. The second of the two, raised over the usability of that head’s uncertainty output, is left unanswered by that stability, since the two uncertainty channels are not separated by the parameterisation. In a single-head NIG actor, the epistemic estimate is obtained by dividing the aleatoric estimate by the evidence parameter, so the two separate only where that divisor varies with the state. No clearly defined supervisory signal for it is available in model-free

reinforcement learning, since a sampled action does not serve as its own label, and the constant value measured in Section 4.3.1 is the consequence. A bound therefore falls on what the architecture’s uncertainty outputs can be used for, and the bound is a property of the parameterisation rather than of this particular training run.

The bound applies to those outputs alone and leaves the value of the two architectural layers open. Neither the covariance input nor the evidential head is worth retaining unless separately justified, so the two are varied independently in the two-by-two ablation designed in Section 3.8, as mandated by objective O3. The design is what converts the gain reported in Section 4.9 into an attribution, since the same covariance input supplied to a standard Gaussian head moves success by an amount indistinguishable from zero. The benefit is thus carried by neither layer on its own but by the pairing, the interaction established in Section 4.4.2, and an architecture adopting the covariance input without an actor able to vary its action distribution from state to state should not expect to recover it.

The vehicle brakes in time with the live EKF standard deviation, which is the uncertainty-conditioned behaviour objective O4 requires the gain to operate through. The timing is caused by the three covariance dimensions and by nothing else the observation carries, since the covariance-blind arm shares every other input and shows no such response. Nothing in the reward specified the behaviour either, the formulation given in Section 3.6 carrying no uncertainty term of any kind. A policy rewarded for parking alone will therefore acquire caution as an instrumental strategy where the observation supports it, which is the emergence argued in Section 4.6.3.

5.1.2 Contributions

This work makes five contributions:

1. The first use of raw EKF covariance features as policy observations for ego-vehicle parking control, closing the input-side absence identified in Section 2.3. The ablation of objective O3 establishes the benefit as causal, and as conditional on pairing those features with an uncertainty-capable head.
2. The first actor-side NIG evidential head on a continuous-control policy-gradient agent, closing the output-side absence identified in Section 2.3 against the prior placements tabulated in Table 2.2. A variance-collapse pathology is inherent to the placement, diagnosed against the DER limitations catalogued in Section 2.1.3 and mitigated by the three-part fix designed in Section 3.5. The repaired head is the uncertainty-capable output on which the covariance benefit of the first contribution depends.

3. Evidence that uncertainty-conditioned behaviour can be emergent. No uncertainty term appears anywhere in the reward and no slow-down is scripted. Under the trained policy the vehicle nonetheless brakes more, steers more smoothly and moves less abruptly as EKF uncertainty rises, with the timing and effort measured in Section 4.6.
4. A negative result with a mechanism: in a single-head NIG actor, the epistemic and aleatoric channels conflate because the evidence parameter is unsupervised in model-free reinforcement learning, as measured by the failure-prediction test conducted in Section 4.3.1.
5. A reproducible evaluation asset: a de-confounded four-condition sweep under controlled GNSS degradation in CARLA, with declarative condition configurations, a fixed three-seed protocol and a deterministic analysis suite, all set out in the experimental setup of Section 4.1. Reconstruction from a clean checkout is documented in the reproduction guide of Appendix D.

5.2 Future Work

Every item below follows from a specific finding of Chapter 4 or from a bound placed on its scope. The ordering runs by proximity to the present system, from what it can already support to what would require rebuilding it.

1. *Real-world validation.* The deployment pipeline is unit-tested but never run on hardware, the bound recorded in Section 4.8.3, which leaves timing, actuation response and receiver behaviour unexamined. The staged validation documented in Appendix B is therefore the single most direct extension available, on the parking area introduced in Section 1.2. Validation of the simulated noise levels against logged RTK traces from the physical receiver belongs to the same campaign. The hand-tuned fix-state transition matrix of Section 3.4 would then be replaced by one estimated from logged fix-state sequences, with the correction stream interrupted to induce the degraded tiers rather than left to chance.
2. *A separable epistemic channel.* The collapse measured in Section 4.3.1 is a consequence of the evidence parameter being unconstrained in this particular head. An unfamiliar state could therefore still be identified under an architecture in which that parameter is tied to the training distribution, which is a second head rather than a repair to the one under test. The most straightforward approach is to make the parameter a function of a density over the latent state, the

posterior-network construction set out in Section 2.1.3 [63]. The same information is recoverable from an ensemble of evidential heads [19] without a density model, by way of the discrepancies among members, at the per-decision cost established in Section 2.1.2. Both would be trained under the current curriculum, and the failure-gate analysis would have to be repeated. The safety layer would benefit accordingly, since a constant evidence parameter leaves the rule of Section 3.9 thresholding a magnitude rather than the channel it was designed to read. A varying parameter would restore the division of Table 2.1, reserving the handover for an unfamiliar state and meeting a high aleatoric reading by slowing and turning less sharply.

3. *The heteroscedastic-Gaussian control arm.* In the fifth arm identified beyond the factorial of Sections 3.8 and 4.3.3, the single learned constant of the standard Gaussian head is replaced by a log standard deviation computed from the state, after the parameterisation of soft actor-critic [15]. The covariance input is supplied to that same head, and the evidential head’s advantage would then be apportioned between the state-conditioned variance and the NIG evidence mechanism. No new apparatus is required for that comparison, as the arm is trained under the current curriculum and assessed on the same protocol, at one training run per seed.
4. *The cost axis as a trade curve.* The passes column of Table 2.2 separates the single-pass estimators from those requiring K ensemble members, and the saving recorded in that column motivates the head adopted in Section 3.5. A count of passes nonetheless prices the estimators without ranking them, since the task performance purchased at each price is left undetermined. Recovering that missing dimension requires a sweep over ensemble size, trained under the current curriculum and assessed on the same protocol. The column is then converted into a trade curve of success rate against per-decision latency, measured on the deployment hardware against the 200 ms budget fixed in Section 3.3.3. Each member carries a training run per seed and multiplies the compute required by the ablation of Section 3.8 by K , so the sweep is a campaign of its own rather than a further cell in the factorial.
5. *Evidential heads on both actor and critic.* The value-side placement surveyed in Section 2.2.2 and the actor-side placement adopted in Section 3.5 quantify different things, the confidence of the return estimate against that of the action proposed under a given pose. The two are therefore complementary, and no published system carries an evidential head at both placements (Table 2.2). A critic-side head was excluded from this work as a third mechanism demanding its

own cell in the factorial design. The open question is whether the sharper advantages available from an evidential critic improve the manoeuvre obtained under an evidential actor.

6. *Out-of-distribution geometry.* Generalisation beyond the single geometry used for training would require a training distribution over layouts, given the complete failure on the unseen lot reported in Section 4.8.1. A varied training set of that kind is obtainable on the existing stack from the layout generator used to produce the unseen lot. The target of such training would be route-finding rather than parking, since the vehicle fails while traversing the unseen lot rather than during the final manoeuvre. The experiment would then settle whether the covariance-conditioning benefit is retained under a change of geometry.

5.3 Final Remarks

The problem set out in Chapter 1 was a vehicle required to park inside the markings of a single bay while its own position estimate varies in accuracy by two orders of magnitude. Once that estimate's covariance was supplied to the policy, and the actor was given the means to express uncertainty in its own actions, parking success rose by between 26.8 and 36.2 pp on every trainable condition. The behaviour underlying that gain is a response for which no reward term was specified, in which the vehicle brakes as the reported uncertainty rises and waits out the periods of poor localisation.

Neither side of the architecture is specific to the setting that produced it. The covariance is computed by any vehicle that runs an EKF, so a quantity the localisation stack already reports is all the input side requires. Nothing about it depends on the degradation being a GNSS one either, since the same confidence signal is published by a filter fusing any sensor set. The output side is equally portable, as the functional property is identified in Section 4.3.3 as state-dependent action variance rather than the NIG parameterisation itself. Any model conditioning its variance on the state satisfies that property, a heteroscedastic Gaussian included.

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Appendix A Sensor Noise Models: Datasheet Derivations

This appendix records the calculations supporting the condensed noise parameters of Table 3.7, linking each simulated noise magnitude to its original manufacturer specification. The two rows marked † in that table have no corresponding datasheet figure, so the basis of the float-tier estimate and the rationale for the degraded tier are addressed separately in Appendix A.1. Every injection method in this appendix is governed by two master flags (`enable_gnss_noise`, `enable_imu_noise`). Noise on real hardware originates from the sensors themselves, so both flags are set to false for deployment and each mechanism reduces to a pass-through, as specified in Appendix B. The following derivations therefore describe only the simulated configuration, sensor by sensor.

A.1 RTK-GNSS: u-blox ZED-F9P

The GNSS datasheet [1] expresses horizontal position accuracy as CEP, so recovering a spread parameter requires a distributional assumption. Under a circularly symmetric zero-mean Gaussian with per-axis standard deviation σ , the radial error follows a Rayleigh distribution characterised by two standard quantities. Its median is $\text{CEP} = \sqrt{2 \ln 2} \sigma \approx 1.1774 \sigma$, whereas its root-mean-square radial error, denoted by Distance Root Mean Square (DRMS), is $\sqrt{2} \sigma$. Dividing the datasheet value by $\sqrt{\ln 2}$ therefore recovers the latter:

$$\text{DRMS} = \frac{\text{CEP}}{0.8326} = \sqrt{2} \sigma \quad (\text{A.1})$$

where the divisor $\sqrt{\ln 2} \approx 0.8326$ is analytic rather than an engineering rule of thumb, so the conversion is exact wherever the circular assumption holds. The resulting DRMS is radial, yet it is adopted as the standard deviation of the zero-mean Gaussian injected on each axis independently. Substituting a radial figure for a per-axis one is deliberate, placing every tier a factor $\sqrt{2}$ above the strict reading. The justification rests on the gap between the datasheet's test conditions and a car park, as Section 3.4.1 establishes, and the specifics of those conditions indicate how generous the resulting margin is. The quoted accuracy comes from 24 hours of static operation [1, Table 3], during which the centimetre figure depends on the carrier-phase ambiguities remaining resolved and the correction stream running without interruption. Both conditions are disrupted in a car park by the multipath and non-line-of-sight mechanisms established in

Section 2.1.4, so a factor of $\sqrt{2}$ is a modest allowance rather than a conservative one. The four tiers and their origins are given in Table A.1.

Table A.1 The four GNSS tiers and their noise derivations.

Tier	Derivation	Value	Provenance
rtk_fixed	$0.010/0.8326 = 0.0120$ m, raised	0.020 m	Datasheet [1, Table 3]
rtk_float	$0.300/0.8326 = 0.360$ m	0.360 m	Measured decimetre regime [34]
standalone	$1.5/0.8326 = 1.802$ m	1.802 m	Datasheet [1, Table 3]
degraded	Beyond specification	5.0 m	Stress condition

Reading the tiers of Table A.1 in the order of the quality ladder in Section 2.1.4, from most accurate to least, the fixed tier begins with the datasheet value of 0.010 m CEP plus one part per million of baseline length. The baseline-dependent term contributes only 0.0001 m over the 100 m typical of a parking lot, so it is disregarded and the CEP converts to a floor of 0.0120 m. The floor is optimistic, as the datasheet qualifies its figure by excluding antenna phase-centre offset errors [1, Table 3], which a real installation incurs alongside the multipath interference that surrounding vehicles and structures produce [32]. Quantifying either demands an antenna position measured on a physical platform and a surveyed site, neither of which this work instruments, as the mount poses of Appendix B.2 remain site-specific placeholders. The floor is therefore raised conservatively to 0.020 m. One rung lower, the float tier converts a 0.300 m CEP figure, taken from the decimetre level of the quality ladder rather than from any published specification. No manufacturer quotes one, as float accuracy varies with atmospheric conditions and baseline length instead of settling at a representative value, so the measured regime of Section 3.4 supplies the value. The standalone tier is again taken from the datasheet, at 1.5 m CEP for an uncorrected fix, and marks the rung at which the error exceeds the geometry of the task. A 2.5 m bay spans only approximately 1.4σ of the resulting error, so a fix landing outside the target bay is a routine occurrence rather than a rare one. The degraded tier lies beyond the datasheet by construction, serving as the stress condition described in Section 3.4, and its 5.0 m figure models conditions outside the receiver's operational range under severe multipath and non-line-of-sight reception [32].

Converting those metric sigmas into the latitude and longitude published by the receiver, as required by Section 3.4.1, uses the flat-earth value of 111 320 m per degree, held fixed on both axes across the parking area. Such a constant is exact only at the equator and for north-south displacement, so a small scale error enters the injected offset. The discrepancy is immaterial to the filter because the EKF configured in Section 3.4.2 consumes the metric odometry output of the relay node together with

a covariance stamped in metres, both of which remain correct regardless of the degree conversion.

The tier governs more than position, since velocity noise scales in the same way. At the fixed tier, the Doppler velocity standard deviation is set to the datasheet accuracy of 0.05 m s^{-1} [1, Table 1]. It is then multiplied by the same per-tier factors as the position noise (1, 18, 90 and 250 times the base), producing values of 0.05, 0.90, 4.5 and 12.5 m s^{-1} , respectively. The speed estimate therefore coarsens with the fix state.

The Doppler standard deviation also governs the third observable, the direction of travel, which is derived from position rather than from a moving-base receiver. It is obtained as COG from the direction of successive fixes, with the angle calculated as $\text{atan2}(\Delta y, \Delta x)$ over the per-callback displacement. Its reported variance follows from that standard deviation and the current speed:

$$\text{var}(\psi_{\text{COG}}) = \left(\frac{\sigma_{\text{doppler}}}{\max(v, 0.1)} \right)^2 \quad (\text{A.2})$$

where σ_{doppler} denotes the tier's Doppler standard deviation and v is the current speed, floored at 0.1 m s^{-1} to keep the expression bounded near standstill. Scaling the variance by speed reproduces the precision of a Doppler-derived COG, established in Section 2.1.4, which holds while the antenna is in motion and degrades as the vehicle slows. At low enough speeds the heading becomes unreliable altogether rather than merely imprecise, so three conditions gate publication. The displacement over the callback must exceed 0.05 m, no IMU ZUPT may be active, and the candidate variance must fall below $\pi^2/3$. An angle failing any one of them is discarded, and the filter runs without a heading until the next usable one arrives.

The three observables above take their magnitudes from the tier, whereas the error anisotropy and the dropping of fixes are randomised independently of it. Section 3.4.1 establishes what each reproduces, leaving the sampling that implements them to be set out alongside the tiers themselves. The anisotropy is governed by a ratio r , drawn uniformly from $[1.0, 1.5]$ at the start of an episode and divided between the axes as $\sqrt{r}$ in a randomly chosen direction and $1/\sqrt{r}$ in the other. Splitting it by the square root gives two factors with a product of one, so the tier's sigma is preserved as the geometric mean and the error changes shape without growing in magnitude. Dropped fixes are governed by a second and far shorter cycle, a Bernoulli draw with probability 0.02 resolved at every callback, which at 20 Hz costs a fix every 2.5 seconds on average. The rate leaves the fix available across the great majority of callbacks, yet interrupts a manoeuvre often enough for the condition to be encountered repeatedly during training.

A.2 IMU: VectorNav VN-100

The IMU datasheet [78] expresses noise as a spectral density, so a conversion is needed before either channel can be given a variance. For a white-noise process of density S sampled at a rate f , the variance of a single observation is:

$$\sigma^2 = S^2 f \quad (\text{A.3})$$

Evaluating Eq. (A.3) at the 20 Hz pipeline rate converts each datasheet density into the variance of one reading, starting from the gyroscope density of $0.0035 \text{ deg s}^{-1} \text{ Hz}^{-1/2}$:

$$\begin{aligned} S_{\text{gyro}} &= 0.0035 \text{ deg s}^{-1} \text{ Hz}^{-1/2} = 6.10865 \times 10^{-5} \text{ rad s}^{-1} \text{ Hz}^{-1/2} \\ \sigma_{\text{gyro}}^2 &= S_{\text{gyro}}^2 \cdot 20 = 7.46314 \times 10^{-8} \text{ rad}^2 \text{ s}^{-2} \end{aligned}$$

The accelerometer density of $0.14 \text{ mg Hz}^{-1/2}$ is treated the same way, once converted out of g units with $g = 9.81 \text{ m s}^{-2}$:

$$\begin{aligned} S_{\text{accel}} &= 0.14 \text{ mg Hz}^{-1/2} = 1.37340 \times 10^{-3} \text{ m s}^{-2} \text{ Hz}^{-1/2} \\ \sigma_{\text{accel}}^2 &= S_{\text{accel}}^2 \cdot 20 = 3.77246 \times 10^{-5} \text{ m}^2 \text{ s}^{-4} \end{aligned}$$

Both variances are recomputed at full floating-point precision rather than read back from the rounded figures above, and each serves two purposes. The first is to parameterise the additive Gaussian noise that corrupts the reading. The second is to populate the covariance field of the outgoing message, which the simulator would otherwise report as zero. The downstream filter therefore receives a corrupted reading together with an honest account of how far it may have been displaced. Expressed as standard deviations, which are easier to compare with the measured quantities, the two values are $2.73 \times 10^{-4} \text{ rad s}^{-1}$ for the gyroscope and $6.14 \times 10^{-3} \text{ m s}^{-2}$ for the accelerometer.

Both figures describe scatter redrawn independently at every sample, one of the three error terms carried by a real inertial sensor. The other two persist across many samples and are therefore modelled separately from the white noise. The first is in-run bias stability, treated as a fixed offset that holds for an entire episode and is redrawn at each boundary. Its value on every axis is sampled from a uniform distribution symmetric about zero and bounded by the stability figure specified in the sensor datasheet, which gives 5° h^{-1} for the gyroscope, the lower end of the typical 5 to 7 range, and 0.04 mg for the accelerometer. Neither is quoted in the units required by the model, so converting the first from degrees per hour and the second from g gives the

bounds actually sampled:

$$b_{\text{gyro}}^{\text{max}} = 5 \cdot \pi / 180 / 3600 = 2.42407 \times 10^{-5} \text{ rad s}^{-1}$$

$$b_{\text{accel}}^{\text{max}} = 0.04 \times 10^{-3} \cdot 9.81 = 3.924 \times 10^{-4} \text{ m s}^{-2}$$

Alongside that offset, the second persistent term is a residual scale-factor error of $\pm 0.5\%$ per axis, taken from the factory calibration specification. It is applied multiplicatively rather than additively and is resampled together with the bias at every episode boundary. Combining both terms with the white noise above, each measured channel follows:

$$z = s x + b + \eta \quad (\text{A.4})$$

where x is the true value, s is the scale factor, b is the per-episode bias and η is the white noise.

The channel model of Eq. (A.4) corrupts whatever motion the simulator reports. A stationary vehicle is not reported as exactly stationary, as the physics solver leaves small residual rates on a body at rest, and the channel model would corrupt those residuals as though they were genuine motion. A ZUPT guard limits the drift that would otherwise accumulate. The yaw rate $\dot{\psi}$ is set to zero when $|\dot{\psi}| < 0.015 \text{ rad s}^{-1}$, and the planar accelerations are set to zero when the gyroscope ZUPT is active and both magnitudes fall below 0.2 m s^{-2} . Noise injection is skipped for every channel clamped in this manner, ensuring that the guard and injection remain consistent.

A.3 2D LiDAR: SICK TiM571

The LiDAR datasheet [73] separates the range error of the scanner into an offset that persists and a scatter that does not, quoting $\pm 60 \text{ mm}$ for the systematic component and below 20 mm at 90% remission for the statistical one. The systematic term varies between units and between mountings while holding constant during a session, so it is represented by a range bias sampled uniformly from $\pm 0.060 \text{ m}$ once per episode. The statistical term instead describes per-point Gaussian range noise with a standard deviation of 0.020 m , taken from the higher 90% remission figure rather than the 0.010 m figure quoted at 10% remission. Car-park surfaces span the full remission range, so the larger value is the more cautious choice for the clearance features.

Both terms act on range and are applied in the polar form in which the scan is published, so a true range r is reported as:

$$r' = r + b_r + \mathcal{N}(0, \sigma_r^2) \quad (\text{A.5})$$

where r' is the reported range, b_r is the per-episode bias and σ_r is the per-point standard deviation. Returns below the sensor's 0.05 m minimum operational range are then discarded. The simulator provides a per-sensor noise attribute of its own, which is set to zero at spawn so that Eq. (A.5) remains the only corruption applied to the scan.

A.4 Extended Kalman Filter Process Noise

The process-noise covariance Q determines the rate at which reported uncertainty rises during a lost fix. The entries for position, yaw and yaw rate that govern that rise are quoted in Section 3.4.2, and this section records the remaining estimated states alongside them, justifying each value relative to the others.

Those relative values are interpretable only against the structure of the matrix, which the filter holds as a continuous-time spectral density over a fifteen-element state. Three positions and three orientations describe the pose, their six time derivatives form the twist, and three linear accelerations complete the vector. Each prediction step adds the matrix to the propagated covariance in proportion to the timestep, so a larger diagonal entry inflates the uncertainty of its associated state more rapidly between corrections. Only part of that state is live, as the planar mode described in Section 3.4.2 estimates the pose $[x, y, \psi]$ together with the two planar velocities and the yaw rate $\dot{\psi}$, leaving the remaining rows inert. The shipped diagonal is listed in Table A.2.

Table A.2 Diagonal of the EKF process-noise covariance Q .

State	Role	Value	Rationale
x, y	Active	1.0×10^{-2}	Residual constant-velocity model error on position
ψ (yaw)	Active	1.0×10^{-4}	Orientation drift consistent with the VN-100 noise floor
$\dot{\psi}$ (yaw rate)	Active	1.0×10^{-3}	Gyro noise density; dominant source of between-fix growth
Planar velocities	Active	1.0×10^{-4}	Constrained by the Doppler update each tick
z , roll, pitch	Suppressed	1.0×10^{-6}	Positive-definite floor; not estimated in planar mode

The yaw-rate entry $\dot{\psi}$, the largest of the rate terms, provides the reference point for the entire diagonal. Its prominence arises because yaw rate is the only fused IMU channel, so the uncertainty admitted by this entry propagates to the reported yaw and, through kinematic coupling, to the reported position. Its value is tuned well above the VN-100 gyroscope noise density derived in Appendix A.2, so that the filter admits not only the noise of the sensor but the far larger error of propagating a single rate channel through a kinematic model for the duration of a lost fix. The three remaining active groups are then sized against it by the measurement each one depends on. The yaw

entry ψ sits an order of magnitude lower, as the yaw rate already supplies most of the orientation inflation and only a small additional drift is required. The two planar velocities are held at that same level, both being corrected by the Doppler speed that arrives with every fix. The position entries are the largest of the estimated states, depending on the least reliable correction of the three, as the fix that constrains them is the one lost during a degraded tier, and their value therefore covers the drift of the constant-velocity motion model across an unbounded gap. Collectively, these entries determine how rapidly the posterior expands between corrections and thereby establish the slope of the rising edges in the sawtooth covariance profile of Fig. 3.5. On the vehicle described in Appendix B, only the yaw-rate and yaw entries would be retuned, against the measured noise floor of the fitted IMU in place of the VN-100 figures of Appendix A.2. The position and velocity entries quantify instead how far the motion model of Eq. (2.10) drifts from the true trajectory between fixes, which no change of hardware affects, so those values would transfer to the vehicle as they stand.

Appendix B Real-World Deployment

This appendix describes the sim-to-real pathway outlined in Chapter 3, which forms the basis for the first item of future work in Section 5.2. The pathway is an implemented and unit-tested software pipeline that has not yet been run on hardware (Section 4.8.3). The work remaining before an initial trial therefore concerns measuring the site-specific physical quantities expected by the pipeline, including the survey of the parking area and calibration of the vehicle’s actuators. Those quantities are supplied through a single machine-verified interface, a set of configuration files gathered under one deployment directory in the repository layout of Fig. D.1.

B.1 Code-Path Parity

Deployment is restricted to one modification, the replacement of the publishers of two topics. The position of that substitution within the data flow is shown in Fig. B.1. The CARLA bridge publishes the GNSS fix and IMU stream in simulation, whereas the vehicle’s hardware driver nodes publish them on the real platform. Everything downstream of those two topics belongs to the localisation and decision pipeline of Section 3.1, which carries over unchanged and differs between the two settings only through the master-switch values of Appendix B.2. The remaining difference is at the far end of the flow, where the resulting command is applied either to the simulated vehicle in CARLA or to the physical actuators through the calibration of Appendix B.4.

Two properties follow from that parity. First, the observation supplied to the policy is produced by the same builder and scaled according to the same predetermined physical ranges used during training, so the learned weights transfer unchanged. Second, simulator ground truth never enters the observation path, ensuring that every input consumed by the policy in simulation is also available on the real vehicle.

B.2 Robot Operating System Interface Parity

The interface between the hardware and the pipeline consists of two standard message types. The relay nodes expect `sensor_msgs/NavSatFix` and `sensor_msgs/Imu`, formats already published by off-the-shelf ROS 2 drivers such as `ublox_ros` and `nmea_navsat_driver`. The two sides therefore connect directly without custom driver code to translate between them.

The interface is bound to the running system by a real-vehicle launch description that forms part of the repository. It starts the same relay executable as the simulation stack and constructs the static sensor transforms under conventional frame

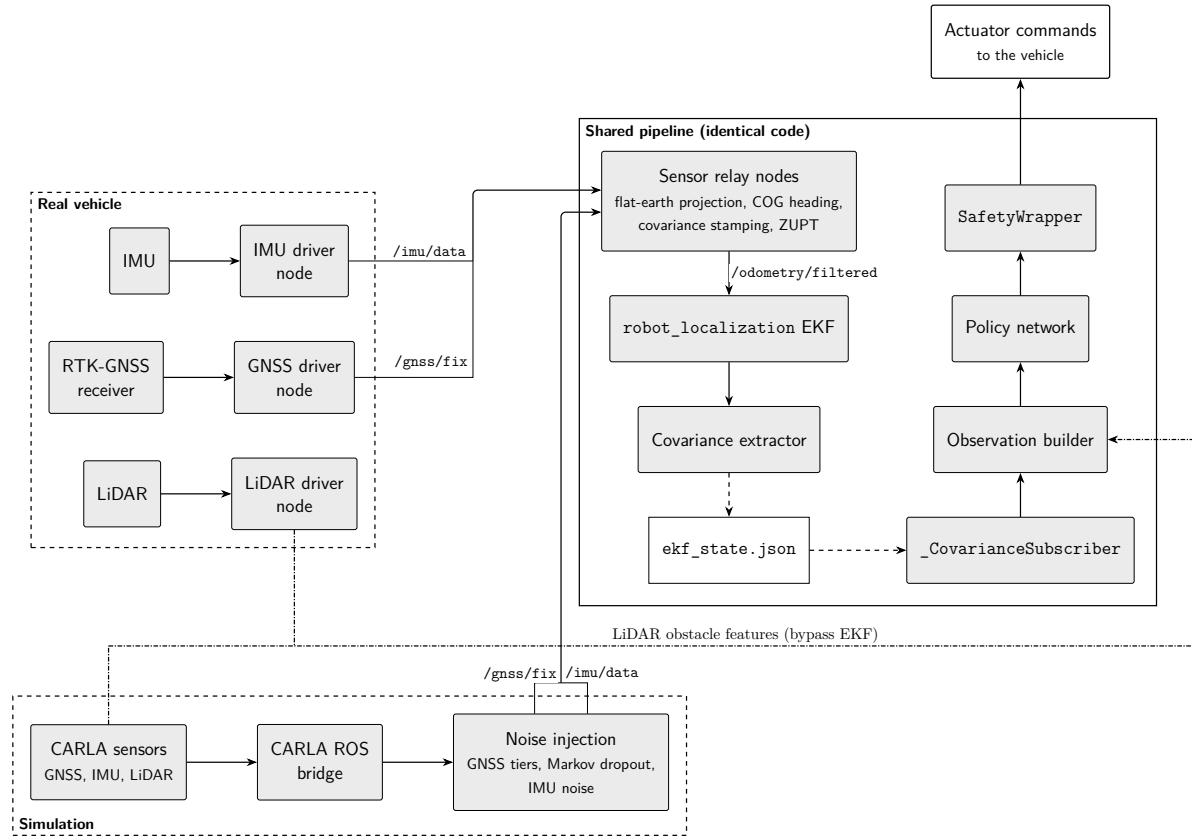


Figure B.1 Sensor data flow in simulation and on the real vehicle.

names (`base_link`, `imu_link`, `lidar_link`, `gnss_link`). The mounting poses encoded in those transforms are read from `sensor_config.yaml`, which also positions the sensors on the simulated vehicle. A mount measured on the physical platform is therefore recorded once and governs both the simulator and the real vehicle.

The work of the relays divides under deployment into what a real receiver renders redundant and what it requires as much as a simulated receiver. That division is settled by the launch description, which sets the master switches of Appendix B.1 to their deployment values. The noise, Markov-transition and anisotropy flags are set to false and the dropout probability to zero, disabling every noise mechanism derived in Appendix A. The signal conditioning established in Section 3.4 remains, comprising the flat-earth projection, COG derivation, covariance stamping and ZUPT clamping, none of which is an artefact of simulation. Real receiver values therefore pass through unchanged in every other respect.

B.3 Extended Kalman Filter Frame Calibration

The EKF reports poses in an odometry frame anchored at the node's most recent initialisation. The bays and boundaries targeted by the policy are instead specified in

the world coordinates of the layout files. Both frames describe the same physical ground, so distances and angles are preserved between them and only their origins and headings differ. Connecting them is therefore a matter of calibration, using one known correspondence to solve for the two-dimensional rigid transform (t_x, t_y, r) satisfying:

$$\begin{aligned}x_{\text{world}} &= \cos r \cdot x_{\text{odom}} - \sin r \cdot y_{\text{odom}} + t_x \\y_{\text{world}} &= \sin r \cdot x_{\text{odom}} + \cos r \cdot y_{\text{odom}} + t_y \\ \psi_{\text{world}} &= \psi_{\text{odom}} + r\end{aligned}\tag{B.1}$$

where $(x_{\text{odom}}, y_{\text{odom}}, \psi_{\text{odom}})$ is a pose in the EKF odometry frame and $(x_{\text{world}}, y_{\text{world}}, \psi_{\text{world}})$ is the equivalent pose in world coordinates. The translation (t_x, t_y) between the origins and yaw offset r between the headings account for the three unknowns. Observing the vehicle once at a location where both its odometry and world poses are known therefore fixes all three simultaneously, and obtaining those readings is the task of the calibration procedure.

The odometry reading is obtained once the EKF pose has converged, a condition met when consecutive measurements remain within defined position and yaw stability thresholds. Its y follows the ROS coordinate convention REP-103, which takes y as positive to the left, so both the inference loop and simulation environment negate it to align the pose with the lot frame. The corrected pose is then matched with a known reference pose, and together they supply the correspondence from which Eq. (B.1) is solved for the offset (t_x, t_y, r) . In simulation that reference pose is the spawn position specified in the layout file, whereas on the vehicle it is a physical marker at the lot datum, surveyed on site and recorded in `real_world_datum.yaml`. The record locates the marker in the lot frame, where its position and heading form the reference pose against which the converged odometry pose is matched. The same point is also recorded in global coordinates as latitude and longitude on the WGS84 ellipsoid, which the sensor relay uses as its projection origin. That same position is retained once more as an easting and northing in the Universal Transverse Mercator projection, which no node reads and which serves instead as a cross-check that exposes errors in the surveyed figures before deployment.

B.4 Actuation Calibration

Physical actuators respond non-linearly to normalised policy outputs. That mismatch takes a different form on the bipolar steering axis than on the two non-negative pedal axes of Section 3.3.3, so the actuation-calibration layer holds a separate parameter set per channel. Let u_{policy} denote the raw policy output on a channel and c the input at

which its deadband is centred. An actuator that responds asymmetrically about its neutral position takes a c displaced from zero. The shifted output $s = u_{\text{policy}} - c$ is the signed distance of the request from that centre, from which the physical command is obtained as:

$$u_{\text{physical}} = \text{clip}(v, u_{\min}, u_{\max}), \quad v = \begin{cases} b & \text{if } |s| < d \\ g \cdot s + b & \text{otherwise} \end{cases} \quad (\text{B.2})$$

where v is the unclipped command, g is the gain, b is the bias, d is the deadband half-width about c , and $[u_{\min}, u_{\max}]$ is the physical output range imposed on each channel. The deadband branch holds the command at the bias regardless of the input, suppressing jitter around the centre, while the clip bounds the affine branch to the range physically admitted by the channel.

Each of those four parameters describes a physical response that can be established only by measuring the actuator, which leaves all four unmeasured until the vehicle is available. Establishing them per channel requires commanding a range of policy outputs and recording the physical response to each, which traces the response curve of that channel. All four are obtained from a single run of that procedure, each being read off a distinct feature of the same curve. The gain is derived from its slope and the deadband from the smallest input that initiates motion. Any asymmetry between the two directions then determines the deadband centre, leaving the bias to be read from any residual output at zero input.

No such run has yet been possible, so every parameter in `actuation_calibration.yaml` retains the identity value for its role. The gain is unity, the bias and deadband are zero, and the output bounds remain at the full physical range. This combination reduces Eq. (B.2) to its input, as does the complete absence of the file. An uncalibrated vehicle therefore transmits each command as issued by the policy rather than applying a correction unsupported by measurement.

Delivering the calibrated command presumes a drive-by-wire connection to the vehicle, marking the end of the symmetry with the sensor interface of Appendix B.2. An incoming fix arrives in a standard message type already published by off-the-shelf drivers, whereas an outgoing command is written to a Controller Area Network (CAN) bus through a suitable interface adapter. The frames it carries are defined by the manufacturer for that platform alone. No such layer exists in the repository because it can be written only for a selected vehicle. The actuation calibration therefore marks the boundary of what transfers without one.

B.5 Mission and Pre-Deployment Sequence

The parity of the code paths and the two calibrations establish the conditions under which the vehicle may be driven, while the objective of a particular deployment run is stated separately. The objective is recorded in `mission.yaml`, which is read only on the vehicle and names the target bay identifier alongside the site's layout file. The deployment entry point reads both fields, loads the layout through the same cached loader used by the simulator and resolves the bay within it. Either step raises an immediate error if its field is unset, as does a bay identifier absent from the layout. A misconfigured mission is therefore rejected before the vehicle moves. Correct behaviour on hardware remains unproven by a valid configuration alone, and the five staged tests below are intended to establish it before any autonomous run, with a driver at the wheel for the first three so that each component is observed while the policy has no control of the vehicle:

1. Verify the EKF pipeline in isolation against its covariance output.
2. Verify the target-bay pose pipeline against the relative target observations.
3. Verify the actuation calibration against known inputs.
4. Run the first closed-loop test in a single bay under RTK fixed conditions.
5. Test progressively across fix-state tiers and conditions.

Appendix C Hyperparameter Reference

Every arm and seed was trained under the same set of hyperparameter values, summarised in Table 3.13. Those values were fixed in advance rather than searched over, since a separate search per arm would make the configuration a fifth variable and leave any measured difference partly attributable to four searches converging unevenly, as Section 3.8.1 sets out. This appendix therefore records the committed values under the parameter names of the configuration files, so that every figure reported in Chapter 3 can be traced to the entry governing it.

C.1 Committed Configuration

Every value governing a training run is declared in a single configuration file, `train_config.yaml`, and its committed contents are reproduced in Table C.1. Those entries restate the summary of Table 3.13, differing in one respect only, as the four evidential coefficients are listed individually rather than in pairs.

Table C.1 Committed values of `train_config.yaml`.

Parameter	Value	Note
<code>n_steps</code>	2048	Steps per update
<code>batch_size</code>	256	Minibatch size for the epoch loop
<code>n_epochs</code>	5	Optimisation passes per rollout
<code>gamma</code>	0.997	Discount
<code>gae_lambda</code>	0.95	GAE bias/variance trade-off
<code>clip_range</code>	0.2	Policy trust region; <code>clip_range_vf</code> null (no value-function clipping)
<code>learning_rate</code>	0.0003 $\rightarrow$ 0.00001	Stage-owned linear decay; see Table C.2
<code>ent_coef</code>	0.02 $\rightarrow$ 0.006	Stage-owned linear decay; see Table C.2
<code>vf_coef</code>	0.5	Value-loss weight
<code>max_grad_norm</code>	0.5	Global gradient-norm clip
<code>target_kl</code>	0.04	Early-stops epochs above this KL; backup to the clip
<code>net_arch</code>	[256, 256], ReLU	Actor and critic hidden layers
<code>log_std_init</code>	-0.6	Initial action $\sigma \approx 0.55$; standard head only
<code>lambda_reg</code>	0.02	Prior-anchor coefficient; linear warm-up over 50 000 steps
<code>aleatoric_floor</code>	0.04	Sampling standard deviation floored at $\sqrt{0.04} = 0.2$

Two tabulated entries, the learning rate and entropy coefficient, carry only a base value in that file. Every curriculum stage of Section 3.7.2 supplies its own

schedule in place of that base. Each stage therefore contains a `standard_overrides` block for the vanilla and input-uncertainty arms alongside an `evidential_overrides` block for the output-uncertainty and full-method arms. The blocks differ only in the entropy coefficient and are restricted to schedule keys by the governing allowlist. Their values across the curriculum are consequently presented separately in Table C.2, which reproduces the budget and learning-rate columns of Table 3.11.

Table C.2 Per-stage schedules. Each value decays linearly from start to final over the stage budget.

Stage	Budget (10^6 decisions)	Learning rate (start $\rightarrow$ final)	Entropy, standard (start $\rightarrow$ final)	Entropy, evidential (start $\rightarrow$ final)
1	1.5	0.0003 $\rightarrow$ 0.00005	0.003 $\rightarrow$ 0.0005	0.003 $\rightarrow$ 0.0015
2	1.5	0.00015 $\rightarrow$ 0.00005	0.0015 $\rightarrow$ 0.0005	0.0015 $\rightarrow$ 0.0015
3	2.0	0.00015 $\rightarrow$ 0.00004	0.0015 $\rightarrow$ 0.0005	0.0015 $\rightarrow$ 0.0015
4	1.5	0.00012 $\rightarrow$ 0.00004	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015
5	2.0	0.00012 $\rightarrow$ 0.00003	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015
6	1.5	0.0001 $\rightarrow$ 0.00002	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015

The pattern of those schedules follows from two design choices made in Section 3.7.2. The learning rate restarts each stage at approximately three times the final value of the preceding one, a level low enough that a nearly converged policy is not destabilised at a difficulty boundary. The entropy floor then varies by head type, with the standard blocks descending to 0.0005 in every stage while the evidential blocks remain at 0.0015. The evidential schedules also begin at a higher value from stage 4 onwards. Exploration on the evidential head is tied to the aleatoric estimate, so both of those departures are required to keep it from collapsing.

No entry of Table C.1 is conditioned on an arm, so the one declaration governs all four and no contrast reported in Chapter 4 can be traced to a configuration difference between them. The entropy floor of Table C.2 is the single exception, and it is set from the head type alone, being the guard against variance collapse on the evidential head rather than a value any arm earned through its measured performance.

Appendix D Reproduction Guide

This appendix documents the procedure for reproducing the result tables of Chapter 4 from a freshly cloned copy of the source, covering the container stack, the seed conventions, the make targets that constitute the pipeline and the repository structure they operate on. Those targets span every workflow in the project and are documented in full, with their variables, in the repository’s `COMMANDS.md`, from which the subset required for reproduction is restated in Appendix D.3. Both that file and the full source it accompanies are available [on GitHub](#).

D.1 Container Stack

All operations occur within Docker, so the host requires only Git, Docker Engine, Make and the NVIDIA Container Toolkit, with no local installation of Python, CARLA [4] or ROS. Everything those operations depend on is instead pinned inside the images, where `pyproject.toml` fixes the Python dependencies to the exact versions captured from the working training image. Code directories are the exception, being bind-mounted into each container from the host so that an edit takes effect without a rebuild. Table D.1 lists the three services across which this arrangement is orchestrated.

Table D.1 The three container services orchestrated by `docker-compose`.

Service	Base image	Contents
<code>carla-server</code>	CARLA 0.9.16	Headless simulator with GPU passthrough
<code>ros2-bridge</code>	ROS 2 Jazzy	CARLA bridge, <code>robot_localization</code> EKF, sensor relay nodes
<code>training</code>	NGC PyTorch 24.10	Stable-Baselines3 2.8.0, Gymnasium 1.2.3, the evidential networks, <code>rcipy</code>

Parallel data collection is also available across these three services, scaled by replicating the `carla-server` and `ros2-bridge` pair rather than the `training` service. The number of replicas is set by the `parallel_workers` key of `train_config.yaml`, from which N simulator-and-bridge pairs are brought up by the worker scripts. Each pair is isolated on three counts, with a distinct port block, a separate DDS domain and its own EKF state file, so that no traffic passes between concurrent workers.

D.2 Seeds and Determinism

The single declaration each run draws its seed from is the `seed` entry of `configs/deployment/agent_config.yaml`, read by both the training process and the

separate noise container rather than set independently in each. Changing the seed of a run is therefore a single edit, with no possibility of the network initialisation and the sensor-noise realisations falling out of step.

The three seed legs of Section 3.8.2 are held apart by the output layout, each written under a `seed_<N>` directory, with N read from the checkpoint leaf and the directory placed one level above the per-arm folders. The records of one seed therefore cannot be overwritten by another. The wider hierarchy this layer sits in is detailed in Appendix D.3.

D.3 Pipeline Reference

Table D.2 lists the make targets that reproduce the reported results, in pipeline order. Two variables recur across them, `BASELINE` designating one of the four arms and `CHECKPOINT` identifying a single run. Neither is a path, both being bare identifiers, from which the full location within the hierarchical output structure is reconstructed by the recipes.

Table D.2 Make targets for the full pipeline.

Target	Purpose
<code>docker-build</code>	Build the three images
<code>docker-up</code> / <code>docker-down</code>	Start or stop the worker pairs and the training stack
<code>docker-train</code> <code>STAGE=N</code> <code>BASELINE=<arm></code>	One curriculum stage; stages 2 to 6 add <code>CHECKPOINT=<leaf></code> to resume from the previous stage
<code>docker-eval</code> <code>BASELINE=<arm></code> <code>CHECKPOINT=<leaf></code>	The 200-episode sweep over the four conditions; <code>NO_SAFETY=1</code> produces the without-wrapper variant
<code>run-seed-leg</code>	One seed's full leg: every arm through all six stages, the final-stage evaluations and the per-seed analysis suite; idempotent, so re-running resumes a crash
<code>analyse-ablation</code> / <code>analyse-gate</code>	Per-seed cross-arm contrast and gate ROC tables (<code>STAGE=6</code> <code>SEED=<s></code>)
<code>analyse-calibration</code> / <code>handover-timing</code>	Per-checkpoint EKF calibration and handover timing (<code>ARM=<arm></code> <code>CHECKPOINT=<leaf></code>)
<code>analyse-cross-seed</code>	Pools every seed's stage-6 evaluation into the headline tables
<code>docker-test-unit</code> / <code>docker-verify</code>	The test suitekw inside the training container

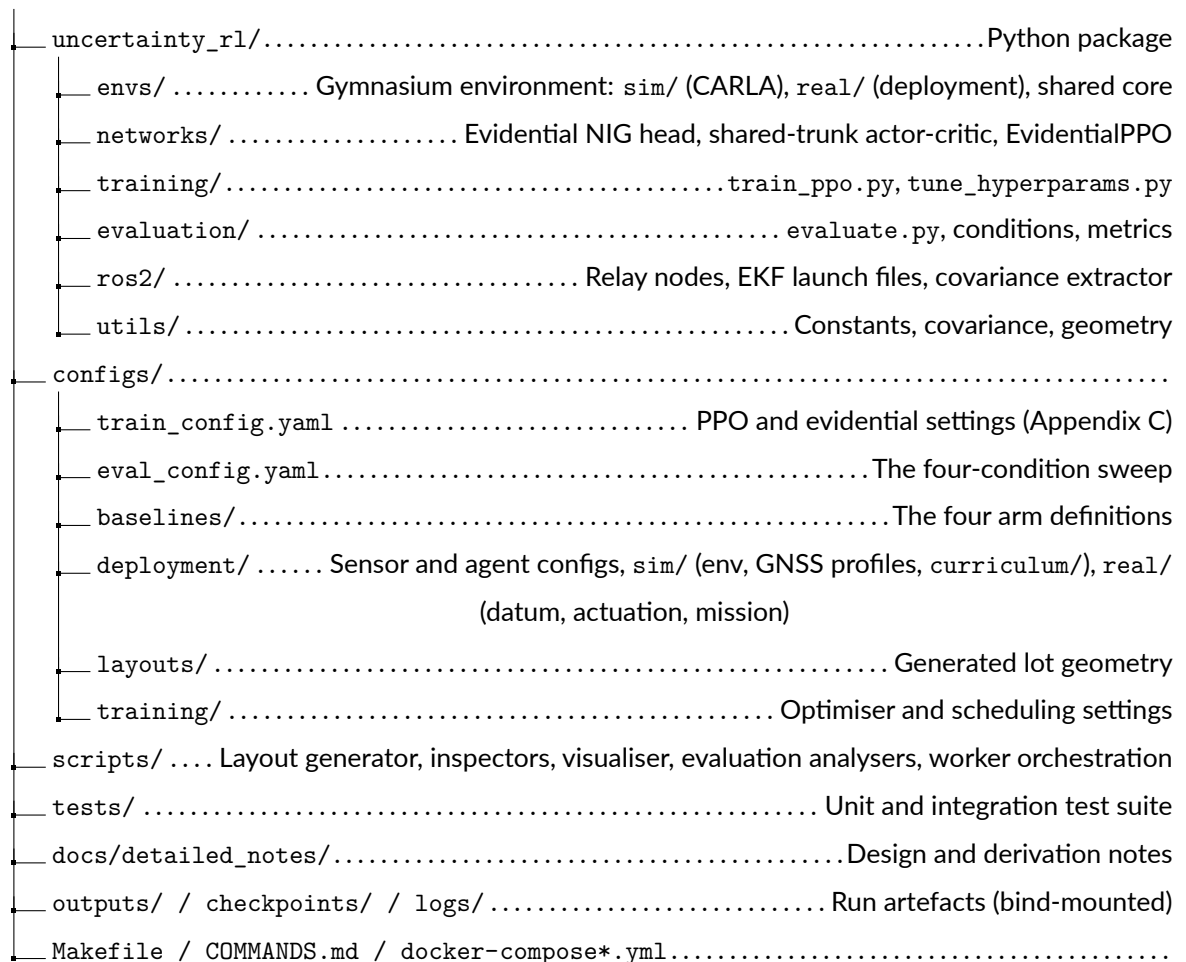
Training and evaluation artefacts are held apart within that hierarchy, a trained run being recorded in `checkpoints/<baseline>/<leaf>/` while its evaluation is written to `outputs/evaluation_results/seed_<N>/<baseline>/<leaf>/`, where the leading `seed_<N>` layer provides the per-seed separation of Appendix D.2. Each evaluation leaf is then divided once more, into a `with_wrapper` and a `without_wrapper` directory,

according to the mode of the safety wrapper. The protocol of Section 4.1.3 is run with the wrapper disabled, so every result in Chapter 4 is read from the without-wrapper directory.

D.4 Repository Structure

The directory tree that every path named in this appendix belongs to is given in Fig. D.1. Its top level keeps three things apart, the Python package, the configuration that parameterises it, and the artefacts a run produces, the last of these being written to directories absent from a fresh clone. The contents of each subpackage and script directory are documented in a README alongside it.

Figure D.1 Repository layout.



Appendix E Per-Seed Evaluation Results

The per-seed success rates and mean final position errors underlying the pooled task figures of Chapter 4 are recorded in Table E.1, so that each step of aggregation can be checked rather than taken on trust. Each row corresponds to one combination of arm, condition and training seed, summarising the 200 deterministic evaluation episodes run under it. Both are derived from these rows, as the pooled figures of Table 4.2 are the mean over the three seeds of a given arm and condition, and the per-seed ranges of Section 4.7 are the extrema.

Table E.1: Per-seed success rate and mean final position error
($n = 200$ episodes per row).

Arm	Condition	Seed	Success (%)	Error (m)
vanilla_ppo	anchor_deployment	7	10.0	11.59
vanilla_ppo	anchor_deployment	42	32.5	8.72
vanilla_ppo	anchor_deployment	123	29.0	11.43
vanilla_ppo	anchor_empty	7	18.5	4.82
vanilla_ppo	anchor_empty	42	41.5	2.36
vanilla_ppo	anchor_empty	123	40.0	1.87
vanilla_ppo	gnss_degrade_one_way	7	12.5	9.36
vanilla_ppo	gnss_degrade_one_way	42	28.0	11.33
vanilla_ppo	gnss_degrade_one_way	123	41.5	7.65
vanilla_ppo	ood_irregular_rtk_fixed	7	0.0	36.01
vanilla_ppo	ood_irregular_rtk_fixed	42	0.0	19.22
vanilla_ppo	ood_irregular_rtk_fixed	123	0.0	23.44
input_uncertainty	anchor_deployment	7	36.5	11.25
input_uncertainty	anchor_deployment	42	23.0	10.45
input_uncertainty	anchor_deployment	123	16.0	10.24
input_uncertainty	anchor_empty	7	21.0	1.75
input_uncertainty	anchor_empty	42	43.5	1.03

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Table E.1: Per-seed success rate and mean final position error ($n = 200$ episodes per row). (Continued)

Arm	Condition	Seed	Success (%)	Error (m)
input_uncertainty	anchor_empty	123	27.5	1.98
input_uncertainty	gnss_degrade_one_way	7	41.5	9.08
input_uncertainty	gnss_degrade_one_way	42	24.5	8.71
input_uncertainty	gnss_degrade_one_way	123	19.5	7.19
input_uncertainty	ood_irregular_rtk_fixed	7	0.0	25.11
input_uncertainty	ood_irregular_rtk_fixed	42	0.0	21.32
input_uncertainty	ood_irregular_rtk_fixed	123	0.0	25.08
output_uncertainty	anchor_deployment	7	18.0	4.80
output_uncertainty	anchor_deployment	42	22.0	5.42
output_uncertainty	anchor_deployment	123	17.0	6.13
output_uncertainty	anchor_empty	7	32.0	1.80
output_uncertainty	anchor_empty	42	19.5	1.54
output_uncertainty	anchor_empty	123	34.0	1.44
output_uncertainty	gnss_degrade_one_way	7	23.0	4.76
output_uncertainty	gnss_degrade_one_way	42	20.5	5.58
output_uncertainty	gnss_degrade_one_way	123	16.5	6.49
output_uncertainty	ood_irregular_rtk_fixed	7	0.0	18.65
output_uncertainty	ood_irregular_rtk_fixed	42	0.0	23.31
output_uncertainty	ood_irregular_rtk_fixed	123	0.0	23.11
full_method	anchor_deployment	7	39.5	7.54
full_method	anchor_deployment	42	47.0	4.08
full_method	anchor_deployment	123	51.0	12.45
full_method	anchor_empty	7	48.0	0.37
full_method	anchor_empty	42	78.5	0.71

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Table E.1: Per-seed success rate and mean final position error
($n = 200$ episodes per row). (Continued)

Arm	Condition	Seed	Success (%)	Error (m)
full_method	anchor_empty	123	67.5	0.52
full_method	gnss_degrade_one_way	7	40.0	8.30
full_method	gnss_degrade_one_way	42	55.0	3.13
full_method	gnss_degrade_one_way	123	62.5	9.55
full_method	ood_irregular_rtk_fixed	7	0.0	19.64
full_method	ood_irregular_rtk_fixed	42	0.0	19.37
full_method	ood_irregular_rtk_fixed	123	0.0	20.46